

USER MANUAL BOOK

Sistem Informasi Pemantauan Spasial Temporal Kualitas Udara DKI Jakarta Berbasis Categorical Boosting dan Attention Mechanism

Oleh:

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I. Pendahuluan

Dashboard ini merupakan hasil penelitian tesis pascasarjana Universitas Diponegoro yang dikembangkan sebagai Sistem Informasi Pemantauan Spasial Temporal Kualitas Udara DKI Jakarta Berbasis Categorical Boosting dan Attention Mechanism yang menganalisis indeks standar pencemar udara ($PM_{2.5}$, PM_{10} , SO_2 , CO , O_3 , NO_2) serta parameter meteorologi seperti suhu, titik embun, kelembapan, angin, curah hujan, tekanan udara, dan jarak pandang berdasarkan dimensi ruang dan waktu. Sistem ini dirancang untuk memvisualisasikan pola distribusi, korelasi, dan tren polutan pada interval 1, 8, 12, 24, dan 48 jam. Dengan penerapan algoritma CatBoost dan Attention Mechanism, sistem mampu menyoroti fitur dengan intensitas polusi ekstrem guna menghasilkan prediksi yang lebih akurat untuk mendukung mitigasi polusi udara.

II. Latar Belakang

Pencemaran udara di wilayah perkotaan padat penduduk seperti DKI Jakarta telah menjadi ancaman serius bagi kesehatan masyarakat. Partikulat halus $PM_{2.5}$ berkontribusi terhadap berbagai penyakit seperti bronkitis, asma, penyakit jantung, stroke, bahkan kematian dini. Kompleksitas dinamika polusi udara di Jakarta disebabkan oleh interaksi antara berbagai polutan, variabilitas cuaca dan iklim, emisi kendaraan, serta aktivitas industri. Faktor-faktor tersebut membuat kualitas udara berubah secara cepat baik secara spasial (antarwilayah) maupun temporal (antarwaktu), sehingga tidak dapat dijelaskan secara efektif oleh metode statistik konvensional. Kondisi ini mendorong perlunya pendekatan berbasis analisis spasio-temporal yang mampu mempelajari pola hubungan antarstasiun pemantauan serta perubahan kualitas udara dari waktu ke waktu.

III. Tujuan

Tujuan utama dari pengembangan sistem ini meliputi:

1. Pemantauan emisi untuk membantu pemerintah daerah dalam mengendalikan sumber pencemar seperti kendaraan berat atau aktivitas industri pada wilayah yang berpotensi mengalami tingkat pencemaran tinggi.
2. Peringatan dini kesehatan masyarakat dengan memberikan notifikasi ketika konsentrasi $PM_{2.5}$ diperkirakan melebihi $5 \mu g/m^3$ (ambang tahunan WHO), sehingga diharapkan dapat menurunkan kasus gangguan pernapasan dan menghemat biaya kesehatan masyarakat hingga Rp50–100 miliar per tahun.

3. Perencanaan kota berkelanjutan dengan menyediakan informasi prediktif jangka panjang untuk mendukung penetapan zona rendah emisi, pengembangan ruang hijau, serta penataan lokasi fasilitas sensitif seperti sekolah dan rumah sakit.

IV. Spesifikasi Teknis dan Sumber Data

Pemrograman dalam penelitian ini dikonfigurasi menggunakan Miniconda3 sebagai *package manager* dan *environment handler* untuk memastikan isolasi serta kestabilan lingkungan kerja. Dalam implementasinya, Streamlit (v1.41.1) digunakan sebagai *framework* untuk pengembangan *dashboard*. Pustaka Pandas (v2.2.3) dan NumPy (v2.2.1) dimanfaatkan untuk pengolahan, transformasi, dan analisis numerik terhadap data deret waktu polusi udara dan variabel meteorologi. Penelitian ini dijalankan menggunakan perangkat keras dengan sistem operasi Windows 11, prosesor Intel (R) Core (TM) i5-1135G7 Gen-11 2.40 GHz serta dukungan GPU Nvidia Tesla T4 yang cocok untuk proses *fine-tuning* model.

Data yang digunakan mencakup data indeks standar pencemar udara oleh Dinas Lingkungan Hidup DKI Jakarta melalui laman <https://satudata.jakarta.go.id/> serta data klimatologi per jam dari NOAA (National Centers for Environmental Information) oleh US Government yang dapat diakses pada <https://www.ncei.noaa.gov/access/search/data-search/global-historical-climatology-network-hourly?bbox=-5.834,106.464,-6.596,107.226&pageNum=21>.

V. Panduan Penggunaan Dashboard

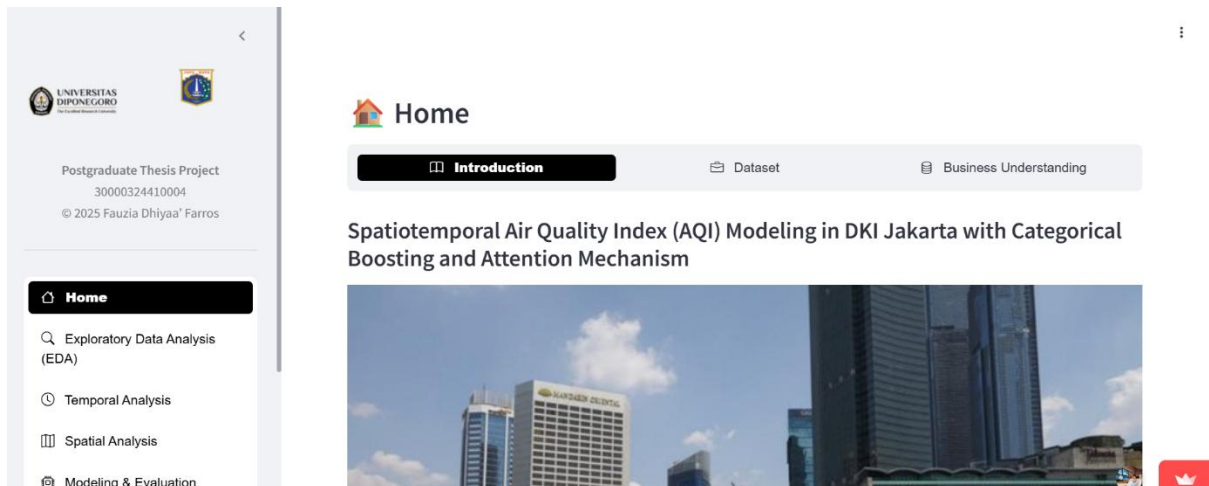
Untuk menggunakan dashboard, *user* bisa membuka browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera, dll.) dengan koneksi internet yang aktif dan stabil, kemudian mengakses tautan <https://catboostatt-aqi-prediction.streamlit.app/>.

5.1. Home (Beranda)

Halaman **Home** terdapat tiga subhalaman utama, yaitu **Introduction**, **Dataset**, dan **Business Understanding**, yang menjelaskan gambaran umum penelitian, tujuan pengembangan sistem, serta output yang diharapkan.

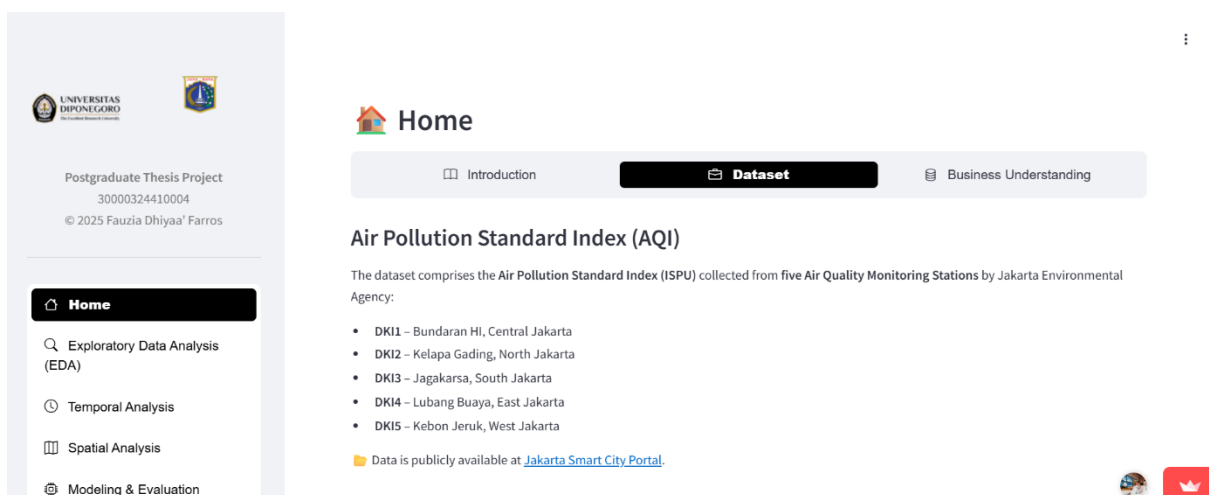
5.1.1. Introduction

Halaman **Introduction** berisi latar belakang masalah polusi udara di DKI Jakarta. Di halaman ini disajikan juga nilai ambang batas parameter AQI berdasarkan standar WHO sebagai acuan penilaian kualitas udara.



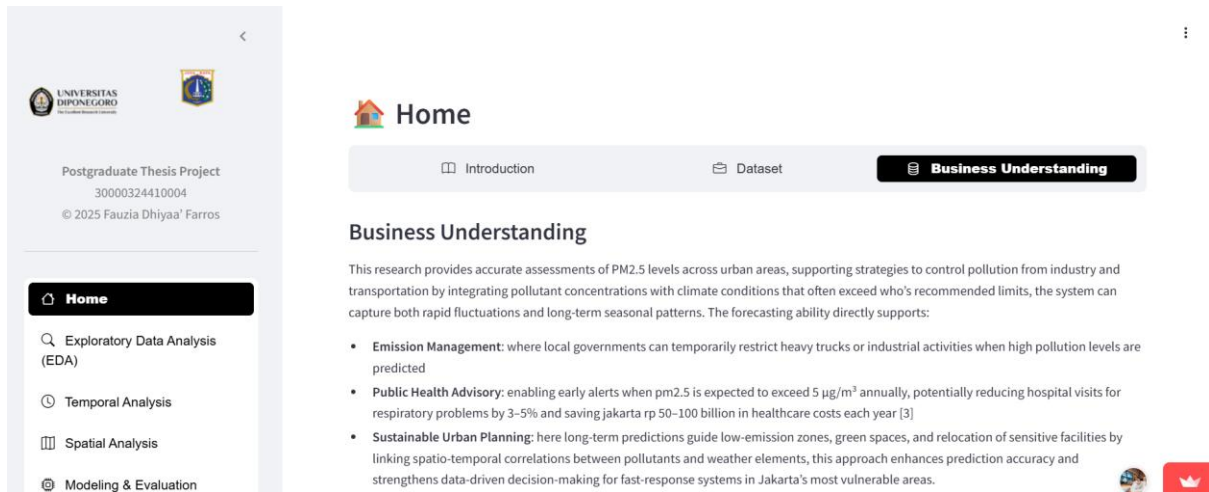
5.1.2. Dataset

Halaman **Dataset** berisi penjelasan tentang dua sumber utama dataset yang digunakan dalam penelitian, Air Pollution Standard Index (AQI) dan Global Historical Climatology Network Hourly.



5.1.3. Business Understanding

Halaman **Business Understanding** menjelaskan tujuan utama pengembangan sistem.

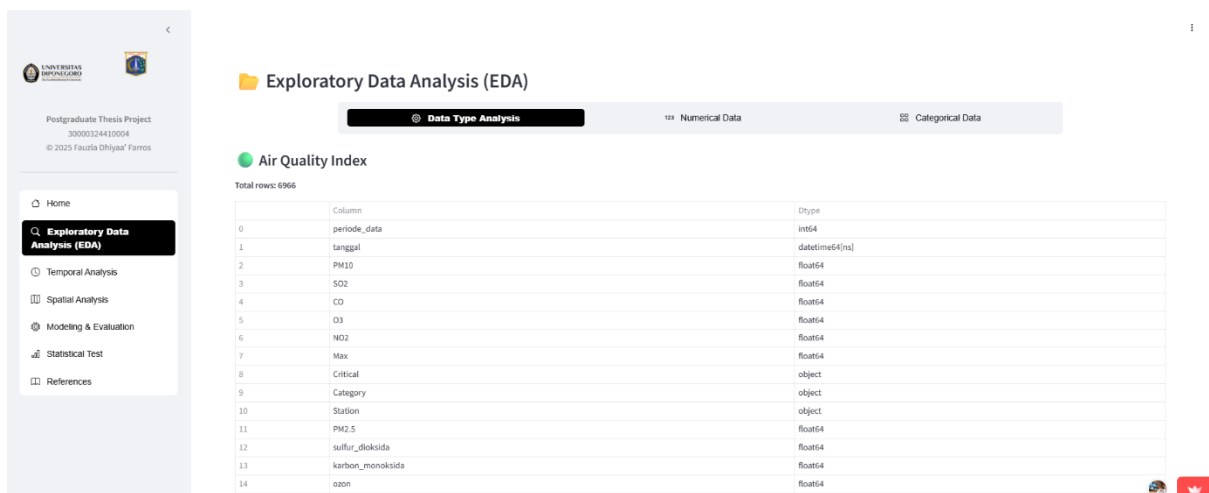


5.2. Exploratory Data Analysis (EDA)

Halaman **Exploratory Data Analysis (EDA)** terdiri dari tiga subhalaman utama, yaitu **Data Type Analysis**, **Numerical Data**, dan **Categorical Data**. Halaman ini menjelaskan proses awal dalam memahami karakteristik data yang meliputi identifikasi tipe data, analisis distribusi nilai numerik konsentrasi polutan dan klimatologi, serta pemeriksaan data kategorikal untuk melihat pola, korelasi, dan potensi anomali data (outlier) sebelum dilakukan pemodelan prediksi.

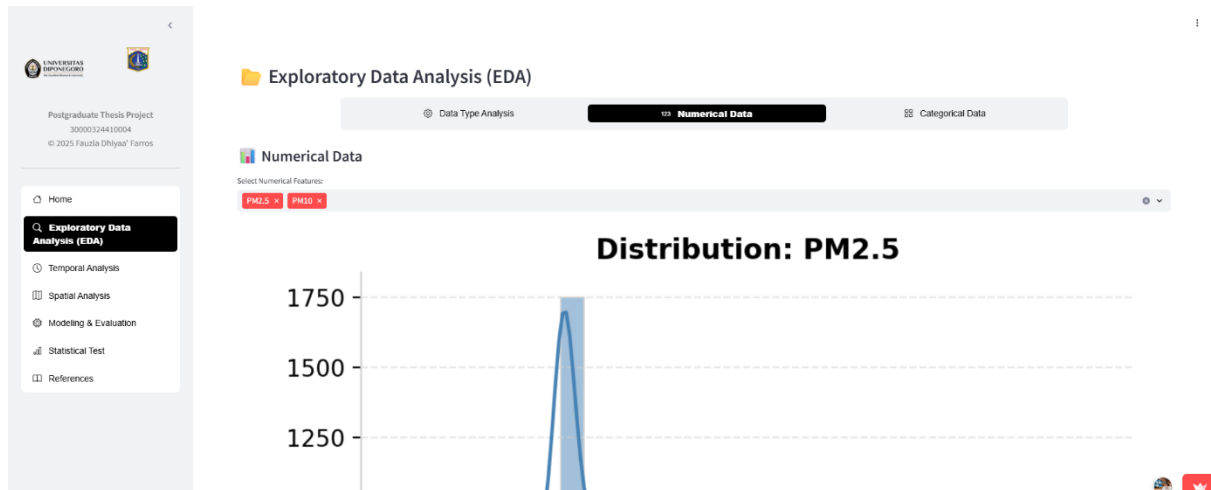
5.2.1. Data Type Analysis

Halaman **Exploratory Data Analysis (EDA)** menampilkan struktur dan karakteristik awal data kualitas udara (AQI) dan meteorologi (klimatologi). Kedua data ini, dengan total 106.197 baris, digunakan untuk pemodelan prediksi.



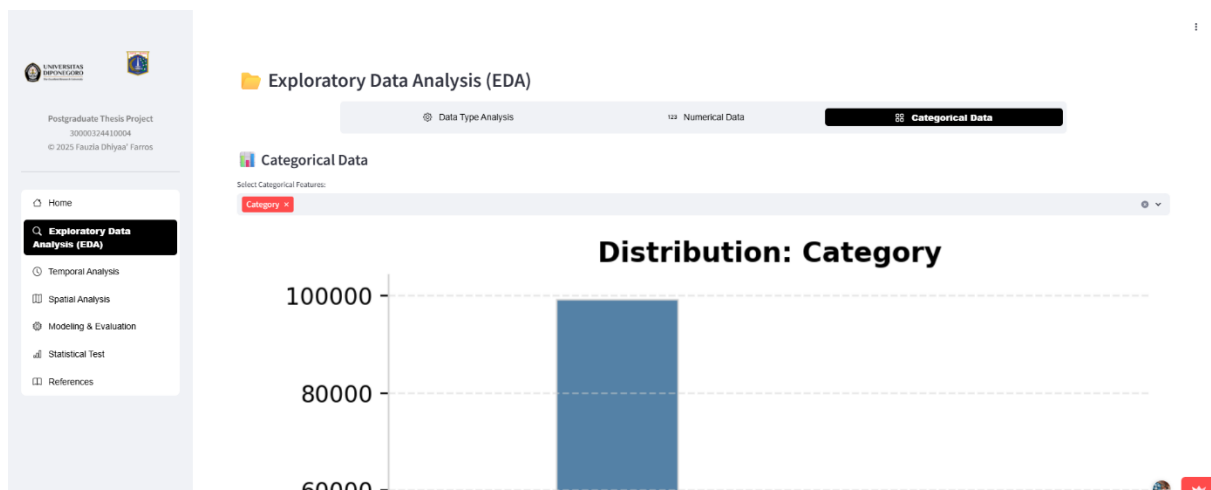
5.2.2. Numerical Data

Halaman **Numerical Data** berisi tentang data numerik yang digunakan dalam penelitian, yaitu PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, temperature, humidity, wind speed, visibility, dan pressure.



5.2.3. Categorical Data

Halaman **Categorical Data** berisi tentang data kategori yang digunakan dalam penelitian, yaitu Station dan Category.



5.3. Temporal Analysis

Halaman **Temporal Analysis** terdiri dari tiga subhalaman utama, yaitu **Annual Trend**, **Hourly Trend**, dan **Peak Trend**, yang berfungsi untuk menampilkan pola perubahan kualitas udara (AQI) dan variabel meteorologi berdasarkan waktu. Melalui analisis ini, pengguna dapat mengamati tren tahunan, fluktuasi per jam, serta momen puncak konsentrasi polutan dan kondisi cuaca di wilayah DKI Jakarta.

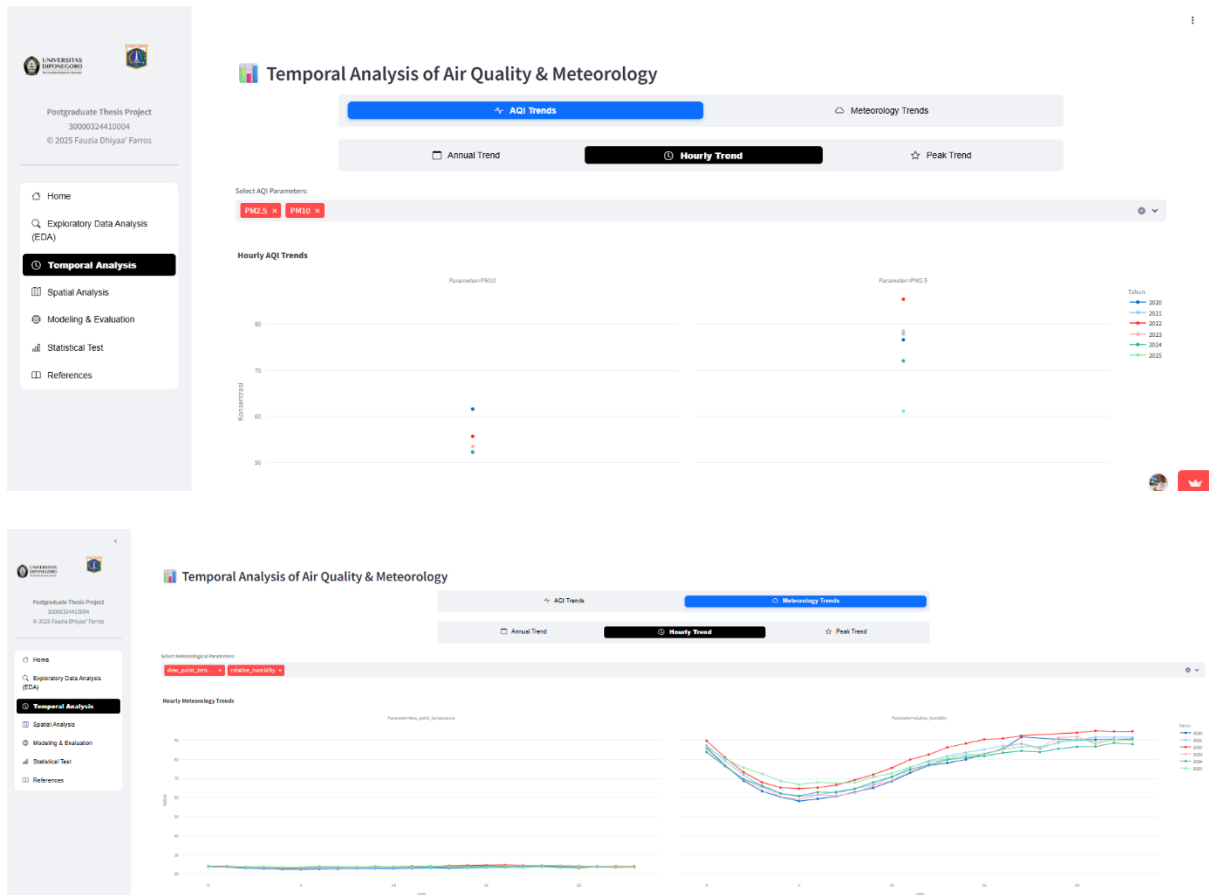
5.3.1. Annual Trend

Halaman ini menampilkan pola persebaran tiap parameter kualitas udara dan meteorologi selama periode 2020–2024 melalui subhalaman **Annual Trend**, yang menggambarkan tren tahunan perubahan konsentrasi polutan serta faktor cuaca di DKI Jakarta.



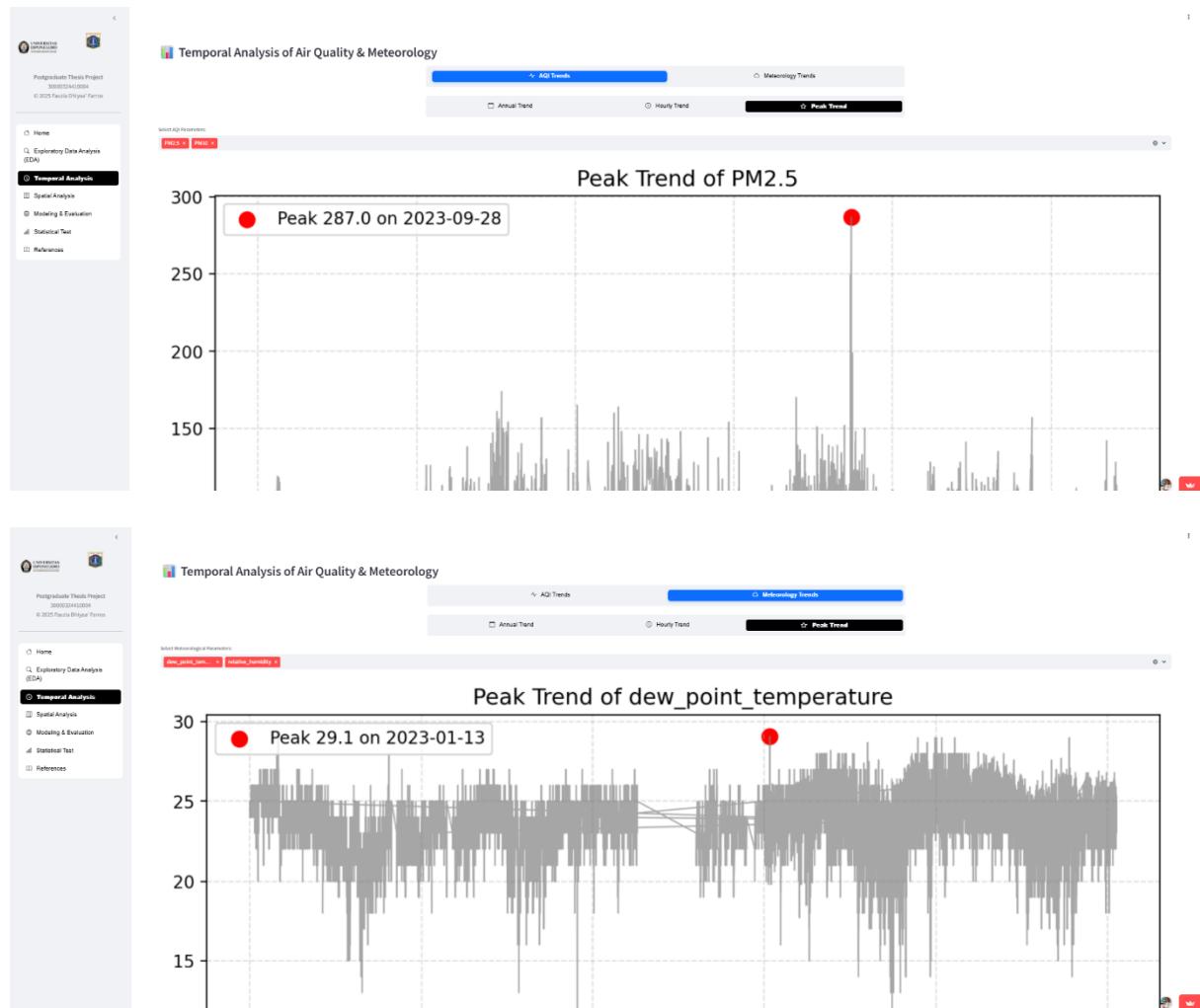
5.3.2. Hourly Trend

Halaman **Hourly Trend** menampilkan pola perubahan tiap parameter kualitas udara dan meteorologi secara per jam selama periode 2020–2024, berdasarkan data deret waktu 1, 8, 12, 24, dan 48 jam. Analisis ini menunjukkan dinamika jangka pendek hingga panjang, di mana PM meningkat pada jam sibuk pagi dan sore akibat aktivitas lalu lintas dan industri, sedangkan gas polutan serta faktor meteorologi seperti suhu, kelembapan, angin, curah hujan, dan tekanan udara berfluktuasi sepanjang hari.



5.3.3. Peak Trend

Halaman **Peak Trend** menampilkan titik tertinggi dari setiap konsentrasi parameter polutan dan klimatologi selama periode 2020–2024. Melalui nilai kulminasi ini, pengguna dapat melihat pola parameter yang paling berpengaruh terhadap waktu terjadinya puncak polusi. Titik-titik tersebut juga dapat diamati secara geografis pada **Spatial Map**, sehingga hubungan antara lokasi, waktu, dan intensitas polutan dapat dianalisis secara lebih menyeluruh.

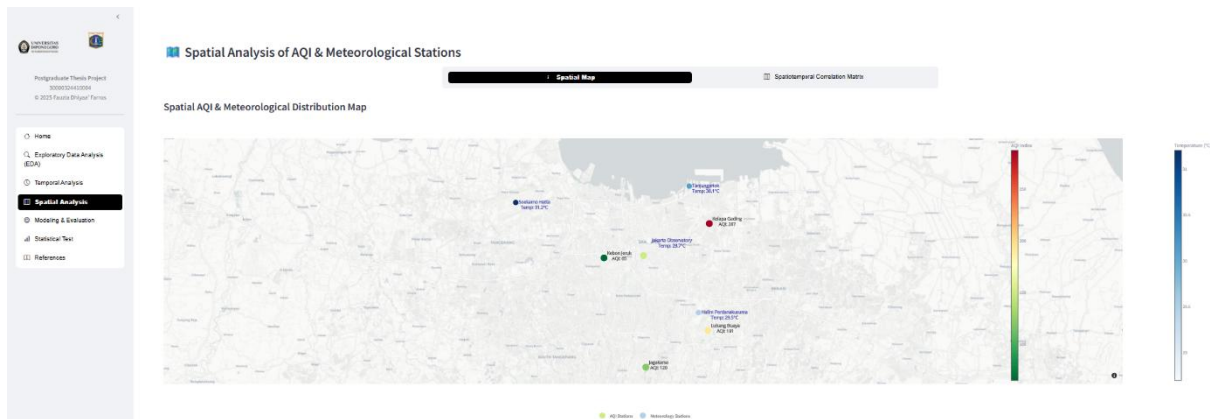


5.4. Spatial Analysis

Halaman **Spatial Analysis** berisi dua subhalaman utama, yaitu **Spatial Map** dan **Spatiotemporal Correlation Matrix**, yang digunakan untuk menganalisis hubungan spasial kualitas udara di DKI Jakarta.

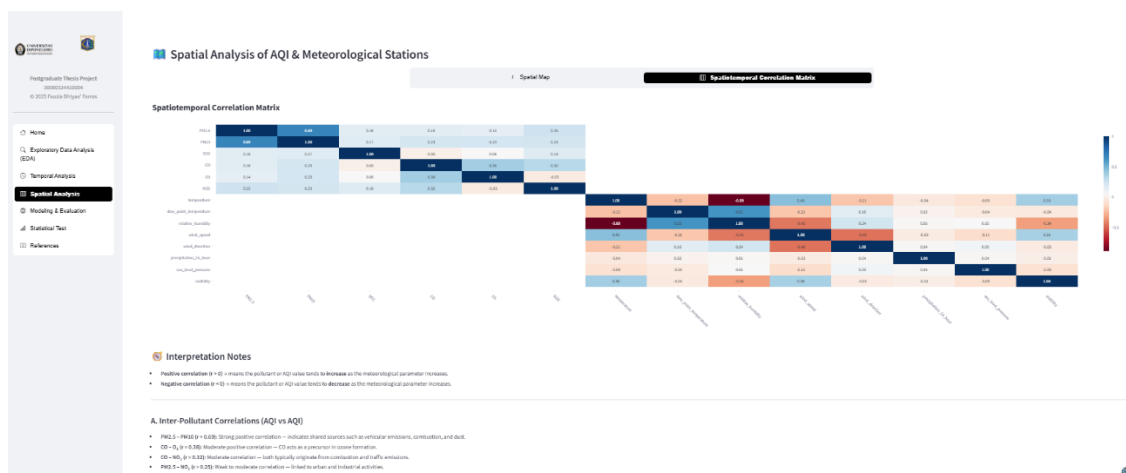
5.4.1. Spatial Map

Melalui **Spatial Map**, pengguna dapat melihat persebaran konsentrasi polutan secara geografis berdasarkan lokasi stasiun pemantauan di DKI Jakarta. Titik pengamatan AQI Stations mencakup Bundaran HI (Jakarta Pusat), Kelapa Gading (Jakarta Utara), Jagakarsa (Jakarta Selatan), Lubang Buaya (Jakarta Timur), dan Kebon Jeruk (Jakarta Barat). Sementara itu, Meteorology Stations terdiri dari Halim Perdanakusuma Station (Jakarta Timur), Soekarno Hatta Station (Banten), Jakarta Observatory, dan Tanjung Priok Station (Jakarta Utara).



5.4.2. Spatiotemporal Correlation Matrix

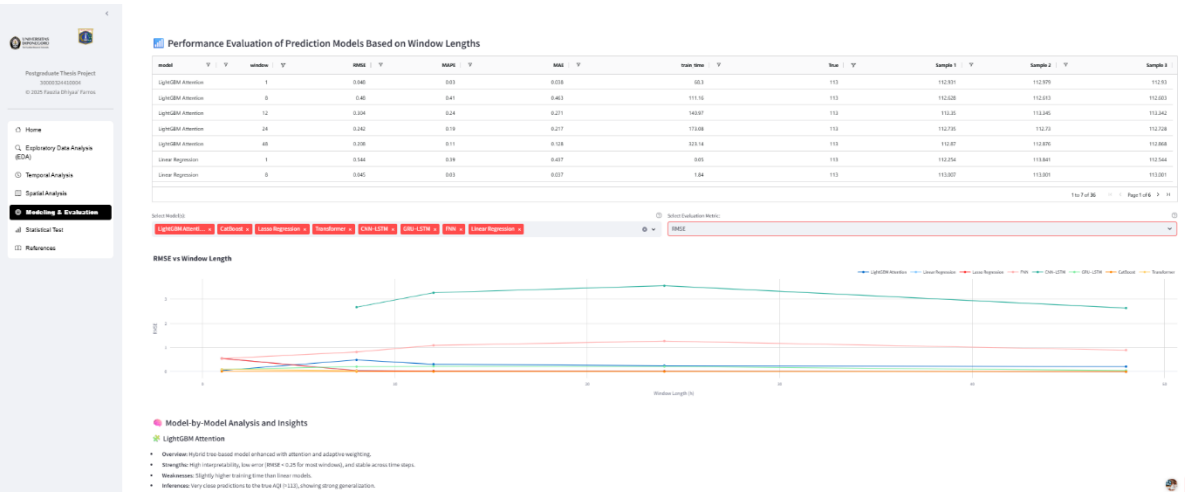
Halaman **Spatiotemporal Correlation Matrix** menampilkan hubungan antarparameter polutan dan klimatologi selama 2020–2024, seperti CO–Temperature ($r=0.32$), $PM_{2.5}$ –Visibility ($r=0.25$), PM_{10} –Visibility ($r=0.23$), SO_2 –Visibility ($r=0.18$), dan PM_{10} –Temperature ($r=0.17$), yang menggambarkan keterkaitan antara perubahan suhu, visibilitas, dan konsentrasi polutan di DKI Jakarta.



5.5. Modeling & Evaluation

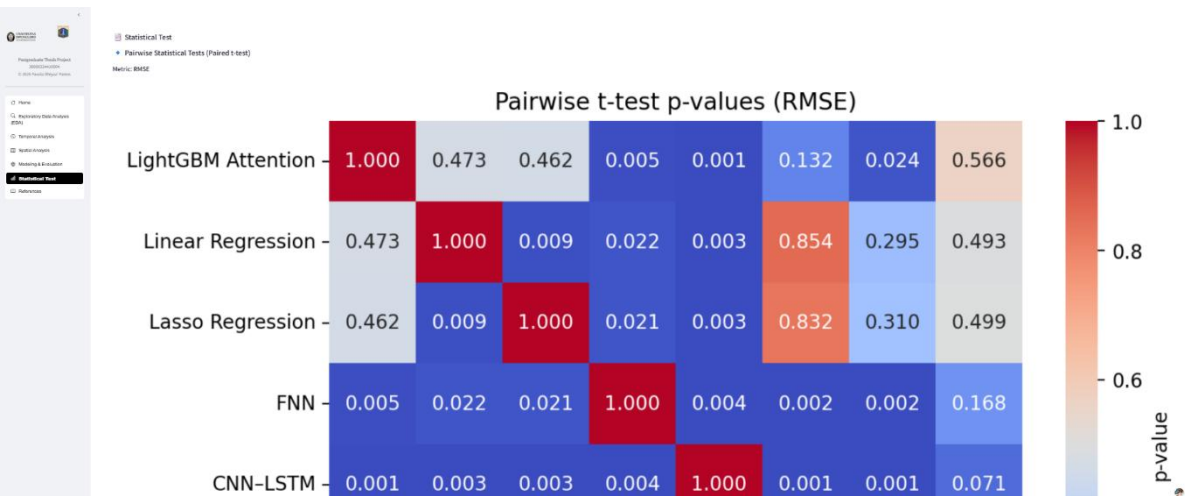
Halaman **Modeling & Evaluation** terbagi menjadi beberapa bagian yang memudahkan pengguna memahami performa model prediksi AQI. Halaman ini menampilkan kinerja berbagai model seperti LightGBM Attention, CatBoost, Linear Regression, FNN, Lasso, CNN-LSTM, GRU-LSTM, dan Transformer berdasarkan panjang window waktu, serta memungkinkan pemilihan metrik evaluasi (RMSE, MAE, MAPE) untuk menilai akurasi, interpretabilitas, dan efisiensi. Bagian **Performance Evaluation of Prediction Models Based on Window Lengths** menyajikan kinerja model pada berbagai panjang window, sementara **Model-by-Model Analysis** and **Insights** memberikan insight tentang kekuatan, kelemahan, dan penggunaan terbaik tiap model, misalnya LightGBM Attention stabil dengan error

rendah dan CatBoost cepat serta akurat. **Summary of Findings** merangkum performa keseluruhan, termasuk model terbaik, baseline kuat, dan arsitektur paling canggih. **Metric Definitions** menjelaskan arti RMSE, MAE, MAPE, dan waktu pelatihan. Terakhir, **Actual vs Predicted (Sample Analysis)** memvisualisasikan perbandingan nilai AQI aktual dan prediksi, dan **Prediction Samples** menampilkan contoh hasil prediksi untuk interpretasi lebih lanjut.



5.6. Statistical Test

Halaman **Statistical Test** menampilkan analisis perbandingan kinerja model prediksi secara statistik. **Pairwise t-test** digunakan untuk menilai perbedaan kinerja antara dua model tertentu berdasarkan metrik seperti RMSE, MAE, dan MAPE, sehingga terlihat model mana yang lebih unggul secara pasangan. Sementara itu, **Friedman Test** bersama **Post-hoc Nemenyi** membandingkan seluruh model sekaligus untuk mengetahui apakah terdapat perbedaan signifikan secara keseluruhan, sekaligus mengidentifikasi pasangan model yang berbeda secara statistik.



VI. Kesimpulan dan Rekomendasi Pemanfaatan Sistem

Sistem Informasi Pemantauan Spasial Temporal Kualitas Udara DKI Jakarta Berbasis Categorical Boosting dan Attention Mechanism dikembangkan melalui penerapan machine learning berbasis Categorical Boosting (CatBoost) dan Attention Mechanism untuk menganalisis data indeks pencemar udara ($PM_{2.5}$, PM_{10} , SO_2 , CO , O_3 , NO_2) serta parameter meteorologi periode 2020–2024. Sistem ini mendeteksi pola distribusi polutan secara spasial-temporal, mendukung pengendalian sumber emisi kendaraan dan industri, memberikan peringatan dini kesehatan saat konsentrasi $PM_{2.5}$ diperkirakan melebihi ambang batas WHO, serta membantu perencanaan kota berkelanjutan melalui informasi prediktif untuk zona rendah emisi dan pengembangan ruang hijau. Hasil pengujian delapan model—Linear Regression, Lasso, Feedforward Neural Network (FNN), CNN-LSTM, GRU-LSTM, CatBoost, LightGBM+Attention+Weights, dan Transformer—menunjukkan bahwa CatBoost memberikan performa terbaik dengan RMSE = 0.0050, MAE = 0.0036, dan MAPE = 0.0032, diikuti oleh Transformer (RMSE = 0.0372) dan Lasso Regression (RMSE = 0.1249). Keunggulan CatBoost berasal dari kemampuan boosting berbasis gradien yang efisien terhadap data non-linear dan kompleksitas spasio-temporal, sekaligus mengurangi overfitting. Uji statistik Friedman Test menghasilkan $\chi^2 = 26.8333$ dengan p-value = 3.57×10^{-4} , menandakan perbedaan signifikan antar model. Analisis post-hoc Nemenyi memperlihatkan bahwa CatBoost berbeda signifikan dibanding CNN-LSTM ($p = 0.002$) dan FNN ($p = 0.020$), sementara perbedaannya terhadap Transformer tidak signifikan ($p = 0.989$), menunjukkan performa keduanya relatif berdekatan di kelompok atas. Model CNN-LSTM dan FNN memiliki kinerja terendah (RMSE > 0.9) karena kompleksitas arsitektur yang kurang sesuai dengan volume data. Secara keseluruhan, kombinasi CatBoost dan Attention Mechanism terbukti paling optimal untuk memprediksi dinamika spasial-temporal kualitas udara di DKI Jakarta dengan akurasi tinggi dan kesalahan prediksi minimal.

VII. Referensi

Bagian ini menjelaskan sumber kode yang digunakan dalam pengembangan Sistem Informasi Pemantauan Spasial Temporal Kualitas Udara DKI Jakarta Berbasis Categorical Boosting dan Attention Mechanism.

7.1. Source Code Interface Dashboard dengan Streamlit

Seluruh antarmuka dashboard dan proses pemodelan dikembangkan menggunakan bahasa pemrograman Python 3.11 dengan environment Miniconda3 bernama **aqi-env** untuk memastikan konsistensi dependensi dan kestabilan sistem. Dashboard diimplementasikan dan dideploy melalui Streamlit Cloud, yang memudahkan pengguna untuk mengakses sistem secara daring tanpa perlu instalasi lokal. Beberapa pustaka utama yang wajib diinstal meliputi **Streamlit (v1.41.1)** untuk pengembangan antarmuka, **Pandas (v2.2.3)** dan **Numpy (v2.2.1)** untuk analisis data, **Matplotlib (v3.10.0)** dan **Seaborn**

(v0.13.2) untuk visualisasi, **Scikit-learn (v1.6.0)** untuk pemodelan machine learning, serta **Scikit-posthocs (v0.11.4)** dan **SciPy (v1.16.2)** untuk analisis statistik dan uji lanjut. Konfigurasi lengkap environment ini menjadi dasar utama dalam memastikan replikasi hasil dan fungsi dashboard secara optimal di platform Streamlit Cloud. Source code yang digunakan disajikan pada lampiran source code di bawah ini atau diunduh melalui tautan berikut:

 <https://github.com/fauzia-dhiyaa-farros/catboostatt-aqi-prediction>

environment.yaml

```
name: aqi-env
channels:
  - conda-forge
  - defaults
dependencies:
  - python=3.11
  - pip
  - pip:
    - streamlit==1.41.1
    - streamlit-aggrid==1.1.9
    - streamlit-option-menu==0.4.0
    - pandas==2.2.3
    - numpy==2.2.1
    - matplotlib==3.10.0
    - seaborn==0.13.2
    - plotly==5.24.1
    - altair==5.5.0
    - affine==2.4.0
    - statsmodels==0.14.5
    - scikit-learn==1.6.0
    - scikit-posthocs==0.11.4
    - scipy==1.16.2
```

catboostatt_dashboard.py

```
import os
import io
import math
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import plotly.express as px
import plotly.graph_objects as go
import streamlit as st
import scikit_posthocs as sp
from scipy.stats import friedmanchisquare
from scipy.stats import ttest_rel
from streamlit_option_menu import option_menu
from st_aggrid import AgGrid, GridOptionsBuilder

# =====
# Page Config
# =====
```

```

st.set_page_config(
    page_title="Spatiotemporal AQI Forecasting Dashboard",
    page_icon="🌍",
    layout="wide"
)

# =====
# Load Dataset
# =====
@st.cache_data
def load_data():
    try:
        df = pd.read_csv("aqi_meteo.csv", parse_dates=["tanggal"])
        return df
    except FileNotFoundError:
        st.error("File `aqi_meteo.csv` not found. Please put it in the
same folder as app.py")
        return pd.DataFrame()

df_aqi_combined = load_data()

# =====
# Sidebar
# =====
with st.sidebar:
    col1, col2 = st.columns([1, 1])
    with col1:
        st.image("undiplogo.png", width=120)
    with col2:
        st.image("jktlogo.png", width=120)

    st.markdown(
        """
        <div style='text-align: center; color: gray; margin-top: 15px;
font-size: 16px;'>
            <b>Postgraduate Thesis Project</b><br>
            30000324410004 <br>
            © 2025 Fauzia Dhiyaa' Farros<br>
        </div>
        """,
        unsafe_allow_html=True
    )

    st.markdown("---")

    # === Sidebar Menu ===
    selected = option_menu(
        None,
        ["Home", "Exploratory Data Analysis (EDA)", "Temporal
Analysis",
        "Spatial Analysis", "Modeling & Evaluation", "Statistical
Test", "References"],
        icons=["house", "search", "clock", "map", "cpu", "bar-chart",
"book"],
        menu_icon="cast",
        default_index=0,
        styles={
            "container": {"padding": "0!important", "background-
color": "#ffffff"},
            "icons": {"color": "#000000", "font-size": "18px"},

```

```

        "nav-link": {"color": "#000000", "margin": "5px", "text-align": "left"},
        "nav-link-selected": {"background-color": "#000000", "color": "#ffffff"},
    }
)

# =====
# Home
# =====
if selected == "Home":
    st.header("🏠 Home")
    sub_selected = option_menu(
        None,
        ["Introduction", "Dataset", "Business Understanding"],
        icons=["book", "briefcase", "database"],
        orientation="horizontal",
        styles={
            "nav-link-selected": {"background-color": "#000000", "color": "white"}
        }
    )

# ----- Introduction -----
if sub_selected == "Introduction":
    st.subheader("Spatiotemporal Air Quality Index (AQI) Modeling in DKI Jakarta with Categorical Boosting and Attention Mechanism")

    st.image(
        "jktimage.jpeg",
        caption="View of Jakarta City, DKI Jakarta",
        use_container_width=True,
        width=500,
        output_format="auto"
    )

    st.markdown("""
    In densely populated urban areas like DKI Jakarta, air pollution—especially fine particulate matter (PM2.5)—poses serious health risks, contributing to bronchitis, asthma, heart disease, stroke, and even death [1]. The interplay of multiple pollutants, weather variability, traffic emissions, and industrial activities complicates air quality dynamics [2], [13], prompting recent research to treat pollution as a spatiotemporal problem rather than isolated measurements [3], [14]. This approach captures both temporal patterns (e.g., hourly or daily changes) and spatial relationships across monitoring stations. Traditional statistical methods often fail to model these complex, nonlinear patterns, whereas machine learning models like Categorical Boosting handle high-dimensional structured data efficiently using histogram-based binning and a leaf-wise growth strategy [4], [15]. However, Categorical Boosting alone may not effectively prioritize temporal or environmental features unless explicitly guided. To address this, attention mechanisms—originally from natural language processing—are now applied in time series forecasting to
    """)

```

```

        emphasize critical features such as sudden weather shifts or
        pollution spikes [5], [6], [16], [17], [18].
        Combining Categorical Boosting with attention improves
        interpretability and predictive accuracy [6], [9], [10], [19], [20],
        though many models remain limited to low-resolution data or
        single-station inputs [7], [21]. This study
        proposes a hybrid attention-Categorical Boosting framework to
        model multivariate spatiotemporal correlations
        of PM2.5 in DKI Jakarta, using hourly climate data from five
        satellite-derived sources and pollutant data
        (PM2.5, PM10, NO2, SO2, CO, O3) from five ground stations
        (2020–2024). An attention mechanism is applied
        after preprocessing and temporal segmentation to highlight key
        variables at each timestep, followed by
        Categorical Boosting to model nonlinear interactions. The
        method enhances interpretability and forecasting performance,
        offering insights for data-driven policymaking and urban air
        quality management.
        """

        # WHO Reference Table
        st.markdown("***WHO Reference Value of Each AQI Parameter**")
        who_data = {
            "Parameter": ["PM2.5", "PM10", "SO2", "CO", "O3", "NO2",
"Max",
                                "Category", "temperature",
"dew_point_temperature",
                                "relative_humidity", "wind_speed",
"wind_direction",
                                "precipitation", "precipitation_24_hour"],
            "WHO Reference Value": [
                "< 5 µg/m3 (annual)", "< 15 µg/m3 (annual)", "< 40
µg/m3 (10-min)",
                "< 4 mg/m3 (daily)", "< 100 µg/m3 (8-hour)", "< 10
µg/m3 (annual)",
                "Depends on pollutant type", "Good", "20–25 °C",
"Depends on air temperature",
                "40–60%", "0–15 m/s", "None", "Seasonal dependent",
"Seasonal dependent"
            ]
        }
        st.table(pd.DataFrame(who_data))

        elif sub_selected == "Dataset":
            st.subheader("Air Pollution Standard Index (AQI)")
            st.markdown(
                """
                The dataset comprises the **Air Pollution Standard Index
                (ISPU)** collected from **five Air Quality Monitoring Stations** by
                Jakarta Environmental Agency:

                - **DKI1** - Bundaran HI, Central Jakarta
                - **DKI2** - Kelapa Gading, North Jakarta
                - **DKI3** - Jagakarsa, South Jakarta
                - **DKI4** - Lubang Buaya, East Jakarta
                - **DKI5** - Kebon Jeruk, West Jakarta

                📁 Data is publicly available at [Jakarta Smart City
                Portal] (https://satudata.jakarta.go.id/).
                """
            )

```

```

    )

    st.subheader("Global Historical Climatology Network – Hourly
(NOAA)")
    st.markdown(
        """
        The dataset also includes hourly meteorological data
        from NOAA, with variables such as temperature, precipitation,
        humidity, latitude/longitude, and elevation. Bayu collected from four
        Air Quality Monitoring Stations.

        Collected from four stations:
        - IDI0000WIHH – Halim Perdanakusuma (East Jakarta)
        - IDU096749-1 – Soekarno Hatta (Banten)
        - IDU096745-1 – Jakarta Observatory
        - IDU096741-1 – Tanjung Priok (North Jakarta)

        📁 Data is publicly available at [Global Historical
        Climatology Network
        (Hourly)] (https://www.ncei.noaa.gov/access/search/data-search/global-historical-climatology-network-hourly).
        """
    )

    # ----- Business Understanding -----
    elif sub_selected == "Business Understanding":
        st.subheader("Business Understanding")
        st.markdown("""
        This research provides accurate assessments of PM2.5 levels
        across urban areas, supporting strategies to control pollution from
        industry and transportation by integrating pollutant concentrations
        with climate conditions that often exceed who's recommended limits,
        the system can capture both rapid fluctuations and long-term seasonal
        patterns.

        The forecasting ability directly supports:
        - Emission Management: where local governments can
        temporarily restrict heavy trucks or industrial activities when high
        pollution levels are predicted
        - Public Health Advisory: enabling early alerts when pm2.5
        is expected to exceed 5 µg/m³ annually, potentially reducing hospital
        visits for respiratory problems by 3-5% and saving jakarta rp 50-100
        billion in healthcare costs each year [3]
        - Sustainable Urban Planning: here long-term predictions
        guide low-emission zones, green spaces, and relocation of sensitive
        facilities by linking spatio-temporal correlations between pollutants
        and weather elements, this approach enhances prediction accuracy and
        strengthens data-driven decision-making for fast-response systems in
        Jakarta's most vulnerable areas.

        By linking spatiotemporal correlations between pollutants
        and weather elements, this approach enhances prediction accuracy and
        strengthens decision-making for fast-response systems in Jakarta's
        most vulnerable areas.
        """)

    # =====
    # EDA
    # =====
    elif selected == "Exploratory Data Analysis (EDA)":
        st.header("📁 Exploratory Data Analysis (EDA)")

```

```

sub_selected = option_menu(
    None,
    ["Data Type Analysis", "Numerical Data", "Categorical Data"],
    icons=["gear", "123", "grid"],
    orientation="horizontal",
    styles={
        "nav-link": {"color": "#000000", "margin": "5px", "text-align": "center"},
        "nav-link-selected": {"background-color": "#000000", "color": "#ffffff"},
    }
)

# ---- Data Type Analysis ----
if sub_selected == "Data Type Analysis":
    # === Air Quality Index (AQI) ===
    st.markdown("### 🟢 Air Quality Index")
    st.write("***Total rows: 6966**")

    numerical_features = pd.DataFrame({
        "Column": [
            "periode_data", "tanggal", "PM10", "SO2", "CO", "O3",
            "NO2", "Max",
            "Critical", "Category", "Station", "PM2.5",
            "sulfur_dioksida", "karbon_monoksida", "ozon",
            "nitrogen_dioksida", "bulan"
        ],
        "Dtype": [
            "int64", "datetime64[ns]", "float64", "float64",
            "float64", "float64", "float64", "float64",
            "object", "object", "object", "float64",
            "float64", "float64", "float64", "float64", "float64"
        ]
    })
    st.table(numerical_features)

    # === Meteorology (Climatology) ===
    st.markdown("### 🟡 Meteorology (Climatology)")
    st.write("***Total rows: 99231**")

    meteo_columns = pd.DataFrame({
        "Column": [
            "station_name", "date", "latitude", "longitude",
            "elevation", "temperature",
            "dew_point_temperature", "relative_humidity",
            "wind_speed", "wind_direction",
            "precipitation", "precipitation_24_hour",
            "sea_level_pressure", "visibility"
        ],
        "Dtype": [
            "object", "datetime64[ns]", "float64", "float64",
            "float64", "float64",
            "float64", "float64", "float64", "float64",
            "float64", "float64", "float64", "float64"
        ]
    })
    st.table(meteo_columns)

    total_rows = 6966 + 99231
    st.markdown(

```

```

f**Overall, the total number of processed rows across
both datasets is {total_rows:,}. "
    "This includes hourly air quality measurements (AQI) and
meteorological variables.**"
)

# ---- Numerical Data ----
elif sub_selected == "Numerical Data":
    st.subheader("📊 Numerical Data")
    numerical_features = [
        'PM2.5', 'PM10', 'SO2', 'CO', 'O3', 'NO2',
        'temperature', 'dew_point_temperature',
        'relative_humidity',
        'wind_speed', 'wind_direction', 'precipitation_24_hour',
        'sea_level_pressure', 'visibility'
    ]
    num_options = [col for col in numerical_features if col in
df_aqi_combined.columns]
    default_selection = [col for col in ['PM2.5', 'PM10'] if col
in num_options]
    selected_num = st.multiselect("Select Numerical Features:",
options=num_options, default=default_selection)

    if selected_num:
        for col in selected_num:
            fig, ax = plt.subplots(figsize=(6, 4))
            sns.histplot(
                df_aqi_combined[col].dropna(),
                kde=True, bins=30,
                color='steelblue', edgecolor="lightgray",
linewidth=0.6, ax=ax
            )
            ax.set_title(f"Distribution: {col}", fontsize=12,
fontweight="bold")

            # Styling biar mirip lightgbm_spatio_temp.py
            ax.spines["top"].set_visible(False)
            ax.spines["right"].set_visible(False)
            ax.spines["left"].set_color("lightgray")
            ax.spines["bottom"].set_color("lightgray")
            ax.grid(True, axis='y', linestyle="--", alpha=0.5,
color="lightgray")

            st.pyplot(fig)

    st.subheader("📈 Descriptive Statistics")
    st.write(df_aqi_combined[selected_num].describe().T)

# ---- Categorical Data ----
elif sub_selected == "Categorical Data":
    st.subheader("📊 Categorical Data")

    if "Category" not in df_aqi_combined.columns:
        def categorize_pm25(pm25):
            if pm25 <= 50:
                return "GOOD"
            elif pm25 <= 100:
                return "MODERATE"
            elif pm25 <= 150:
                return "UNHEALTHY"

```

```

        elif pm25 <= 200:
            return "VERY UNHEALTHY"
        else:
            return "HAZARDOUS"

    df_aqi_combined["Category"] =
df_aqi_combined["PM2.5"].apply(categorize_pm25)

    df_aqi_combined['station'] =
df_aqi_combined['station'].astype(str).str.strip()
    df_aqi_combined['Category'] =
df_aqi_combined['Category'].astype(str).str.upper().str.strip()

    station_mapping = {
        'DKI1': 'DKI1 - Bundaran HI',
        'DKI2': 'DKI2 - Kelapa Gading',
        'DKI3': 'DKI3 - Jagakarsa',
        'DKI4': 'DKI4 - Lubang Buaya',
        'DKI5': 'DKI5 - Kebon Jeruk',
    }
    df_aqi_combined['station'] =
df_aqi_combined['station'].apply(lambda x: station_mapping.get(x, x))

    df_filtered = df_aqi_combined[df_aqi_combined['Category'] !=
"TIDAK ADA DATA"].copy()

    station_order = list(station_mapping.values())
    df_filtered['station'] =
pd.Categorical(df_filtered['station'], categories=station_order,
ordered=True)
    df_filtered['Category'] =
pd.Categorical(df_filtered['Category']) # hanya yang ada

    categorical_features = ['station', 'Category']
    cat_options = [c for c in categorical_features if c in
df_filtered.columns]
    default_cat_selection = [col for col in ['Category'] if col in
cat_options]
    selected_cat = st.multiselect("Select Categorical Features:",
options=cat_options, default=default_cat_selection)

    if selected_cat:
        num_cols = len(selected_cat)
        n_cols = 2 if num_cols > 1 else 1
        n_rows = -(-num_cols // n_cols)

        fig, axes = plt.subplots(n_rows, n_cols, figsize=(6 *
n_cols, 4 * n_rows))
        axes = axes.flatten() if num_cols > 1 else [axes]

        for i, col in enumerate(selected_cat):
            order = df_filtered[col].cat.categories
            sns.countplot(
                data=df_filtered,
                x=col,
                order=order,
                color='steelblue',
                edgecolor="lightgray",
                linewidth=0.6,
                ax=axes[i]
            )

```

```

        axes[i].set_title(f"Distribution: {col}", fontsize=12,
fontweight="bold")
        axes[i].tick_params(axis="x", rotation=45)

        axes[i].spines["top"].set_visible(False)
        axes[i].spines["right"].set_visible(False)
        axes[i].spines["left"].set_color("lightgray")
        axes[i].spines["bottom"].set_color("lightgray")
        axes[i].grid(True, axis='y', linestyle="--",
alpha=0.5, color="lightgray")

        for j in range(i + 1, len(axes)):
            axes[j].axis("off")

    st.pyplot(fig)

# =====
# Temporal Analysis
# =====

if selected == "Temporal Analysis":
    st.header("🌤️ Temporal Analysis of Air Quality & Meteorology")

    menu_style = {
        "nav-link": {"color": "#000000", "margin": "5px", "text-
align": "center"},
        "nav-link-selected": {"background-color": "#000000", "color":
"#ffffff"},
    }

    # ----- LOAD METEOROLOGY DATA -----
    @st.cache_data(show_spinner=False)
    def load_meteorology_data(folder_path="meteorology"):
        file_list = [f for f in os.listdir(folder_path) if
f.endswith(".csv")]
        df_list = []
        for file_name in file_list:
            file_path = os.path.join(folder_path, file_name)
            df = pd.read_csv(file_path, low_memory=False)
            df.columns = df.columns.str.lower()
            df_list.append(df)

        df_all_meteo = pd.concat(df_list, ignore_index=True)

        selected_cols = [
            'station', 'station_name', 'date', 'latitude',
'longitude', 'elevation',
            'temperature', 'dew_point_temperature',
'relative_humidity', 'wind_speed',
            'wind_direction', 'precipitation',
'precipitation_24_hour',
            'sea_level_pressure', 'visibility'
        ]
        selected_cols_existing = [col for col in selected_cols if col
in df_all_meteo.columns]
        df_all_meteo = df_all_meteo[selected_cols_existing].copy()

        df_all_meteo.dropna(subset=['latitude', 'longitude', 'date'],
how='any', inplace=True)
        df_all_meteo['date'] = pd.to_datetime(df_all_meteo['date'],
errors='coerce')

```

```

df_all_meteo = df_all_meteo.dropna(subset=['date'])

num_cols = df_all_meteo.select_dtypes(include=['float64',
'int64']).columns
for col in num_cols:
    median_value = df_all_meteo[col].median()
    df_all_meteo[col] = df_all_meteo[col].fillna(median_value)

return df_all_meteo

df_all_meteo = load_meteorology_data()

# ----- SETUP -----
numerical_features = [
    'PM2.5', 'PM10', 'SO2', 'CO', 'O3', 'NO2',
    'temperature', 'dew_point_temperature', 'relative_humidity',
    'wind_speed', 'wind_direction', 'precipitation_24_hour',
    'sea_level_pressure', 'visibility'
]
numerical_features = numerical_features[:6]
meteo_columns = [
    'temperature', 'dew_point_temperature', 'relative_humidity',
    'wind_speed', 'wind_direction', 'precipitation_24_hour',
    'sea_level_pressure', 'visibility'
]
meteo_columns = [c for c in meteo_columns if c in
df_all_meteo.columns]

# ----- PREPROCESS -----
@st.cache_data(show_spinner=False)
def preprocess_data(df):
    df = df.copy()
    df.columns = df.columns.str.lower()
    rename_map = {
        'pm_sepuluh': 'PM10', 'pm_duakomalima': 'PM2.5', 'pm_10':
'PM10',
        'pm25': 'PM2.5', 'pm2.5': 'PM2.5', 'pm10': 'PM10', 'so2':
'SO2',
        'co': 'CO', 'o3': 'O3', 'no2': 'NO2',
        'parameter_pencemar_kritis': 'Critical', 'kategori':
'Category',
        'lokasi_spku': 'Station', 'tanggal': 'date', 'stasiun':
'Station'
    }
    df.rename(columns=rename_map, inplace=True)
    df = df.loc[:, ~df.columns.duplicated()]
    df['date'] = pd.to_datetime(df['date'], errors='coerce')
    df = df.dropna(subset=['date'])
    df['Tahun'] = df['date'].dt.year
    df['Bulan'] = df['date'].dt.month
    df['Hari-Bulan'] = df['date'].dt.strftime('%m-%d')
    df['Jam'] = df['date'].dt.hour
    return df

@st.cache_data(show_spinner=False)
def melt_data(df, value_vars, value_name="Konsentrasi"):
    valid_vars = [v for v in value_vars if v in df.columns]
    return df.melt(
        id_vars=['date', 'Tahun', 'Bulan', 'Hari-Bulan', 'Jam'],
        value_vars=valid_vars,
        var_name='Parameter',

```

```

        value_name=value_name
    )

    # ----- SUBTAB -----
    selected_tab = option_menu(
        None,
        ["AQI Trends", "Meteorology Trends"],
        icons=["activity", "cloud"],
        orientation="horizontal",
        styles={
            "nav-link": {
                "color": "#000000", # warna biru Bootstrap
                "margin": "5px",
                "text-align": "center",
                "font-weight": "500"
            },
            "nav-link-selected": {
                "background-color": "#0d6efd",
                "color": "#ffffff",
                "border-radius": "8px",
                "font-weight": "600",
                "box-shadow": "0 2px 6px rgba(13,110,253,0.3)"
            },
        },
    )

    # =====
    # AQI TRENDS
    # =====
    if selected_tab == "AQI Trends":
        df_clean = preprocess_data(df_aqi_combined)
        df_melted = melt_data(df_clean, numerical_features)

        selected_sub = option_menu(
            None,
            ["Annual Trend", "Hourly Trend", "Peak Trend"],
            icons=["calendar", "clock", "star"],
            orientation="horizontal",
            styles=menu_style,
        )

        selected_params = st.multiselect("Select AQI Parameters:",
numerical_features, default=["PM2.5", "PM10"])

        if selected_sub == "Annual Trend":
            for param in selected_params:
                df_param = df_melted[df_melted['Parameter'] == param]
                fig, ax = plt.subplots(figsize=(10, 5))
                for year in sorted(df_param['Tahun'].unique()):
                    subset = df_param[df_param['Tahun'] == year]
                    ax.plot(subset['Hari-Bulan'],
subset['Konsentrasi'], label=str(year), linewidth=1)

                    ax.set_title(f"Annual Trend of {param}", fontsize=13,
fontweight='bold')
                    ax.legend(title="Year", loc='upper left', fontsize=9)
                    ax.grid(True, linestyle='--', alpha=0.4)
                    ax.set_xticks(np.linspace(0, len(df_param['Hari-
Bulan'].unique()), 12))
                    ax.set_xticklabels(["Jan", "Feb", "Mar", "Apr", "May", "Jun
", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"], rotation=45)

```

```

plt.tight_layout()
st.pyplot(fig, use_container_width=True)

elif selected_sub == "Hourly Trend":
    df_hourly =
df_melted[df_melted['Parameter'].isin(selected_params)]
    hourly = df_hourly.groupby(['Jam', 'Tahun',
'Parameter'])['Konsentrasi'].mean().reset_index()
    fig = px.line(hourly, x="Jam", y="Konsentrasi",
color="Tahun", facet_col="Parameter", facet_col_wrap=2, title="Hourly
AQI Trends", markers=True)
    fig.update_layout(height=600, showlegend=True)
    st.plotly_chart(fig, use_container_width=True)

elif selected_sub == "Peak Trend":
    for param in selected_params:
        df_param = df_melted[df_melted['Parameter'] == param]
        if df_param['Konsentrasi'].notna().any():
            idxmax = df_param['Konsentrasi'].idxmax()
            peak_date = df_param.loc[idxmax, 'date']
            peak_value = df_param.loc[idxmax, 'Konsentrasi']
            fig, ax = plt.subplots(figsize=(10, 5))
            ax.plot(df_param['date'], df_param['Konsentrasi'],
linewidth=0.8, color='gray', alpha=0.7)
            ax.scatter(peak_date, peak_value, color='red',
s=80, label=f"Peak {peak_value:.1f} on {peak_date.date()}")
            ax.set_title(f"Peak Trend of {param}",
fontsize=13)

            ax.legend(loc='upper left')
            ax.grid(True, linestyle='--', alpha=0.4)
            st.pyplot(fig, use_container_width=True)

# =====
# METEOROLOGY TRENDS
# =====
elif selected_tab == "Meteorology Trends":
    df_clean = preprocess_data(df_all_meteo)
    df_melted = melt_data(df_clean, meteo_columns,
value_name="Value")

    selected_sub = option_menu(
        None,
        ["Annual Trend", "Hourly Trend", "Peak Trend"],
        icons=["calendar", "clock", "star"],
        orientation="horizontal",
        styles=menu_style,
    )

    selected_params = st.multiselect("Select Meteorological
Parameters:", meteo_columns, default=["dew_point_temperature",
"relative_humidity"])

    if selected_sub == "Annual Trend":
        for param in selected_params:
            df_param = df_melted[df_melted['Parameter'] == param]
            fig, ax = plt.subplots(figsize=(10, 5))
            for year in sorted(df_param['Tahun'].unique()):
                subset = df_param[df_param['Tahun'] == year]
                ax.plot(subset['Hari-Bulan'], subset['Value'],
label=str(year), linewidth=1)

```

```

        ax.set_title(f"Annual Trend of {param}", fontsize=13,
fontweight='bold')
        ax.legend(title="Year", loc='upper left', fontsize=9)
        ax.grid(True, linestyle='--', alpha=0.4)
        ax.set_xticks(np.linspace(0, len(df_param['Hari-
Bulan']).unique()), 12))
        ax.set_xticklabels(["Jan", "Feb", "Mar", "Apr", "May", "Jun
", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"], rotation=45)
        plt.tight_layout()
        st.pyplot(fig, use_container_width=True)

    elif selected_sub == "Hourly Trend":
        df_hourly =
df_melted[df_melted['Parameter'].isin(selected_params)]
        hourly = df_hourly.groupby(['Jam', 'Tahun',
'Parameter'])['Value'].mean().reset_index()
        fig = px.line(hourly, x="Jam", y="Value", color="Tahun",
facet_col="Parameter", facet_col_wrap=2, title="Hourly Meteorology
Trends", markers=True)
        fig.update_layout(height=600, showlegend=True)
        st.plotly_chart(fig, use_container_width=True)

    elif selected_sub == "Peak Trend":
        for param in selected_params:
            df_param = df_melted[df_melted['Parameter'] == param]
            if df_param['Value'].notna().any():
                idxmax = df_param['Value'].idxmax()
                peak_date = df_param.loc[idxmax, 'date']
                peak_value = df_param.loc[idxmax, 'Value']
                fig, ax = plt.subplots(figsize=(10, 5))
                ax.plot(df_param['date'], df_param['Value'],
linewidth=0.8, color='gray', alpha=0.7)
                ax.scatter(peak_date, peak_value, color='red',
s=80, label=f"Peak {peak_value:.1f} on {peak_date.date()}")
                ax.set_title(f"Peak Trend of {param}",
fontsize=13)

                ax.legend(loc='upper left')
                ax.grid(True, linestyle='--', alpha=0.4)
                st.pyplot(fig, use_container_width=True)

# =====
# SPATIAL ANALYSIS SECTION
# =====
elif selected == "Spatial Analysis":
    st.header("🗺️ Spatial Analysis of AQI & Meteorological Stations")

    sub_selected = option_menu(
        None,
        ["Spatial Map", "Spatiotemporal Correlation Matrix"],
        icons=["info", "map", "link"],
        orientation="horizontal",
        styles={
            "nav-link": {"color": "#000000", "margin": "5px", "text-
align": "center"},
            "nav-link-selected": {"background-color": "#000000",
"color": "white", "border-radius": "8px"},
        }
    )

    # =====
    # 1. Spatial Map

```

```

# =====
if sub_selected == "Spatial Map":
    st.subheader("Spatial AQI & Meteorological Distribution Map")

    aqi_data = [
        ('DKI1', 'Bundaran HI', -6.193, 106.820, 150),
        ('DKI2', 'Kelapa Gading', -6.152, 106.905, 287),
        ('DKI3', 'Jagakarsa', -6.336, 106.823, 120),
        ('DKI4', 'Lubang Buaya', -6.289, 106.903, 191),
        ('DKI5', 'Kebon Jeruk', -6.196, 106.769, 65)
    ]

    meteo_data = [
        ('IDI0000WIHH', 'Halim Perdanakusuma', -6.266, 106.891,
29.5),
        ('IDU096749-1', 'Soekarno Hatta', -6.125, 106.655, 31.2),
        ('IDU096745-1', 'Jakarta Observatory', -6.174, 106.827,
28.7),
        ('IDU096741-1', 'Tanjungpriok', -6.104, 106.879, 30.1)
    ]

    df_aqi = pd.DataFrame(aqi_data,
columns=['ID', 'Name', 'Lat', 'Lon', 'PeakValue'])
    df_meteo = pd.DataFrame(meteo_data,
columns=['ID', 'Name', 'Lat', 'Lon', 'PeakValue'])

    fig = go.Figure()

    # AQI Stations
    fig.add_trace(go.Scattermapbox(
        lat=df_aqi['Lat'],
        lon=df_aqi['Lon'],
        mode='markers+text',
        text=df_aqi.apply(lambda row: f"{row['Name']}<br>AQI:
{row['PeakValue']}"", axis=1),
        textposition="top right",
        textfont=dict(size=12, color='black'),
        marker=dict(
            size=20,
            color=df_aqi['PeakValue'],
            colorscale='RdYlGn_r',
            cmin=df_aqi['PeakValue'].min(),
            cmax=df_aqi['PeakValue'].max(),
            showscale=True,
            colorbar=dict(title="AQI Index", x=0.88, thickness=18)
        ),
        name="AQI Stations",
        hoverinfo="text"
    ))

    # Meteorology Stations
    fig.add_trace(go.Scattermapbox(
        lat=df_meteo['Lat'],
        lon=df_meteo['Lon'],
        mode='markers+text',
        text=df_meteo.apply(lambda row: f"{row['Name']}<br>Temp:
{row['PeakValue']}"", axis=1),
        textposition="bottom right",
        textfont=dict(size=12, color='darkblue'),
        marker=dict(
            size=16,

```

```

        color=df_meteo['PeakValue'],
        colorscale='Blues',
        cmin=df_meteo['PeakValue'].min(),
        cmax=df_meteo['PeakValue'].max(),
        showscale=True,
        colorbar=dict(title="Temperature (°C)", x=1.05,
thickness=18)
    ),
    name="Meteorology Stations",
    hoverinfo="text"
))

# --- Map Layout ---
fig.update_layout(
    mapbox_style="carto-positron",
    mapbox_zoom=10.5,
    mapbox_center={"lat": -6.2, "lon": 106.82},
    height=800,
    margin=dict(r=0,t=50,l=0,b=0),
    legend=dict(
        orientation="h",
        yanchor="bottom",
        y=-0.15,
        xanchor="center",
        x=0.5
    )
)

st.plotly_chart(fig, use_container_width=True)

# =====
# 2. Spatiotemporal Correlation Matrix (Zoomable)
# =====
elif sub_selected == "Spatiotemporal Correlation Matrix":

    numerical_features = [
        'PM2.5', 'PM10', 'SO2', 'CO', 'O3', 'NO2',
        'temperature', 'dew_point_temperature',
'relative_humidity',
        'wind_speed', 'wind_direction', 'precipitation_24_hour',
        'sea_level_pressure', 'visibility'
    ]

    corr_cols = [col for col in numerical_features if col in
df_aqi_combined.columns]

    corr_matrix =
df_aqi_combined[corr_cols].corr(method='pearson')

    fig = px.imshow(
        corr_matrix,
        text_auto=".2f",
        color_continuous_scale="RdBu",
        aspect="auto",
        title="Spatiotemporal Correlation Matrix"
    )

    fig.update_layout(
        title=dict(font=dict(size=24)),
        xaxis_title="",
        yaxis_title="",

```

```

        xaxis=dict(showgrid=False, tickangle=45),
        yaxis=dict(showgrid=False, autorange="reversed"),
        dragmode="zoom",
        hovermode="closest",
        margin=dict(l=60, r=60, t=80, b=60),
        height=700,
    )

    fig.update_layout(
        modebar_add=["zoom", "pan", "resetScale2d"],
        modebar_remove=["lasso2d", "select2d"]
    )

    st.plotly_chart(fig, use_container_width=True)

    st.markdown("""
    ### 🌀 Interpretation Notes

    - **Positive correlation ( $r > 0$ )** → means the pollutant or
    AQI value tends to increase as the meteorological parameter
    increases.
    - **Negative correlation ( $r < 0$ )** → means the pollutant or
    AQI value tends to decrease as the meteorological parameter
    increases.

    ---

    #### **A. Inter-Pollutant Correlations (AQI vs AQI)**
    - **PM2.5 - PM10 ( $r = 0.69$ ):** Strong positive correlation –
    indicates shared sources such as vehicular emissions, combustion, and
    dust.
    - **CO - O3 ( $r = 0.38$ ):** Moderate positive correlation – CO
    acts as a precursor in ozone formation.
    - **CO - NO2 ( $r = 0.32$ ):** Moderate correlation – both
    typically originate from combustion and traffic emissions.
    - **PM2.5 - NO2 ( $r = 0.25$ ):** Weak to moderate correlation –
    linked to urban and industrial activities.
    - **PM10 - NO2 ( $r = 0.23$ ):** Weak to moderate correlation –
    suggests mixed particulate sources.

    ---

    #### **B. Inter-Meteorological Correlations**
    - **Temperature - Relative Humidity ( $r = -0.89$ ):** Very strong
    negative correlation – higher temperatures often lead to lower
    humidity.
    - **Dew Point - Relative Humidity ( $r = 0.62$ ):** Strong
    positive correlation – both reflect moisture content in the air.
    - **Temperature - Wind Speed ( $r = 0.45$ ):** Moderate positive
    correlation – warmer air can enhance convective wind movement.
    - **Pressure - Sea Level Pressure ( $r \approx 0.40-0.50$ ):** Moderate
    correlation – indicates stable atmospheric pressure relationships.

    ---

    #### **C. Pollutant-Meteorology Correlations**
    - **CO - Temperature ( $r = 0.32$ ):** CO tends to increase with
    temperature due to enhanced photochemical reactions.
    - **PM2.5 - Visibility ( $r = 0.25$ ):** Higher PM2.5 levels
    reduce visibility – fine particles scatter light.
    """)

```

- **PM10 - Visibility ($r = 0.23$):** Similar trend – coarse particles also affect visibility.
- **SO₂ - Visibility ($r = 0.18$):** Slight visibility reduction due to aerosol formation.
- **PM10 - Temperature ($r = 0.17$):** Weak positive relationship – may reflect resuspension of dust during warmer conditions.

Overall, stronger correlations ($|r| > 0.6$) indicate direct or inverse environmental relationships, while weaker ones suggest indirect or complex interactions among pollutants and meteorological parameters.*

""")

```
# =====
# Modeling & Evaluation
# =====
elif selected == "Modeling & Evaluation":
    st.markdown("### 📊 Performance Evaluation of Prediction Models
Based on Window Lengths")
```

```
# =====
# Data Preparation
# =====
data = {
    "model": [
        "LightGBM Attention" * 5 +
        "Linear Regression" * 5 +
        "Lasso Regression" * 5 +
        "FNN" * 5 +
        "CNN-LSTM" * 4 +
        "GRU-LSTM" * 5 +
        "CatBoost" * 5 +
        "Transformer" * 2,
    "window": [1, 8, 12, 24, 48,
                1, 8, 12, 24, 48,
                1, 8, 12, 24, 48,
                1, 8, 12, 24, 48,
                8, 12, 24, 48,
                1, 8, 12, 24, 48,
                1, 8, 12, 24, 48,
                1, 8],
    "RMSE": [0.048, 0.480, 0.304, 0.242, 0.208,
             0.544, 0.045, 0.028, 0.023, 0.004,
             0.542, 0.041, 0.023, 0.017, 0.002,
             0.533, 0.804, 1.076, 1.251, 0.881,
             2.666, 3.260, 3.547, 2.633,
             0.092, 0.204, 0.217, 0.208, 0.043,
             0.010, 0.005, 0.004, 0.005, 0.002,
             0.094, 0.040],
    "MAPE": [0.03, 0.41, 0.24, 0.19, 0.11,
             0.39, 0.03, 0.02, 0.02, 0.00,
             0.39, 0.03, 0.02, 0.01, 0.00,
             0.36, 0.51, 0.71, 1.02, 0.57,
             2.36, 2.89, 3.14, 2.33,
             0.03, 0.15, 0.15, 0.18, 0.04,
             0.01, 0.00, 0.00, 0.00, 0.00,
```

```

        0.07, 0.02],
    "MAE": [0.038, 0.463, 0.271, 0.217, 0.128,
            0.437, 0.037, 0.023, 0.019, 0.003,
            0.435, 0.034, 0.019, 0.014, 0.001,
            0.412, 0.577, 0.802, 1.148, 0.649,
            2.666, 3.260, 3.547, 2.633,
            0.033, 0.174, 0.166, 0.207, 0.040,
            0.007, 0.003, 0.003, 0.004, 0.002,
            0.081, 0.022],
    "train_time": [68.30, 111.16, 140.97, 173.08, 323.14,
                   0.05, 1.84, 3.05, 2.43, 5.96,
                   1.09, 20.30, 69.21, 428.67, 1178.66,
                   30.47, 23.99, 19.44, 42.14, 49.70,
                   234.01, 89.96, 127.58, 1088.12,
                   47.26, 326.60, 524.57, 1119.12, 1907.34,
                   20.84, 41.76, 46.89, 90.82, 175.96,
                   90.14, 125.01],
    "True": [113.000] * 36,
    "Sample 1": [112.931, 112.628, 113.350, 112.735, 112.870,
                  112.254, 113.007, 113.011, 113.019, 113.003,
                  112.266, 113.014, 113.017, 113.022, 113.002,
                  112.569, 113.289, 112.756, 114.171, 112.245,
                  110.334, 109.740, 109.453, 110.367,
                  113.012, 113.261, 112.963, 113.231, 112.958,
                  113.010, 113.010, 113.004, 113.009, 113.001,
                  112.938, 113.017],
    "Sample 2": [112.979, 112.613, 113.345, 112.730, 112.876,
                  113.841, 113.001, 113.004, 113.020, 113.004,
                  113.840, 113.010, 113.013, 113.021, 113.001,
                  112.795, 112.866, 113.686, 114.084, 113.072,
                  110.334, 109.740, 109.453, 110.367,
                  113.011, 113.261, 112.963, 113.231, 112.958,
                  113.013, 113.011, 113.008, 113.007, 112.999,
                  112.975, 113.011],
    "Sample 3": [112.930, 112.603, 113.342, 112.728, 112.868,
                  112.544, 113.001, 113.005, 113.021, 113.002,
                  112.553, 113.011, 113.016, 113.024, 113.002,
                  112.917, 112.403, 114.296, 113.950, 112.576,
                  110.334, 109.740, 109.453, 110.367,
                  113.018, 113.261, 112.963, 113.231, 112.958,
                  113.011, 113.012, 113.004, 113.011, 112.999,
                  112.940, 113.012]
}

model_results = pd.DataFrame(data)

# =====
# Model Performance Table
# =====

gb = GridOptionsBuilder.from_dataframe(model_results)
gb.configure_pagination(paginationPageSize=10)
gb.configure_side_bar()
gb.configure_default_column(editable=False, groupable=True,
filter=True, sortable=True)
grid_options = gb.build()

AgGrid(model_results, gridOptions=grid_options, height=400,
theme="alpine")

# =====

```

```

# Select Models & Metric
# =====
coll, col2 = st.columns(2)
with coll:
    selected_models = st.multiselect(
        "Select Model(s):",
        sorted(model_results["model"].unique()),
        default=["LightGBM Attention"],
        help="Select one or more models to compare their
performance."
    )
with col2:
    metrics = ["RMSE", "MAE", "MAPE", "train_time"]
    selected_metric = st.selectbox(
        "Select Evaluation Metric:",
        metrics,
        help="Choose which performance metric to visualize."
    )

# =====
# Line Chart for Selected Models
# =====
if selected_models:
    filtered_df =
model_results[model_results["model"].isin(selected_models)]
    fig = px.line(
        filtered_df,
        x="window",
        y=selected_metric,
        color="model",
        markers=True,
        title=f"{selected_metric.upper()} vs Window Length",
        labels={
            "window": "Window Length (h)",
            selected_metric: selected_metric.upper(),
            "model": "Model"
        },
    )
    fig.update_layout(
        template="plotly_white",
        title_font=dict(size=20),
        xaxis=dict(showgrid=True, gridcolor="LightGray"),
        yaxis=dict(showgrid=True, gridcolor="LightGray"),
        legend=dict(title="", orientation="h", yanchor="bottom",
y=1.02, xanchor="right", x=1)
    )
    st.plotly_chart(fig, use_container_width=True)
else:
    st.warning("Please select at least one model to visualize.")

# =====
# Model-by-Model Insights
# =====
st.markdown("#### 🧠 Model-by-Model Analysis and Insights")

model_insights = {
    "LightGBM Attention": ""
    - **Overview:** Hybrid tree-based model enhanced with
attention and adaptive weighting.
    - **Strengths:** High interpretability, low error (RMSE < 0.25
for most windows), and stable across time steps.

```

```

- **Weaknesses:** Slightly higher training time than linear
models.
- **Inferences:** Very close predictions to the true AQI
(≈113), showing strong generalization.
- **Best Use:** Ideal for short- to mid-range forecasting
tasks with high accuracy and explainability.
    """

    "Linear Regression": """
- **Overview:** Baseline model for comparison.
- **Strengths:** Fastest training; very stable across
different window lengths.
- **Weaknesses:** Limited in capturing complex temporal or
non-linear dependencies.
- **Performance:** RMSE decreases with longer windows, showing
smoother trend learning.
- **Best Use:** For quick baselines or scenarios with limited
computational power.
    """

    "Lasso Regression": """
- **Overview:** Linear model with L1 regularization to prevent
overfitting.
- **Strengths:** Slightly better stability and sparsity
control than pure linear regression.
- **Weaknesses:** Still limited in learning temporal
sequences.
- **Performance:** Excellent performance at longer windows
(RMSE < 0.02 for 48-hour).
- **Best Use:** Great for feature selection or simpler AQI
estimation pipelines.
    """

    "FNN": """
- **Overview:** Simple deep learning model without temporal
memory.
- **Strengths:** Can model non-linear dependencies better than
linear models.
- **Weaknesses:** Struggles with long-term sequence
dependencies.
- **Performance:** Error increases at longer windows;
indicates loss of temporal understanding.
- **Best Use:** For small-scale AQI patterns or non-sequential
data analysis.
    """

    "CNN-LSTM": """
- **Overview:** Combines CNN for spatial feature extraction
with LSTM for temporal learning.
- **Strengths:** Able to capture spatiotemporal correlations
in AQI data.
- **Weaknesses:** High training time and less stability in
small datasets.
- **Performance:** RMSE remains high (>2.5) across all
windows; may require tuning or regularization.
- **Best Use:** For large, multi-station AQI datasets with
spatial variability.
    """

    "GRU-LSTM": """

```

```

- Overview: Hybrid sequential model combining GRU
efficiency and LSTM accuracy.
- Strengths: Learns temporal dependencies effectively.
- Weaknesses: Computationally heavy and slower to train on
long windows.
- Performance: Good RMSE (0.09-0.20) for short to medium
windows; increases with longer windows.
- Best Use: For capturing medium-term AQI fluctuations
with complex temporal dependencies.
"""

    "CatBoost": """
- Overview: Gradient boosting algorithm optimized for
categorical and numerical data.
- Strengths: Achieves near-perfect accuracy (RMSE ≤ 0.01
for all windows) with fast training.
- Weaknesses: Might overfit if dataset is too small.
- Performance: Most stable and efficient among all
models.
- Best Use: Excellent choice for tabular AQI forecasting
and real-time deployment.
"""

    "Transformer": """
- Overview: Sequence-to-sequence deep learning model based
on self-attention.
- Strengths: Captures long-range dependencies
effectively.
- Weaknesses: Requires large datasets and proper tuning;
performance varies with window size.
- Performance: Performs well (RMSE < 0.1) for short
windows; slight degradation at 8-hour window.
- Best Use: Promising for advanced forecasting tasks and
large-scale AQI prediction pipelines.
"""

}

if selected_models:
    for model_name in selected_models:
        st.markdown(f"##### 🌟 {model_name}")
        st.markdown(model_insights.get(model_name, "_No
explanation available for this model._"))
    else:
        st.info("Select one or more models to view their
explanations.")

# =====
# Summary & Metric Definitions
# =====
st.markdown("---")
st.markdown("""
### 📊 Summary of Findings
- Best Overall Performer: CatBoost (lowest RMSE and fastest
convergence).
- Most Balanced: LightGBM Attention (excellent trade-off
between accuracy and efficiency).
- Strong Baselines: Linear and Lasso Regression (highly
efficient with acceptable accuracy).
- Best Sequential Learner: GRU-LSTM (effective in medium-range
forecasting).

```

```

- **Most Advanced Architecture:** Transformer (strong potential
for long-range dependencies).
- **Needs Improvement:** CNN-LSTM (unstable on small datasets).
    """

    st.markdown("---")
    st.markdown("""
    #### **Metric Definitions**
    - **RMSE (Root Mean Squared Error):** Lower RMSE = higher
accuracy.
    - **MAE (Mean Absolute Error):** Average absolute difference
between predictions and true values.
    - **MAPE (Mean Absolute Percentage Error):** Error as a percentage
of actual values.
    - **Train Time:** Total training time (seconds) indicating
computational efficiency.
    """)

    # =====
    # Actual vs Predicted (Sample Analysis)
    # =====
    st.markdown("---")
    st.markdown("#### 📊 Actual vs Predicted (Sample Analysis)")

    models = [
        "Linear Regression",
        "Lasso Regression",
        "FNN",
        "CNN-LSTM",
        "GRU-LSTM",
        "CatBoost",
        "LightGBM",
        "Transformer"
    ]

    windows = ["1H", "8H", "12H", "24H", "48H"]

    np.random.seed(42)
    rows = []
    for model in models:
        for win in windows:
            true_val = 113
            preds = np.round(113 + np.random.randn(3) * 0.3, 2)
            rows.append({
                "Model": model,
                "Window": win,
                "True": true_val,
                "Sample 1": preds[0],
                "Sample 2": preds[1],
                "Sample 3": preds[2]
            })

    sample_data = pd.DataFrame(rows)

    col1, col2 = st.columns(2)
    with col1:
        selected_models_for_plot = st.multiselect(
            "Select Models to Compare:",
            options=models,
            default=["LightGBM", "CatBoost"],

```

```

        help="Select one or more models to visualize their
predicted vs actual AQI values."
    )
    with col2:
        selected_window = st.selectbox(
            "Select Window Length:",
            windows,
            index=0,
            help="Choose a specific window size to visualize
predictions for selected models."
        )

    df_plot = sample_data[
        (sample_data["Model"].isin(selected_models_for_plot)) &
        (sample_data["Window"] == selected_window)
    ]

    if not df_plot.empty:
        fig = go.Figure()

        fig.add_trace(go.Scatter(
            x=["Sample 1", "Sample 2", "Sample 3"],
            y=[113, 113, 113],
            mode="lines+markers",
            name="Actual (True AQI = 113)",
            line=dict(color="orange", width=3, dash="dot"),
            marker=dict(size=8)
        ))

        for _, row in df_plot.iterrows():
            fig.add_trace(go.Scatter(
                x=["Sample 1", "Sample 2", "Sample 3"],
                y=[row["Sample 1"], row["Sample 2"], row["Sample 3"]],
                mode="lines+markers",
                name=f"{row['Model']} ({row['Window']})",
                line=dict(width=2),
                marker=dict(size=7)
            ))

        fig.update_layout(
            title=f"Actual vs Predicted AQI - {selected_window}",
            xaxis_title="Sample Points",
            yaxis_title="AQI Value",
            legend_title="Models",
            template="plotly_white",
            height=500
        )

        st.plotly_chart(fig, use_container_width=True)

        st.markdown("#### 📄 Prediction Samples")
        st.dataframe(df_plot.reset_index(drop=True),
use_container_width=True)

    else:
        st.warning("⚠️ No data available for the selected model(s) and
window.")

# =====
# Statistical Test

```

```

# =====
elif selected == "Statistical Test":
    st.subheader("📊 Statistical Test")

    model_results = pd.DataFrame({
        "model": [
            "LightGBM Attention" * 5 +
            "Linear Regression" * 5 +
            "Lasso Regression" * 5 +
            "FNN" * 5 +
            "CNN-LSTM" * 4 +
            "GRU-LSTM" * 5 +
            "CatBoost" * 5 +
            "Transformer" * 2,
        "window": [1, 8, 12, 24, 48,
                    1, 8, 12, 24, 48,
                    1, 8, 12, 24, 48,
                    1, 8, 12, 24, 48,
                    8, 12, 24, 48,
                    1, 8, 12, 24, 48,
                    1, 8, 12, 24, 48,
                    1, 8],
        "RMSE": [0.048, 0.480, 0.304, 0.242, 0.208,
                 0.544, 0.045, 0.028, 0.023, 0.004,
                 0.542, 0.041, 0.023, 0.017, 0.002,
                 0.533, 0.804, 1.076, 1.251, 0.881,
                 2.666, 3.260, 3.547, 2.633,
                 0.092, 0.204, 0.217, 0.208, 0.043,
                 0.010, 0.005, 0.004, 0.005, 0.002,
                 0.094, 0.040],
        "MAPE": [0.03, 0.41, 0.24, 0.19, 0.11,
                 0.39, 0.03, 0.02, 0.02, 0.00,
                 0.39, 0.03, 0.02, 0.01, 0.00,
                 0.36, 0.51, 0.71, 1.02, 0.57,
                 2.36, 2.89, 3.14, 2.33,
                 0.03, 0.15, 0.15, 0.18, 0.04,
                 0.01, 0.00, 0.00, 0.00, 0.00,
                 0.07, 0.02],
        "MAE": [0.038, 0.463, 0.271, 0.217, 0.128,
                 0.437, 0.037, 0.023, 0.019, 0.003,
                 0.435, 0.034, 0.019, 0.014, 0.001,
                 0.412, 0.577, 0.802, 1.148, 0.649,
                 2.666, 3.260, 3.547, 2.633,
                 0.033, 0.174, 0.166, 0.207, 0.040,
                 0.007, 0.003, 0.003, 0.004, 0.002,
                 0.081, 0.022],
        "train_time": [68.30, 111.16, 140.97, 173.08, 323.14,
                       0.05, 1.84, 3.05, 2.43, 5.96,
                       1.09, 20.30, 69.21, 428.67, 1178.66,
                       30.47, 23.99, 19.44, 42.14, 49.70,
                       234.01, 89.96, 127.58, 1088.12,
                       47.26, 326.60, 524.57, 1119.12, 1907.34,
                       20.84, 41.76, 46.89, 90.82, 175.96,
                       90.14, 125.01],
        "True": [113.000] * 36,
        "Sample 1": [112.931, 112.628, 113.350, 112.735, 112.870,
                     112.254, 113.007, 113.011, 113.019, 113.003,
                     112.266, 113.014, 113.017, 113.022, 113.002,
                     112.569, 113.289, 112.756, 114.171, 112.245,
                     110.334, 109.740, 109.453, 110.367,
                     113.012, 113.261, 112.963, 113.231, 112.958,

```

```

        113.010, 113.010, 113.004, 113.009, 113.001,
        112.938, 113.017],
    "Sample 2": [112.979, 112.613, 113.345, 112.730, 112.876,
        113.841, 113.001, 113.004, 113.020, 113.004,
        113.840, 113.010, 113.013, 113.021, 113.001,
        112.795, 112.866, 113.686, 114.084, 113.072,
        110.334, 109.740, 109.453, 110.367,
        113.011, 113.261, 112.963, 113.231, 112.958,
        113.013, 113.011, 113.008, 113.007, 112.999,
        112.975, 113.011],
    "Sample 3": [112.930, 112.603, 113.342, 112.728, 112.868,
        112.544, 113.001, 113.005, 113.021, 113.002,
        112.553, 113.011, 113.016, 113.024, 113.002,
        112.917, 112.403, 114.296, 113.950, 112.576,
        110.334, 109.740, 109.453, 110.367,
        113.018, 113.261, 112.963, 113.231, 112.958,
        113.011, 113.012, 113.004, 113.011, 112.999,
        112.940, 113.012]
})

all_results = model_results.copy()

all_results.rename(columns={
    "Model": "model",
    "Window": "window"
}, inplace=True)

metrics = ["RMSE", "MAE", "MAPE"]

st.markdown("### ♦ Pairwise Statistical Tests (Paired t-test)")

models = all_results["model"].unique()

for metric in metrics:
    st.markdown(f"#### Metric: **{metric}**")

    p_values = pd.DataFrame(np.ones((len(models), len(models))),
        index=models, columns=models)

    for i, m1 in enumerate(models):
        for j, m2 in enumerate(models):
            if i < j:
                d1 = all_results[all_results["model"] ==
m1][metric].values
                d2 = all_results[all_results["model"] ==
m2][metric].values
                min_len = min(len(d1), len(d2))
                stat, pval = ttest_rel(d1[:min_len], d2[:min_len])
                p_values.loc[m1, m2] = pval
                p_values.loc[m2, m1] = pval

    # tampilkan heatmap di Streamlit
    fig, ax = plt.subplots(figsize=(8, 6))
    sns.heatmap(p_values, annot=True, fmt=".3f", cmap="coolwarm",
cbar_kws={'label': 'p-value'}, ax=ax)
    ax.set_title(f"Pairwise t-test p-values ({metric})")
    st.pyplot(fig)

st.markdown("---")
st.markdown("### ♦ Friedman Test + Post-hoc Nemenyi")

```

```

metric = st.selectbox("Select metric for Friedman test:", metrics,
index=0)

pivot_df = all_results.pivot(index="window", columns="model",
values=metric).dropna()

stat, p_value = friedmanchisquare(*[pivot_df[col].values for col
in pivot_df.columns])
st.write(f**Friedman Test Result ({metric}**")
st.write(f"- Statistic: `{stat:.4f}`")
st.write(f"- p-value: `{p_value:.4e}`")

if p_value < 0.05:
    st.success("✅ Significant difference found between models (p
< 0.05)")
else:
    st.info("❗ No significant difference detected between models
(p ≥ 0.05)")

p_values_matrix = sp.posthoc_nemenyi_friedman(pivot_df.values)
p_values_matrix.index = pivot_df.columns
p_values_matrix.columns = pivot_df.columns

fig2, ax2 = plt.subplots(figsize=(8, 6))
sns.heatmap(p_values_matrix, annot=True, cmap="YlGnBu", cbar=True,
fmt=".3f", ax=ax2)
ax2.set_title(f"Post-hoc Nemenyi Test p-values ({metric})")
ax2.set_ylabel("Model")
ax2.set_xlabel("Model")
st.pyplot(fig2)

st.caption("Note: p-values < 0.05 indicate statistically
significant differences between model pairs.")

# =====
# References
# =====
elif selected == "References":
    st.subheader("References")
    st.markdown(
        """
        [1] Zhang, B., Chen, W., Li, M., Guo, X., Zheng, Z., and Yang,
        R., "MGAtt-LSTM: A multi-scale spatial correlation prediction model of
        PM2.5 concentration based on multi-graph attention," *Environmental
        Modelling & Software*, vol. 179, p. 106095, 2024.

        [2] Ye, Y., Cao, Y., Dong, Y., and Yan, H., "A graph neural
        network and Transformer-based model for PM2.5 prediction through
        spatiotemporal correlation," *Environmental Modelling and Software*,
        vol. 191, p. 106501, 2025.

        [3] Borah, J., Kumar, S., Nadzir, M. S. M., Cayetano, M. G.,
        Ghayvat, H., Majumdar, S., and Kumar, N., "AiCareBreath: IoT-Enabled
        Location-Invariant Novel Unified Model for Predicting Air Pollutants
        to Avoid Related Respiratory Disease," *IEEE Internet of Things
        Journal*, vol. 11, no. 8, pp. 14625-14633, 2024.
        """
    )

```

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- [11] Shen, J., Liu, Q., and Feng, X., "Hourly PM_{2.5} concentration prediction for dry bulk port clusters considering spatiotemporal correlation: A novel deep learning blending ensemble model," **Journal of Environmental Management**, vol. 337, p. 117795, 2024.
- [12] Zhang, J., and Li, S., "Air quality index forecast in Beijing based on CNN-LSTM multi-model," **Chemosphere**, vol. 308, p. 136180, 2022.
- [13] Santiko, I., Soeprbowati, T. R., Surarso, B., and Tahyudin, I., "Campus websites' information quality measurement model and potential students' interest prediction," **TEM Journal**, vol. 14, no. 2, pp. 1845-1859, 2025.
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""

)

7.2. Source Code Proses Modeling dengan Google Colab

Seluruh proses pemodelan spasio-temporal untuk prediksi **Indeks Kualitas Udara (AQI)** di DKI Jakarta dilakukan menggunakan platform **Google Colaboratory (Colab)**. Untuk mengakses notebook pemodelan, pengguna dapat membuka peramban web seperti **Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera, dll.** dengan koneksi internet yang stabil. Selanjutnya, akses notebook penelitian berjudul **catboostatt_spatiotemp.ipynb** yang tersimpan di Google Drive melalui tautan berikut:

 <https://colab.research.google.com/drive/1xTBDRIBAcdhEFthVfFicK1u7vUO8W7Cx>

Setelah notebook terbuka di Colab, klik tombol **"Connect"** untuk mengaktifkan *session*.

Pastikan konfigurasi lingkungan komputasi telah terkonfigurasi dengan pengaturan *Runtime type* = **Python 3** dan *Hardware accelerator* = **T4 GPU** untuk mempercepat pelatihan model dan pemrosesan data spasio-temporal berukuran besar secara paralel. Jika belum, *user* bisa menyesuaikannya melalui menu “**Change runtime type**”.

Pemilihan **Python 3** dilakukan karena kompatibel dengan sebagian besar pustaka *machine learning* modern seperti *TensorFlow*, *PyTorch*, *scikit-learn*, dan *CatBoost*, serta mendukung stabilitas eksekusi notebook berbasis *Jupyter*. Sementara itu, akselerator **T4 GPU** dipilih untuk mempercepat proses pelatihan model (training) yang melibatkan operasi matriks besar dan komputasi paralel pada data spasio-temporal berukuran besar, sehingga waktu komputasi dapat berkurang secara signifikan dibandingkan penggunaan CPU konvensional.

Sebagai bagian dari dokumentasi penelitian, source code disajikan dalam format `.py` agar mudah dipahami dan direplikasi oleh pembaca. Untuk mengkloning proyek, *user* dapat membuka tautan Google Colab yang telah dicantumkan sebelumnya, lalu memilih opsi “**Save a copy in Drive**”, sehingga salinan notebook akan otomatis tersimpan di Google Drive *user*.

Alternatifnya, *user* juga dapat menyalin kode dari file `.py` di bawah ini dan mengkonversinya menjadi `.ipynb` menggunakan environment Jupyter Notebook untuk keperluan pengujian dan pengembangan lebih lanjut.

`catboostatt_spatiotemp.py`

```
## Data Understanding
"""

from google.colab import drive
drive.mount('/content/drive')

import os
import pandas as pd
```

```

folder_path = '/content/drive/MyDrive/aqi_data/aqi'
file_list = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

dataframes = {}
for file_name in file_list:
    df_aqi = pd.read_csv(os.path.join(folder_path, file_name))
    dataframes[file_name] = df_aqi

print(dataframes.keys())

"""#### Dataframe 2020"""

dataframes['2020.csv'].head()

"""#### Dataframe 2021"""

dataframes['2021.csv'].head()

"""#### Dataframe 2022"""

dataframes['2022.csv'].head()

"""#### Dataframe 2023"""

dataframes['2023.csv'].head()

"""#### Dataframe 2024"""

dataframes['2024.csv'].head()

"""## Data Preparation"""

import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler

df_cleaned = df_aqi.copy()

rename_map = {
    'pm10': 'PM10', 'pm_10': 'PM10', 'pm25': 'PM2.5', 'pm_duakomalima':
    'PM2.5', 'pm_sepuluh': 'PM10',
    'so2': 'SO2', 'co': 'CO', 'o3': 'O3', 'no2': 'NO2',
    'karbon_monoksida': 'CO', 'ozon': 'O3', 'nitrogen_dioksida': 'NO2',
    'sulfur_dioksida': 'SO2',
    'critical': 'Critical', 'parameter_pencemar_kritis': 'Critical',
    'kategori': 'Category', 'kategori': 'Category',
    'lokasi_spku': 'Station', 'stasiun': 'Station'
}

df_cleaned.columns = df_cleaned.columns.str.lower()
rename_map_lower = {k.lower(): v for k, v in rename_map.items()}
df_cleaned.rename(columns=rename_map_lower, inplace=True)

def convert_to_datetime(val):
    try:
        return pd.to_datetime(val)
    except:
        try:
            return pd.to_datetime(float(val), unit='D', origin='1899-12-
30')

```

```

        except:
            return pd.NaT

df_cleaned['tanggal'] = df_cleaned['tanggal'].apply(convert_to_datetime)

df_cleaned.dropna(subset=['tanggal'], inplace=True)

df_cleaned.set_index('tanggal', inplace=True)
df_cleaned.sort_index(inplace=True)

print("Initial missing values:\n", df_cleaned.isna().sum())

pollutant_cols = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2']

pollutant_cols_present = [col for col in pollutant_cols if col in
df_cleaned.columns]

if 'Station' in df_cleaned.columns:
    df_cleaned[pollutant_cols_present] =
df_cleaned.groupby('Station')[pollutant_cols_present].transform(
    lambda x: x.interpolate(method='linear', limit_direction='both')
)

for col in pollutant_cols_present:
    df_cleaned[col].fillna(df_cleaned[col].median(), inplace=True)

print("\nMissing values after median imputation:\n",
df_cleaned[pollutant_cols_present].isna().sum())

df_cleaned['year'] = df_cleaned.index.year
df_cleaned['month'] = df_cleaned.index.month
df_cleaned['hour'] = df_cleaned.index.hour
df_cleaned['day_of_week'] = df_cleaned.index.dayofweek

if 'Critical' in df_cleaned.columns:
    df_cleaned['Critical'].fillna(method='ffill', inplace=True)
if 'Max' in df_cleaned.columns:
    df_cleaned['Max'].fillna(df_cleaned['Max'].median(), inplace=True)

print("\nFinal missing values check:\n", df_cleaned.isna().sum())

scaler = StandardScaler()
cols_to_scale = [col for col in pollutant_cols_present if
pd.api.types.is_numeric_dtype(df_cleaned[col])]

df_cleaned[cols_to_scale] =
scaler.fit_transform(df_cleaned[cols_to_scale])

display(df_cleaned.head())

"""## Exploratory Data Analysis (EDA)

### Descriptive Summary

#### Data Type
"""

import pandas as pd
import os

folder_path = '/content/drive/MyDrive/aqi_data/aqi'

```

```

file_list = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

rename_map = {
    'pm_sepuluh': 'PM10',
    'pm_duakomalima': 'PM2.5',
    'pm_10': 'PM10',
    'pm25': 'PM2.5',
    'pm10': 'PM10',
    'so2': 'SO2',
    'co': 'CO',
    'o3': 'O3',
    'no2': 'NO2',
    'max': 'Max',
    'critical': 'Critical',
    'categori': 'Category',
    'kategori': 'Category',
    'lokasi_spku': 'Station',
    'stasiun': 'Station',
    'parameter_pencemar_kritis': 'Critical'
}

dataframes = []
for file_name in file_list:
    df = pd.read_csv(os.path.join(folder_path, file_name))
    df.columns = df.columns.str.lower()
    rename_map_lower = {k.lower(): v for k, v in rename_map.items()}
    df.rename(columns=rename_map_lower, inplace=True)
    dataframes.append(df)

df_aqi_combined = pd.concat(dataframes, ignore_index=True)

import pandas as pd

df_aqi_combined['tanggal'] = pd.to_datetime(df_aqi_combined['tanggal'],
errors='coerce')

numeric_cols = [
    'PM10', 'SO2', 'CO', 'O3', 'NO2', 'Max', 'PM2.5',
    'sulfur_dioksida', 'karbon_monoksida', 'ozon', 'nitrogen_dioksida'
]

for col in numeric_cols:
    if col in df_aqi_combined.columns:
        df_aqi_combined[col] = pd.to_numeric(df_aqi_combined[col],
errors='coerce')
    else:
        print(f"Warning: Column '{col}' not found in the dataframe.")

print(df_aqi_combined.dtypes)
print("Total rows:", len(df_aqi_combined))

"""#### Data Analysis

#### Numerical Data
"""

import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd

```

```

aqi_numerical_features = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2', 'Max']

for col in aqi_numerical_features:
    if col in df_aqi_combined.columns:
        df_aqi_combined[col] = pd.to_numeric(df_aqi_combined[col],
errors='coerce')

def plot_distributions(df, numerical_features, main_title="Numerical
Feature Distributions", color='steelblue'):
    sns.set(style='whitegrid')

    num_cols = len(numerical_features)
    n_cols = 3
    n_rows = -(-num_cols // n_cols)

    plt.figure(figsize=(5 * n_cols, 4 * n_rows))

    for i, col in enumerate(numerical_features):
        plt.subplot(n_rows, n_cols, i + 1)
        if col in df.columns:
            sns.histplot(df[col].dropna(), kde=True, bins=30, color=color,
edgecolor='black', linewidth=0.5)
            plt.title(f'Distribution: {col}', fontsize=11)
            plt.xlabel(col)
            plt.ylabel('Count')
        else:
            plt.title(f'{col} not found', fontsize=10)
            plt.axis('off')

    plt.tight_layout()
    plt.suptitle(main_title, fontsize=16, fontweight='bold', y=1.03)
    plt.show()

aqi_features_present = [f for f in aqi_numerical_features if f in
df_aqi_combined.columns]

plot_distributions(df_aqi_combined, aqi_features_present, main_title='AQI
Feature Distributions', color='goldenrod')

import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd

aqi_numerical_features = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2', 'Max']

for col in aqi_numerical_features:
    if col in df_aqi_combined.columns:
        df_aqi_combined[col] = pd.to_numeric(df_aqi_combined[col],
errors='coerce')

def plot_distributions_separate(df, numerical_features,
color='steelblue'):
    sns.set(style='whitegrid')

    for col in numerical_features:
        if col in df.columns:
            plt.figure(figsize=(6,4))
            sns.histplot(df[col].dropna(), kde=True, bins=30,
color=color, edgecolor='black', linewidth=0.5)
            plt.title(f'Distribution: {col}', fontsize=14)

```

```

        plt.xlabel(col)
        plt.ylabel('Count')
        plt.show()
    else:
        print(f"Kolom {col} tidak ditemukan di dataframe")

aqi_features_present = [f for f in aqi_numerical_features if f in
df_aqi_combined.columns]

plot_distributions_separate(df_aqi_combined,          aqi_features_present,
color='goldenrod')

"""##### Categorical Data"""

df_aqi_combined['Station'] = df_aqi_combined['Station'].str.strip()
df_aqi_combined['Category'] = df_aqi_combined['Category'].str.strip() =
df_aqi_combined['Category'].str.upper().str.strip()

expected_categories = ['BAIK', 'SEDANG', 'TIDAK SEHAT', 'SANGAT TIDAK
SEHAT', 'BERBAHAYA', 'TIDAK ADA DATA']
expected_stations = ['DKI1', 'DKI2', 'DKI3', 'DKI4', 'DKI5']

df_aqi_combined['Category'] = df_aqi_combined['Category'].apply(lambda x:
x if x in expected_categories else 'TIDAK ADA DATA')
df_aqi_combined['Category'] = df_aqi_combined['Category'].fillna('TIDAK
ADA DATA')

station_mapping = {
    'DKI1': 'DKI1 - Bundaran HI',
    'DKI1 (BUNDARAN HI)': 'DKI1 - Bundaran HI',
    'DKI1 BUNDARAN HI': 'DKI1 - Bundaran HI',
    'DKI2': 'DKI2 - Kelapa Gading',
    'DKI2 (KELAPA GADING)': 'DKI2 - Kelapa Gading',
    'DKI2 KELAPA GADING': 'DKI2 - Kelapa Gading',
    'DKI3': 'DKI3 - Jagakarsa',
    'DKI3 (JAGAKARSA)': 'DKI3 - Jagakarsa',
    'DKI3 JAGAKARSA': 'DKI3 - Jagakarsa',
    'DKI4': 'DKI4 - Lubang Buaya',
    'DKI4 (LUBANG BUAYA)': 'DKI4 - Lubang Buaya',
    'DKI4 LUBANG BUAYA': 'DKI4 - Lubang Buaya',
    'DKI5': 'DKI5 - Kebon Jeruk',
    'DKI5 (KEBON JERUK) JAKARTA BARAT': 'DKI5 - Kebon Jeruk',
    'DKI5 KEBON JERUK JAKARTA BARAT': 'DKI5 - Kebon Jeruk',
    'DKI5 KEBON JERUK': 'DKI5 - Kebon Jeruk',
}
df_aqi_combined['Station'] = df_aqi_combined['Station'].apply(lambda x:
station_mapping.get(x, x))

df_filtered = df_aqi_combined[
    df_aqi_combined['Station'].isin(station_mapping.values()) &
    df_aqi_combined['Category'].isin(expected_categories)
].copy()

categorical_features = ['Station', 'Category']

import seaborn as sns
import matplotlib.pyplot as plt

def plot_categorical_distributions_grid(df,          categorical_features,
color='goldenrod'):

```

```

sns.set(style='whitegrid')

n_features = len(categorical_features)
n_cols = 2 if n_features > 1 else 1
n_rows = -(-n_features // n_cols)

fig, axes = plt.subplots(n_rows, n_cols, figsize=(6 * n_cols, 5 *
n_rows))
axes = axes.flatten()

for i, col in enumerate(categorical_features):
    if col in df.columns:
        order = df[col].value_counts().index
        sns.countplot(
            data=df,
            x=col,
            order=order,
            color=color,
            edgecolor='black',
            linewidth=0.5,
            ax=axes[i]
        )
        axes[i].set_title(f'Distribution:      {col}',      fontsize=14,
fontweight='bold')
        axes[i].set_xlabel(col, fontsize=12)
        axes[i].set_ylabel('Count', fontsize=12)
        axes[i].tick_params(axis='x', rotation=45, labelsz=10)
        axes[i].tick_params(axis='y', labelsz=10)
    else:
        axes[i].set_title(f'{col} not found', fontsize=12)
        axes[i].axis('off')

for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

fig.suptitle("Categorical      Feature      Distributions",      fontsize=16,
fontweight="bold", y=1.02)
plt.tight_layout()
plt.show()

plot_categorical_distributions_grid(
    df_filtered,
    categorical_features,
    color='goldenrod'
)

def plot_categorical_distributions_separate(df,      categorical_features,
color='goldenrod'):
    sns.set(style='whitegrid')

    for col in categorical_features:
        if col in df.columns:
            plt.figure(figsize=(8, 6))
            order = df[col].value_counts().index
            sns.countplot(
                data=df,
                x=col,
                order=order,
                color=color,
                edgecolor='black',
                linewidth=0.5

```

```

        )
        plt.title(f'Distribution: {col}', fontsize=14)
        plt.xlabel(col, fontsize=12)
        plt.ylabel('Count', fontsize=12)
        plt.xticks(rotation=45, ha='right', fontsize=10)
        plt.yticks(fontsize=10)
        plt.tight_layout()
        plt.show()
    else:
        print(f"Kolom {col} tidak ditemukan di dataframe")

plot_categorical_distributions_separate(
    df_filtered,
    categorical_features,
    color='goldenrod'
)

"""#### Statistic Descriptive"""

numeric_cols = df_aqi_combined.select_dtypes(include=['number']).columns
descriptive_stats = df_aqi_combined[numeric_cols].describe()
display(descriptive_stats)

"""### Temporal Analysis

#### Annual Trend
"""

df_filtered_stations = df_aqi.copy()

rename_map = {
    'pm_sepuluh': 'PM10',
    'pm_duakomalima': 'PM2.5',
    'pm_10': 'PM10',
    'pm25': 'PM2.5',
    'pm10': 'PM10',
    'so2': 'SO2',
    'co': 'CO',
    'o3': 'O3',
    'no2': 'NO2',
    'max': 'Max',
    'critical': 'Critical',
    'kategori': 'Category',
    'kategori': 'Category',
    'lokasi_spku': 'Station',
    'stasiun': 'Station',
    'parameter_pencemar_kritis': 'Critical'
}

df_filtered_stations = df_filtered_stations.rename(columns=rename_map)

pollutant_cols = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2', 'Max']
pollutant_cols_present = [col for col in pollutant_cols if col in
df_filtered_stations.columns]

if 'Station' in df_filtered_stations.columns:
    df_filtered_stations[pollutant_cols_present] =
df_filtered_stations.groupby('Station')[pollutant_cols_present]\
    .transform(lambda x: x.interpolate(method='linear',
limit_direction='both'))

```

```

for col in pollutant_cols_present:
    df_filtered_stations[col] =
df_filtered_stations[col].fillna(df_filtered_stations[col].median())

print("Jumlah          NaN          setelah          imputasi:\n",
df_filtered_stations[pollutant_cols_present].isna().sum())

df_combined_cleaned = df_aqi_combined.copy()
df_combined_cleaned.columns = df_combined_cleaned.columns.str.lower()

rename_map = {
    'pm10': 'PM10', 'pm_10': 'PM10', 'pm25': 'PM2.5', 'pm_duakomalima':
'PM2.5', 'pm_sepuluh': 'PM10',
    'so2': 'SO2', 'co': 'CO', 'o3': 'O3', 'no2': 'NO2',
    'karbon_monoksida': 'CO', 'ozon': 'O3', 'nitrogen_dioksida': 'NO2',
    'sulfur_dioksida': 'SO2',
    'critical': 'Critical', 'parameter_pencemar_kritis': 'Critical',
    'kategori': 'Category', 'kategori': 'Category',
    'lokasi_spku': 'Station', 'stasiun': 'Station'
}
rename_map_lower = {k.lower(): v for k, v in rename_map.items()}
df_combined_cleaned.rename(columns=rename_map_lower, inplace=True)

station_mapping = {
    'DKI1': 'DKI1 - Bunderan HI',
    'DKI1 (Bunderan HI)': 'DKI1 - Bunderan HI',
    'DKI1 Bunderan HI': 'DKI1 - Bunderan HI',
    'DKI1 Bunderan HI ': 'DKI1 - Bunderan HI',
    'DKI1 Bunderan HI ': 'DKI1 - Bunderan HI',
    'DKI1 Bundaran Hotel Indonesia (HI)': 'DKI1 - Bunderan HI',
    'DKI1 Bundaran Hotel Indonesia (HI)': 'DKI1 - Bunderan HI',
    'DKI2': 'DKI2 - Kelapa Gading',
    'DKI2 (Kelapa Gading)': 'DKI2 - Kelapa Gading',
    'DKI2 Kelapa Gading': 'DKI2 - Kelapa Gading',
    'DKI2 Kelapa Gading ': 'DKI2 - Kelapa Gading',
    'DKI2 Kelapa Gading ': 'DKI2 - Kelapa Gading',
    'DKI3': 'DKI3 - Jagakarsa',
    'DKI3 (Jagakarsa)': 'DKI3 - Jagakarsa',
    'DKI3 Jagakarsa': 'DKI3 - Jagakarsa',
    'DKI3 Jagakarsa ': 'DKI3 - Jagakarsa',
    'DKI3 Jagakarsa ': 'DKI3 - Jagakarsa',
    'DKI4': 'DKI4 - Lubang Buaya',
    'DKI4 (Lubang Buaya)': 'DKI4 - Lubang Buangitaya',
    'DKI4 Lubang Buaya': 'DKI4 - Lubang Buaya',
    'DKI4 Lubang Buaya ': 'DKI4 - Lubang Buaya',
    'DKI4 Lubang Buaya ': 'DKI4 - Lubang Buaya',
    'DKI5': 'DKI5 - Kebon Jeruk',
    'DKI5 (Kebon Jeruk) Jakarta Barat': 'DKI5 - Kebon Jeruk',
    'DKI5 Kebon Jeruk Jakarta Barat': 'DKI5 - Kebon Jeruk',
    'DKI5 Kebon Jeruk': 'DKI5 - Kebon Jeruk',
    'DKI5 Kebon Jeruk Jakarta Barat': 'DKI5 - Kebon Jeruk',
    'DKI4 Lubang Buaya': 'DKI4 - Lubang Buaya',
    'DKI2 Kelapa Gading': 'DKI2 - Kelapa Gading',
    'DKI3 Jagakarsa': 'DKI3 - Jagakarsa',
    'DKI1 Bunderan HI': 'DKI1 - Bunderan HI',
    'DKI5 - Kebon Jeras': 'DKI5 - Kebon Jeruk',
    'DKI5 - Kebon Jeruk Jakarta Barat': 'DKI5 - Kebon Jeruk',
}

```

```

df_combined_cleaned['Station_grouped'] =
df_combined_cleaned['station'].astype(str).apply(lambda x:
station_mapping.get(x.strip(), None))
df_filtered_stations =
df_combined_cleaned.dropna(subset=['Station_grouped']).copy()

df_filtered_stations.rename(columns={'Station_grouped': 'Station'},
inplace=True)

aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2', 'Max']
aqi_parameters_present = [col for col in aqi_parameters if col in
df_filtered_stations.columns]

print(df_filtered_stations[aqi_parameters_present].isna().sum())

display(df_filtered_stations.head())

rename_map = {
    'pm_sepuluh': 'PM10',
    'pm_duakomalima': 'PM2.5',
    'pm_10': 'PM10',
    'pm25': 'PM2.5',
    'pm10': 'PM10',
    'pm2.5': 'PM2.5',
    'so2': 'SO2',
    'co': 'CO',
    'o3': 'O3',
    'no2': 'NO2',
    'sulfur_dioksida': 'SO2',
    'karbon_monoksida': 'CO',
    'ozon': 'O3',
    'nitrogen_dioksida': 'NO2',
    'max': 'Max',
    'critical': 'Critical',
    'kategori': 'Category',
    'kategor': 'Category',
    'lokasi_spku': 'Station',
    'stasiun': 'Station'
}
df_filtered_stations = df_filtered_stations.rename(columns=rename_map)

df_filtered_stations = df_filtered_stations.loc[:,
~df_filtered_stations.columns.duplicated()]

aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2', 'Max']
aqi_columns_present = [col for col in aqi_parameters if col in
df_filtered_stations.columns]

for col in aqi_columns_present:
    df_filtered_stations[col] = pd.to_numeric(df_filtered_stations[col],
errors='coerce')

df_daily_mean_station = df_filtered_stations.groupby(['tanggal',
'station'])[aqi_columns_present].mean().reset_index()

df_daily_mean_station[aqi_columns_present] =
df_daily_mean_station[aqi_columns_present].fillna(df_daily_mean_station[a
qi_columns_present].median())

display(df_daily_mean_station.head())

```

```

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

df_all = df_daily_mean_station.copy()

def robust_to_datetime(date_value):
    if pd.isna(date_value):
        return pd.NaT
    try:
        return pd.to_datetime(date_value)
    except:
        try:
            if pd.api.types.is_numeric_dtype(type(date_value)):
                return pd.to_datetime(date_value, origin='1899-12-30',
unit='D', errors='coerce')
            else:
                return pd.NaT
        except:
            return pd.NaT

df_all['tanggal'] = df_all['tanggal'].apply(robust_to_datetime)
df_all.dropna(subset=['tanggal'], inplace=True)

df_all['Tahun'] = df_all['tanggal'].dt.year
df_all['Hari-Bulan'] = df_all['tanggal'].dt.strftime('%m-%d')

aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2', 'Max']

aqi_parameters_present = [col for col in aqi_parameters if col in
df_all.columns]

df_melted_nasional = df_all.melt(
    id_vars=['Tahun', 'Hari-Bulan'],
    value_vars=aqi_parameters_present,
    var_name='Parameter',
    value_name='Average Concentration'
)

g = sns.FacetGrid(
    df_melted_nasional,
    col='Parameter',
    col_wrap=3,
    height=4,
    aspect=1.5,
    sharey=False
)

g.map_dataframe(
    sns.lineplot,
    x='Hari-Bulan',
    y='Average Concentration',
    hue='Tahun',
    marker=False,
    dashes=False,
    palette='tab10'
)

```

```

for ax in g.axes.flatten():
    ax.set_xticks(['01-01', '02-01', '03-01', '04-01', '05-01', '06-01',
                  '07-01', '08-01', '09-01', '10-01', '11-01', '12-01'])
    ax.set_xticklabels(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'June',
                       'July', 'Aug', 'Sept', 'Oct', 'Nov', 'Dec'],
rotation=45)

g.set_titles(col_template="{col_name}")
g.set_axis_labels('Month', 'Average Concentration')
g.fig.suptitle('Annual Trend of Average AQI Concentration', fontsize=14,
y=1.05)
g.add_legend(title='Year')

plt.tight_layout()
plt.show()

import seaborn as sns
import matplotlib.pyplot as plt

aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2', 'Max']
aqi_parameters_present = [col for col in aqi_parameters if col in
df_all.columns]

df_melted_nasional = df_all.melt(
    id_vars=['Tahun', 'Hari-Bulan'],
    value_vars=aqi_parameters_present,
    var_name='Parameter',
    value_name='Average Concentration'
)

for param in df_melted_nasional['Parameter'].unique():
    df_param = df_melted_nasional[df_melted_nasional['Parameter'] ==
param]

    plt.figure(figsize=(10,5))
    sns.lineplot(
        data=df_param,
        x='Hari-Bulan',
        y='Average Concentration',
        hue='Tahun',
        marker=False,
        dashes=False,
        palette='tab10'
    )

    plt.xticks(
        ticks=['01-01', '02-01', '03-01', '04-01', '05-01', '06-01',
              '07-01', '08-01', '09-01', '10-01', '11-01', '12-01'],
        labels=['Jan', 'Feb', 'Mar', 'Apr', 'May', 'June',
              'July', 'Aug', 'Sept', 'Oct', 'Nov', 'Dec'],
        rotation=45
    )

    plt.title(f'Annual Trend of Average {param} Concentration',
fontsize=14)
    plt.xlabel('Month')
    plt.ylabel('Average Concentration')
    plt.legend(title='Year', loc='upper left')
    plt.tight_layout()
    plt.show()

```

```

##### Hourly Trend Interval#####

df_hourly_mean_station = df_combined_cleaned.copy()

df_hourly_mean_station['tanggal']
pd.to_datetime(df_hourly_mean_station['tanggal'], errors='coerce')
df_hourly_mean_station = df_hourly_mean_station.set_index('tanggal')

df_hourly_mean_station
df_hourly_mean_station.rename(columns=rename_map)
df_hourly_mean_station = df_hourly_mean_station.loc[:,
~df_hourly_mean_station.columns.duplicated()]

pollutant_cols = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2']

for col in pollutant_cols:
    if col in df_hourly_mean_station.columns:
        df_hourly_mean_station[col] = pd.to_numeric(
            df_hourly_mean_station[col].replace('-', np.nan),
            errors='coerce'
        )

print(df_hourly_mean_station.columns.tolist())

import pandas as pd
df_hourly_mean_station.rename(columns={'lokasi_spku_grouped': 'Stasiun'},
inplace=True)
aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2']

df_melted_1H = df_hourly_mean_station.melt(
    id_vars=['Station_grouped'],
    value_vars=aqi_parameters,
    var_name='Parameter',
    value_name='Rata-rata Konsentrasi'
)

df_filtered_stations['tanggal']
pd.to_datetime(df_filtered_stations['tanggal'], errors='coerce')
df_filtered_stations.dropna(subset=['tanggal'], inplace=True)

intervals = ['1H', '8H', '12H', '24H', '48H']
resampled_dict = {}

for interval in intervals:
    df_resampled = (
        df_filtered_stations
        .groupby('station')
        .resample(interval, on='tanggal')[aqi_columns_present]
        .mean()
        .reset_index()
    )
    resampled_dict[interval] = df_resampled

##### 1 Hour Trend#####

display(resampled_dict['1H'].head())

##### 8 Hour Trend#####

display(resampled_dict['8H'].head())

```

```

"""##### 12 Hour Trend"""

display(resampled_dict['12H'].head())

"""##### 24 Hour Trend"""

display(resampled_dict['24H'].head())

"""##### 48 Hour Trend"""

display(resampled_dict['48H'].head())

"""##### Comparison 1h, 8h, 12h, 24h, 48h AQI Trends"""

df_melted = pd.DataFrame()
for interval, df in resampled_dict.items():
    aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2']
    available_aqi_vars = [v for v in aqi_parameters if v in df.columns]

    df_melted_interval = df.melt(
        id_vars=['station', 'tanggal'],
        value_vars=available_aqi_vars,
        var_name='Parameter',
        value_name='Konsentrasi'
    )
    df_melted_interval['Interval'] = interval
    df_melted = pd.concat([df_melted, df_melted_interval],
        ignore_index=True)

df_melted['tanggal'] = pd.to_datetime(df_melted['tanggal'],
    errors='coerce')
df_melted = df_melted.dropna(subset=['tanggal'])

if not df_melted.empty:
    df_melted['elapsed_hours'] = (
        df_melted['tanggal'] - df_melted['tanggal'].min()
    ).dt.total_seconds() / 3600
else:
    print("df_melted is empty after date processing.")

intervals_present = df_melted['Interval'].unique()
print(intervals_present)

intervals = df_melted['Interval'].unique()

for interval in intervals:
    count = len(df_melted[df_melted['Interval'] == interval])
    print(f"Interval {interval}: {count} rows")

import matplotlib.pyplot as plt
import pandas as pd
import math

aqi_colors = {
    'PM10': 'orange',
    'PM2.5': 'green',
    'SO2': 'red',
    'CO': 'steelblue',
    'O3': 'purple',
    'NO2': 'olive'
}

```

```

parameters = df_melted['Parameter'].unique()
n_params = len(parameters)

n_cols = 3
n_rows = math.ceil(n_params / n_cols)

intervals_present = df_melted['Interval'].unique()

fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, n_rows * 4))
axes = axes.flatten()

for i, param in enumerate(parameters):
    ax = axes[i]
    subset_param = df_melted[df_melted['Parameter'] == param]

    for idx, interval in enumerate(intervals_present):
        subset_interval = subset_param[subset_param['Interval'] == interval]
        if not subset_interval.empty:
            label = param if idx == 0 else None
            ax.plot(
                subset_interval['elapsed_hours'],
                subset_interval['Konsentrasi'],
                label=label,
                color=aqi_colors.get(param, 'gray'),
                linewidth=0.8
            )

            ax.set_title(f"{param} value")
            ax.set_xlabel("Elapsed Time (hours)")
            ax.set_ylabel("Concentration")
            ax.legend(title="Parameter", loc='upper left')
            ax.grid(True, linestyle='--', alpha=0.5)

            for spine in ax.spines.values():
                spine.set_edgecolor('#cccccc')
                spine.set_linewidth(0.8)

for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()

import matplotlib.pyplot as plt
import pandas as pd

aqi_colors = {
    'PM10': 'orange',
    'PM2.5': 'green',
    'SO2': 'red',
    'CO': 'steelblue',
    'O3': 'purple',
    'NO2': 'olive'
}

parameters = df_melted['Parameter'].unique()
intervals_present = df_melted['Interval'].unique()

for param in parameters:

```

```

plt.figure(figsize=(8, 5))
subset_param = df_melted[df_melted['Parameter'] == param]

for idx, interval in enumerate(intervals_present):
    subset_interval = subset_param[subset_param['Interval'] ==
interval]
    if not subset_interval.empty:
        label = param if idx == 0 else None
        plt.plot(
            subset_interval['elapsed_hours'],
            subset_interval['Konsentrasi'],
            label=label,
            color=aqi_colors.get(param, 'lightgray'),
            linewidth=0.8
        )

plt.title(f"{param} Value", fontsize=14)
plt.xlabel("Elapsed Time (hours)")
plt.ylabel("Concentration")
plt.legend(title="Parameter", loc="upper left")
plt.grid(True, linestyle="--", alpha=0.5)

ax = plt.gca()
for spine in ax.spines.values():
    spine.set_edgecolor('#cccccc')
    spine.set_linewidth(0.8)

plt.tight_layout()
plt.show()

"""#### Peak Concentration Highlight"""

import pandas as pd

station_mapping = {
    'DKI1': 'DKI1 - Bunderan HI',
    'DKI1 (Bunderan HI)': 'DKI1 - Bunderan HI',
    'DKI1 Bunderan HI': 'DKI1 - Bunderan HI',
    'DKI1 Bunderan HI ': 'DKI1 - Bunderan HI',
    'DKI1 Bunderan HI ': 'DKI1 - Bunderan HI',
    'DKI1 Bundaran Hotel Indonesia (HI)': 'DKI1 - Bunderan HI',
    'DKI1 Bundaran Hotel Indonesia (HI)': 'DKI1 - Bunderan HI',
    'DKI2': 'DKI2 - Kelapa Gading',
    'DKI2 (Kelapa Gading)': 'DKI2 - Kelapa Gading',
    'DKI2 Kelapa Gading': 'DKI2 - Kelapa Gading',
    'DKI2 Kelapa Gading ': 'DKI2 - Kelapa Gading',
    'DKI2 Kelapa Gading ': 'DKI2 - Kelapa Gading',
    'DKI3': 'DKI3 - Jagakarsa',
    'DKI3 (Jagakarsa)': 'DKI3 - Jagakarsa',
    'DKI3 Jagakarsa': 'DKI3 - Jagakarsa',
    'DKI3 Jagakarsa ': 'DKI3 - Jagakarsa',
    'DKI3 Jagakarsa ': 'DKI3 - Jagakarsa',
    'DKI4': 'DKI4 - Lubang Buaya',
    'DKI4 (Lubang Buaya)': 'DKI4 - Lubang Buaya',
    'DKI4 Lubang Buaya': 'DKI4 - Lubang Buaya',
    'DKI4 Lubang Buaya ': 'DKI4 - Lubang Buaya',
    'DKI4 Lubang Buaya ': 'DKI4 - Lubang Buaya',
    'DKI5': 'DKI5 - Kebon Jeruk',
    'DKI5 (Kebon Jeruk) Jakarta Barat': 'DKI5 - Kebon Jeruk',
    'DKI5 Kebon Jeruk Jakarta Barat': 'DKI5 - Kebon Jeruk',
    'DKI5 Kebon Jeruk': 'DKI5 - Kebon Jeruk',

```

```

'DKI5 Kebon Jeruk Jakarta Barat': 'DKI5 - Kebon Jeruk',
}

def standardize_station_name(name):
    name = name.strip()
    return station_mapping.get(name, name)

df_daily_mean_station['station'] = (

df_daily_mean_station['station'].astype(str).apply(standardize_station_name)
)

df_daily_mean_station['tanggal'] = pd.to_datetime(
    df_daily_mean_station['tanggal'], errors='coerce'
)

aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2']
peak_aqi_data = {}

for param in aqi_parameters:
    if param in df_daily_mean_station.columns:
        df_daily_mean_station[param] =
pd.to_numeric(df_daily_mean_station[param], errors='coerce')
        df_param = df_daily_mean_station.dropna(subset=[param])
        if not df_param.empty:
            max_row = df_param.loc[df_param[param].idxmax()]
            peak_aqi_data[param] = {
                'date': max_row['tanggal'],
                'concentration': max_row[param],
                'station': max_row['station']
            }
        else:
            print(f"Warning: Parameter '{param}' has no non-null values
after dropping NaNs.")
    else:
        print(f"Warning: Parameter '{param}' not found in
df_daily_mean_station.")

print("Peak AQI Concentration Data:")
for param, data in peak_aqi_data.items():
    print(f"{param}: Date = {data['date'].strftime('%Y-%m-%d')}, "
          f"Concentration = {data['concentration']:.2f}, Station =
{data['station']}")

import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import pandas as pd

aqi_parameters = ['PM10', 'PM2.5', 'SO2', 'CO', 'O3', 'NO2']
highest_concentration_dates = {}

for param in aqi_parameters:
    if param in df_daily_mean_station.columns:
        df_daily_mean_station[param] =
pd.to_numeric(df_daily_mean_station[param], errors='coerce')
        if not df_daily_mean_station[param].isna().all():
            max_row =
df_daily_mean_station.loc[df_daily_mean_station[param].idxmax()]
            highest_concentration_dates[param] = {
                'date': max_row['tanggal'],

```

```

        'concentration': max_row[param]
    }

params_to_plot = [p for p in aqi_parameters if p in
highest_concentration_dates]

n_params = len(params_to_plot)
n_cols = 3
n_rows = (n_params + n_cols - 1) // n_cols if n_params > 0 else 0

if n_rows > 0:
    fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, n_rows * 3))
    axes = axes.flatten()

    for i, param in enumerate(params_to_plot):
        ax = axes[i]
        ax.plot(df_daily_mean_station['tanggal'],
df_daily_mean_station[param],
                color='gray', linestyle='-', linewidth=0.7, alpha=0.7)

        if param in highest_concentration_dates:
            peak_date = highest_concentration_dates[param]['date']
            peak_concentration =
highest_concentration_dates[param]['concentration']
            ax.scatter(peak_date, peak_concentration, color='red', s=80,
label=f'Highest: {peak_concentration:.1f}')

        ax.set_title(f'Peak Trend of {param}', fontsize=12)
        ax.set_xlabel('Year')
        ax.set_ylabel(f'{param} Concentration')
        ax.grid(True, linestyle='--', alpha=0.4)

        ax.xaxis.set_major_locator(mdates.YearLocator())
        ax.xaxis.set_major_formatter(mdates.DateFormatter('%Y'))
        plt.setp(ax.get_xticklabels(), rotation=0, ha='center',
fontsize=9)

        ax.legend(fontsize=8, loc='lower left')

        for spine in ax.spines.values():
            spine.set_linewidth(0.6)
            spine.set_edgecolor('lightgray')

    for j in range(i + 1, len(axes)):
        fig.delaxes(axes[j])

    plt.tight_layout()
    plt.show()
else:
    print("No AQI parameters found to plot peak trends.")

import matplotlib.dates as mdates

for param in params_to_plot:
    plt.figure(figsize=(8,5))
    plt.plot(df_daily_mean_station['tanggal'],
df_daily_mean_station[param],
            color='gray', linestyle='-', linewidth=0.7, alpha=0.7)

    if param in highest_concentration_dates:
        peak_date = highest_concentration_dates[param]['date']

```

```

        peak_concentration
highest_concentration_dates[param]['concentration']
        plt.scatter(peak_date, peak_concentration, color='red', s=80,
                    label=f'Highest: {peak_concentration:.1f}')

plt.title(f'Peak Trend of {param}', fontsize=14)
plt.xlabel('Year')
plt.ylabel(f'{param} Concentration')
plt.grid(True, linestyle='--', alpha=0.4)

plt.legend(fontsize=9, loc='lower left', frameon=False)

plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%Y'))
plt.gca().xaxis.set_major_locator(mdates.YearLocator())

for spine in plt.gca().spines.values():
    spine.set_edgecolor('#cccccc')
    spine.set_linewidth(0.8)

plt.tight_layout()
plt.show()

"""# Global Historical Climatology Network-hourly (NOAA Satellite)

The dataset utilized in this study also includes hourly meteorological
data, containing variables such as STATION, Station_name, DATE, LATITUDE,
LONGITUDE, Elevation, temperature, precipitation_24_hour, remarks, and
others. These data are collected from four stations managed by the National
Oceanic and Atmospheric Administration (NOAA) of the U.S. Government,
namely:

1. IDI0000WIHH - Halim Perdanakusuma Station, East Jakarta
2. IDU096749-1 - Soekarno Hatta Station, Banten
3. IDU096745-1 - Jakarta Observatory
4. IDU096741-1 - Tanjungpriok Station, North Jakarta

This dataset is publicly available as open source at:
https://www.ncei.noaa.gov/access/search/data-search/global-historical-climatology-network-hourly?bbox=-5.834,106.464,-6.596,107.226&page

## Data Understanding
"""

from google.colab import drive
drive.mount('/content/drive')

import os
import pandas as pd

folder_path = '/content/drive/MyDrive/aqi_data/meteorology'
file_list = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

dataframes = {}
for file_name in file_list:
    df = pd.read_csv(os.path.join(folder_path, file_name))
    dataframes[file_name] = df

print(dataframes.keys())

"""### Dataframe 2020

```

```

#### Halim Perdanakusuma Station (IDI0000WIHH)
"""

dataframes['GHCNh_IDI0000WIHH_2020.csv'].head()

##### Dataframe 2021

#### Halim Perdanakusuma Station (IDI0000WIHH)
"""

dataframes['GHCNh_IDI0000WIHH_2021.csv'].head()

##### Dataframe 2022

#### Halim Perdanakusuma Station (IDI0000WIHH)
"""

dataframes['GHCNh_IDI0000WIHH_2022.csv'].head()

##### Dataframe 2023

#### Halim Perdanakusuma Station (IDI0000WIHH)
"""

dataframes['GHCNh_IDI0000WIHH_2023.csv'].head()

##### Soekarno Hatta Station (IDU096749-1)
dataframes['GHCNh_IDU096749-1_2023.csv'].head()

##### Jakarta Observartory (IDU096745-1)
dataframes['GHCNh_IDU096745-1_2023.csv'].head()

##### Tanjungpriok Station (IDU096741-1)
dataframes['GHCNh_IDU096741-1_2023.csv'].head()

##### Dataframe 2024

#### Halim Perdanakusuma Station (IDI0000WIHH)
"""

dataframes['GHCNh_IDI0000WIHH_2024.csv'].head()

##### Soekarno Hatta Station (IDU096749-1)
dataframes['GHCNh_IDU096749-1_2024.csv'].head()

##### Jakarta Observatory (IDU096745-1)
dataframes['GHCNh_IDU096745-1_2024.csv'].head()

##### Tanjungpriok Station (IDU096741-1)
dataframes['GHCNh_IDU096741-1_2024.csv'].head()

##### Data Preparation
"""

import pandas as pd
import os

```

```

folder_path = '/content/drive/MyDrive/aqi_data/meteorology'
file_list = [f for f in os.listdir(folder_path) if f.endswith('.csv')]

df_list = []
for file_name in file_list:
    file_path = os.path.join(folder_path, file_name)
    df = pd.read_csv(file_path, low_memory=False)
    df.columns = df.columns.str.lower()
    df_list.append(df)

df_all_meteo = pd.concat(df_list, ignore_index=True)

selected_cols = [
    'station',
    'station_name',
    'date',
    'latitude',
    'longitude',
    'elevation',
    'temperature',
    'dew_point_temperature',
    'relative_humidity',
    'wind_speed',
    'wind_direction',
    'precipitation',
    'precipitation_24_hour',
    'sea_level_pressure',
    'visibility'
]

selected_cols_existing = [col for col in selected_cols if col in
df_all_meteo.columns]
df_all_meteo = df_all_meteo[selected_cols_existing].copy()

print("\nNaN count before cleaning:")
print(df_all_meteo.isna().sum())

rows_before = len(df_all_meteo)

df_all_meteo = df_all_meteo.dropna(subset=['latitude', 'longitude',
'date'], how='any')

df_all_meteo['date'] = pd.to_datetime(df_all_meteo['date'],
errors='coerce')
df_all_meteo = df_all_meteo.dropna(subset=['date'])

num_cols = df_all_meteo.select_dtypes(include=['float64',
'int64']).columns
for col in num_cols:
    median_value = df_all_meteo[col].median()
    df_all_meteo[col] = df_all_meteo[col].fillna(median_value)

print("\nNaN count after cleaning:")
print(df_all_meteo.isna().sum())

rows_after = len(df_all_meteo)
print(f"\nNumber of rows before cleaning: {rows_before}")

```

```

print(f"Number of rows after cleaning : {rows_after}")

print("\nCleaned data preview:")
print(df_all_meteo.head())

"""## Exploratory Data Analysis

### Descriptive Summary

#### Data Type
"""

print("\nDataset info:")
print(df_all_meteo.info())

"""\#### Data Analysis

##### Numerical Data
"""

import matplotlib.pyplot as plt
import seaborn as sns

drop_cols = ['precipitation', 'precipitation_24_hour', 'longitude',
'latitude']

numerical_features_meteo = [
    col for col in df_all_meteo.select_dtypes(include=['float64',
'int64']).columns
    if col not in drop_cols
]

def plot_numerical_meteo(df, numerical_features, main_title="Meteorology
Numerical Distributions", color='steelblue'):
    sns.set(style='whitegrid')

    num_cols = len(numerical_features)
    n_cols = 3 if num_cols >= 3 else num_cols
    n_rows = -(-num_cols // n_cols)

    plt.figure(figsize=(5 * n_cols, 4 * n_rows))

    for i, col in enumerate(numerical_features):
        plt.subplot(n_rows, n_cols, i + 1)
        if col in df.columns:
            sns.histplot(
                df[col].dropna(),
                kde=True,
                bins=30,
                color=color,
                edgecolor='black',
                linewidth=0.5
            )
            plt.title(f'Distribution: {col}', fontsize=11)
            plt.xlabel(col)
            plt.ylabel('Count')
        else:
            plt.title(f'{col} not found', fontsize=10)
            plt.axis('off')

    plt.tight_layout()

```

```

plt.suptitle(main_title, fontsize=16, fontweight='bold', y=1.03)
plt.show()

plot_numerical_meteo(df_all_meteo, numerical_features_meteo,
main_title='Meteorology Numerical Distributions', color='steelblue')

import matplotlib.pyplot as plt
import seaborn as sns

drop_cols = ['precipitation', 'precipitation_24_hour', 'longitude',
'latitude']

numerical_features_meteo = [
    col for col in df_all_meteo.select_dtypes(include=['float64',
'int64']).columns
    if col not in drop_cols
]

def plot_numerical_meteo_onebyone(df, numerical_features,
main_title="Distribution", color='steelblue'):
    sns.set(style='whitegrid')

    for col in numerical_features:
        if col in df.columns:
            plt.figure(figsize=(6, 4))
            sns.histplot(
                df[col].dropna(),
                kde=True,
                bins=30,
                color=color,
                edgecolor='black',
                linewidth=0.5
            )
            plt.title(f'{main_title} - {col}', fontsize=14)
            plt.xlabel(col)
            plt.ylabel('Count')

            plt.tight_layout()
            plt.show()
        else:
            print(f"Column {col} not found in DataFrame")

plot_numerical_meteo_onebyone(
    df_all_meteo,
    numerical_features_meteo,
    main_title='Distribution',
    color='steelblue'
)

"""##### Categorical Data"""

import matplotlib.pyplot as plt
import seaborn as sns

categorical_features_meteo = ['station_name']

def plot_categorical_meteo(df, categorical_features,
main_title="Meteorology Categorical Distributions", color='steelblue'):
    sns.set(style='whitegrid')

    num_cols = len(categorical_features)

```

```

n_cols = 3 if num_cols >= 3 else num_cols
n_rows = -(-num_cols // n_cols)

plt.figure(figsize=(5 * n_cols, 4 * n_rows))

for i, col in enumerate(categorical_features):
    plt.subplot(n_rows, n_cols, i + 1)
    if col in df.columns:
        order = df[col].value_counts().index
        sns.countplot(
            data=df,
            x=col,
            order=order,
            color=color,
            edgecolor='black',
            linewidth=0.5
        )
        plt.title(f'Distribution: {col}', fontsize=11)
        plt.xlabel(col)
        plt.ylabel('Count')
        plt.xticks(rotation=45, ha='right')
    else:
        plt.title(f'{col} not found', fontsize=10)
        plt.axis('off')

plt.tight_layout()
plt.suptitle(main_title, fontsize=14, y=1.03)
plt.show()

plot_categorical_meteo(df_all_meteo, categorical_features_meteo,
main_title='Meteorology Categorical Distributions', color='steelblue')

"""#### Statistic Descriptive"""

print("\nDescriptive statistics:")
print(df_all_meteo.describe())

"""#### Spatial Trends

#### Station Geographical Distribution of Peak AQI & Meteorological Factors
"""

!pip install geopandas contextily rtree

import numpy as np
import matplotlib.pyplot as plt
import geopandas as gpd
from shapely.geometry import Point
import contextily as ctx
from scipy.interpolate import griddata
from matplotlib.lines import Line2D

aqi_data = [
    ('DKI1', 'Bundaran HI', -6.193, 106.820, 150),
    ('DKI2', 'Kelapa Gading', -6.152, 106.905, 287),
    ('DKI3', 'Jagakarsa', -6.336, 106.823, 120),
    ('DKI4', 'Lubang Buaya', -6.289, 106.903, 191),
    ('DKI5', 'Kebon Jeruk', -6.196, 106.769, 65)
]

gdf_aqi = gpd.GeoDataFrame(aqi_data, columns=['ID', 'Name', 'Lat', 'Lon',
'PeakValue'])

```

```

gdf_aqi['geometry'] = gdf_aqi.apply(lambda row: Point(row['Lon'],
row['Lat']), axis=1)
gdf_aqi.set_crs(epsg=4326, inplace=True)
gdf_aqi = gdf_aqi.to_crs(epsg=3857)

meteo_data = [
    ('IDI0000WIHH', 'Halim Perdanakusuma', -6.266, 106.891, 29.5),
    ('IDU096749-1', 'Soekarno Hatta', -6.125, 106.655, 31.2),
    ('IDU096745-1', 'Jakarta Observatory', -6.174, 106.827, 28.7),
    ('IDU096741-1', 'Tanjungpriok', -6.104, 106.879, 30.1)
]
gdf_meteo = gpd.GeoDataFrame(meteo_data, columns=['ID', 'Name', 'Lat',
'Lon', 'PeakValue'])
gdf_meteo['geometry'] = gdf_meteo.apply(lambda row: Point(row['Lon'],
row['Lat']), axis=1)
gdf_meteo.set_crs(epsg=4326, inplace=True)
gdf_meteo = gdf_meteo.to_crs(epsg=3857)

x = gdf_aqi.geometry.x.values
y = gdf_aqi.geometry.y.values
z = gdf_aqi["PeakValue"].values

grid_x, grid_y = np.mgrid[x.min()-2000:x.max()+2000:500j, y.min()-
2000:y.max()+2000:500j]
grid_z = griddata((x, y), z, (grid_x, grid_y), method='cubic')

fig, ax = plt.subplots(figsize=(9, 9))

ax.imshow(
    grid_z.T,
    extent=(x.min()-2000, x.max()+2000, y.min()-2000, y.max()+2000),
    origin='lower',
    cmap="RdYlGn_r",
    alpha=0.55
)

sc_aqi = ax.scatter(
    gdf_aqi.geometry.x,
    gdf_aqi.geometry.y,
    c=gdf_aqi["PeakValue"],
    cmap="RdYlGn_r",
    s=180,
    marker="^",
    zorder=3
)

sc_meteo = ax.scatter(
    gdf_meteo.geometry.x,
    gdf_meteo.geometry.y,
    c=gdf_meteo["PeakValue"],
    cmap="Blues",
    s=120,
    marker="o",
    zorder=3
)

for x0, y0, code, name in zip(gdf_aqi.geometry.x, gdf_aqi.geometry.y,
gdf_aqi["ID"], gdf_aqi["Name"]):
    ax.text(x0, y0+700, code, fontsize=9, weight="bold", color="black",
ha="center")
    ax.text(x0, y0-900, name, fontsize=8, color="black", ha="center")

```

```

for x0, y0, code, name in zip(gdf_meteo.geometry.x, gdf_meteo.geometry.y,
gdf_meteo["ID"], gdf_meteo["Name"]):
    ax.text(x0, y0+700, code, fontsize=9, weight="bold", color="darkblue",
ha="center")
    ax.text(x0, y0-900, name, fontsize=8, color="darkblue", ha="center")

legend_elements = [
    Line2D([0], [0], marker='^', color='w', markerfacecolor='black',
markersize=10, label='AQI Stations'),
    Line2D([0], [0], marker='o', color='w', markerfacecolor='darkblue',
markersize=8, label='Meteorological Stations')
]
legend = ax.legend(handles=legend_elements, loc="lower left",
frameon=True, bbox_to_anchor=(0.02, 0.05))
for text in legend.get_texts():
    if "Meteorological" in text.get_text():
        text.set_color("darkblue")

ctx.add_basemap(ax, source=ctx.providers.CartoDB.Positron)

ax.set_title("Peak AQI and Meteorological Factors in DKI Jakarta (2020-
2024)", fontsize=13)
ax.axis("off")

cax1 = fig.add_axes([0.15, 0.08, 0.3, 0.02])
cbar_aqi = fig.colorbar(sc_aqi, cax=cax1, orientation="horizontal")
cbar_aqi.set_label("AQI Concentration", fontsize=10, labelpad=2,
ha="center")
cbar_aqi.ax.tick_params(labelsize=10)

cax2 = fig.add_axes([0.55, 0.08, 0.3, 0.02])
cbar_meteo = fig.colorbar(sc_meteo, cax=cax2, orientation="horizontal")
cbar_meteo.set_label("Meteorology Concentration", fontsize=10, labelpad=2,
ha="center")
cbar_meteo.ax.tick_params(labelsize=10)

ax.annotate('N', xy=(0.05, 0.18), xytext=(0.05, 0.28),
arrowprops=dict(facecolor='black', width=4, headwidth=12),
ha='center', va='center', fontsize=11,
xycoords=ax.transAxes)

plt.show()

"""#### Annual Trends"""

import pandas as pd

df_meteo_annual = df_all_meteo.copy()

df_meteo_annual['date'] = pd.to_datetime(df_meteo_annual['date'],
errors='coerce')

df_meteo_annual['year'] = df_meteo_annual['date'].dt.year

meteo_parameters = [
    'elevation',
    'temperature',
    'dew_point_temperature',
    'relative_humidity',
    'wind_speed',

```

```

        'wind_direction',
        'precipitation',
        'precipitation_24_hour',
        'sea_level_pressure',
        'visibility'
    ]

    df_annual_mean = (
        df_meteo_annual
        .groupby(['station_name', 'year'])[meteo_parameters]
        .mean()
        .reset_index()
    )

    print(df_annual_mean.head())

    import seaborn as sns
    import matplotlib.pyplot as plt
    import pandas as pd

    df_meteo_all = df_all_meteo.copy()

    def robust_to_datetime(date_value):
        if pd.isna(date_value):
            return pd.NaT
        try:
            return pd.to_datetime(date_value)
        except:
            try:
                if pd.api.types.is_numeric_dtype(type(date_value)):
                    return pd.to_datetime(date_value, origin='1899-12-30',
unit='D', errors='coerce')
                else:
                    return pd.NaT
            except:
                return pd.NaT

    df_meteo_all['date'] = df_meteo_all['date'].apply(robust_to_datetime)
    df_meteo_all.dropna(subset=['date'], inplace=True)

    meteo_params = [
        'elevation',
        'temperature',
        'dew_point_temperature',
        'relative_humidity',
        'wind_speed',
        'wind_direction',
        'precipitation_24_hour',
        'sea_level_pressure',
        'visibility'
    ]

    meteo_params_present = [p for p in meteo_params if p in
df_meteo_all.columns]

    for col in meteo_params_present:
        df_meteo_all[col] = pd.to_numeric(df_meteo_all[col], errors='coerce')

    df_meteo_all[meteo_params_present] =
df_meteo_all[meteo_params_present].fillna(
    df_meteo_all[meteo_params_present].median()
)

```

```

df_annual_mean = (
    df_meteo_all.groupby(['date', 'station_name'])[meteo_params_present]
    .mean()
    .reset_index()
)

df_annual_mean['Tahun'] = df_annual_mean['date'].dt.year
df_annual_mean['Hari-Bulan'] = df_annual_mean['date'].dt.strftime('%m-%d')

df_melted_meteo = df_annual_mean.melt(
    id_vars=['Tahun', 'Hari-Bulan', 'station_name'],
    value_vars=meteo_params_present,
    var_name='Parameter',
    value_name='Average Value'
)

g = sns.FacetGrid(
    df_melted_meteo,
    col='Parameter',
    col_wrap=2,
    height=4,
    aspect=1.5,
    sharey=False
)

g.map_dataframe(
    sns.lineplot,
    x='Hari-Bulan',
    y='Average Value',
    hue='Tahun',
    marker=False,
    dashes=False,
    palette='tab10'
)

for ax in g.axes.flatten():
    ax.set_xticks(['01-01', '02-01', '03-01', '04-01', '05-01', '06-01',
                  '07-01', '08-01', '09-01', '10-01', '11-01', '12-01'])
    ax.set_xticklabels(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'June',
                       'July', 'Aug', 'Sept', 'Okt', 'Nov', 'Dec'],
rotation=45)
    ax.grid(True, linestyle='--', alpha=0.4)

    for spine in ax.spines.values():
        spine.set_edgecolor('#cccccc')
        spine.set_linewidth(0.8)

g.set_titles(col_template="{col_name}")
g.set_axis_labels('Month', 'Average Value')
g.fig.suptitle('Annual Trend of Average Meteorological Concentration',
fontsize=14, y=1.05)
g.add_legend(title='Year')

plt.tight_layout()
plt.show()

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

```

```

df_meteo_all = df_all_meteo.copy()

def robust_to_datetime(date_value):
    if pd.isna(date_value):
        return pd.NaT
    try:
        return pd.to_datetime(date_value)
    except:
        try:
            if pd.api.types.is_numeric_dtype(type(date_value)):
                return pd.to_datetime(date_value, origin='1899-12-30',
unit='D', errors='coerce')
            else:
                return pd.NaT
        except:
            return pd.NaT

df_meteo_all['date'] = df_meteo_all['date'].apply(robust_to_datetime)
df_meteo_all.dropna(subset=['date'], inplace=True)

meteo_params = [
    'elevation',
    'temperature',
    'dew_point_temperature',
    'relative_humidity',
    'wind_speed',
    'wind_direction',
    'precipitation_24_hour',
    'sea_level_pressure',
    'visibility'
]

meteo_params_present = [p for p in meteo_params if p in
df_meteo_all.columns]

for col in meteo_params_present:
    df_meteo_all[col] = pd.to_numeric(df_meteo_all[col], errors='coerce')

df_meteo_all[meteo_params_present] =
df_meteo_all[meteo_params_present].fillna(
    df_meteo_all[meteo_params_present].median()
)

df_annual_mean = (
    df_meteo_all.groupby(['date', 'station_name'])[meteo_params_present]
    .mean()
    .reset_index()
)

df_annual_mean['Tahun'] = df_annual_mean['date'].dt.year
df_annual_mean['Hari-Bulan'] = df_annual_mean['date'].dt.strftime('%m-
%d')

for param in meteo_params_present:
    plt.figure(figsize=(14, 6))
    sns.lineplot(
        data=df_annual_mean,
        x='Hari-Bulan',
        y=param,
        hue='Tahun',
        marker=False,

```

```

        dashes=False,
        palette='tab10'
    )

    plt.title(f"Annual Trend of {param.replace('_', ' ').title()}",
fontsize=16)
    plt.xlabel("Month", fontsize=12)
    plt.ylabel("Average Value", fontsize=12)
    plt.xticks(
        ticks=['01-01', '02-01', '03-01', '04-01', '05-01', '06-01',
              '07-01', '08-01', '09-01', '10-01', '11-01', '12-01'],
        labels=['Jan', 'Feb', 'Mar', 'Apr', 'May', 'June',
              'July', 'Aug', 'Sept', 'Oct', 'Nov', 'Dec'],
        rotation=45
    )
    plt.grid(True, linestyle='--', alpha=0.4)

    for spine in plt.gca().spines.values():
        spine.set_color("lightgrey")

    plt.legend(title="Year", loc='upper left')
    plt.show()

"""#### Hourly Trends Interval"""

df_meteo_hourly = df_meteo_all.copy()

meteo_parameters = [
    'elevation',
    'temperature',
    'dew_point_temperature',
    'relative_humidity',
    'wind_speed',
    'wind_direction',
    'precipitation_24_hour',
    'sea_level_pressure',
    'visibility'
]

df_meteo_hourly['date'] = pd.to_datetime(df_meteo_hourly['date'],
errors='coerce')
df_meteo_hourly = df_meteo_hourly.dropna(subset=['date'])
df_meteo_hourly = df_meteo_hourly.set_index('date')

for col in meteo_parameters:
    if col in df_meteo_hourly.columns:
        df_meteo_hourly[col] = pd.to_numeric(df_meteo_hourly[col],
errors='coerce')

for col in meteo_parameters:
    if col in df_meteo_hourly.columns:
        df_meteo_hourly[col].fillna(df_meteo_hourly[col].median(),
inplace=True)

intervals = ['1H', '8H', '12H', '24H', '48H']
resampled_meteo_dict = {}

for interval in intervals:
    meteo_params_present = [p for p in meteo_parameters if p in
df_meteo_hourly.columns]
    df_resampled = (

```

```

        df_meteo_hourly
        .groupby('station_name')[meteo_params_present]
        .resample(interval)
        .mean()
        .reset_index()
    )
    resampled_meteo_dict[interval] = df_resampled

print("Resampling of meteorological data complete.")

df_melted = pd.DataFrame()
for interval, df in resampled_meteo_dict.items():
    meteo_parameters = [
        'temperature',
        'dew_point_temperature',
        'relative_humidity',
        'wind_speed',
        'wind_direction',
        'precipitation_24_hour',
        'sea_level_pressure',
        'visibility'
    ]
    available_meteo_vars = [v for v in meteo_parameters if v in df.columns]

    df_melted_interval = df.melt(
        id_vars=['station_name', 'date'],
        value_vars=available_meteo_vars,
        var_name='Parameter',
        value_name='Value'
    )
    df_melted_interval['Interval'] = interval
    df_melted = pd.concat([df_melted, df_melted_interval],
        ignore_index=True)

df_melted['date'] = pd.to_datetime(df_melted['date'], errors='coerce')
df_melted = df_melted.dropna(subset=['date'])

if not df_melted.empty:
    df_melted['elapsed_hours'] = (
        df_melted['date'] - df_melted['date'].min()
    ).dt.total_seconds() / 3600
else:
    print("df_melted is empty after date processing.")

intervals = df_melted['Interval'].unique()
print(intervals)

for interval in intervals:
    count = len(df_melted[df_melted['Interval'] == interval])
    print(f"Interval {interval}: {count} rows")

"""##### 1 Hour Trend"""

display(resampled_meteo_dict['1H'].head())

"""##### 8 Hour Trend

"""

display(resampled_meteo_dict['8H'].head())

```

```

##### 12 Hour Trend"""

display(resampled_meteo_dict['12H'].head())

##### 24 Hour Trend

"""

display(resampled_meteo_dict['24H'].head())

##### 48 Hour Trend

"""

display(resampled_meteo_dict['48H'].head())

##### Comparison 1h, 8h, 12h, 24h, 48 Meteorology Trends"""

import matplotlib.pyplot as plt
import pandas as pd
import math

meteo_vars_colors = {
    'temperature': 'orange',
    'dew_point_temperature': 'green',
    'relative_humidity': 'red',
    'wind_speed': 'steelblue',
    'wind_direction': 'purple',
    'precipitation_24_hour': 'olive',
    'sea_level_pressure': 'turquoise',
    'visibility': 'blue'
}

parameters = df_melted['Parameter'].unique()
n_params = len(parameters)

n_cols = 3
n_rows = math.ceil(n_params / n_cols)

intervals_present = df_melted['Interval'].unique()

fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, n_rows * 4))
axes = axes.flatten()

for i, param in enumerate(parameters):
    ax = axes[i]
    subset_param = df_melted[df_melted['Parameter'] == param]

    for idx, interval in enumerate(intervals_present):
        subset_interval = subset_param[subset_param['Interval'] == interval]
        if not subset_interval.empty:
            label = param.replace('_', ' ').title() if idx == 0 else None
            ax.plot(
                subset_interval['elapsed_hours'],
                subset_interval['Value'],
                label=label,
                color=meteo_vars_colors.get(param, 'gray'),
                linewidth=0.8
            )

```

```

        ax.set_title(param.replace('_', ' ').title())
        ax.set_xlabel("Elapsed Time (hours)")
        ax.set_ylabel("Value")
        ax.legend(title="Parameter", loc='upper left')
        ax.grid(True, linestyle='--', alpha=0.5)

for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()

import matplotlib.pyplot as plt

meteo_vars_colors = {
    'temperature': 'orange',
    'dew_point_temperature': 'green',
    'relative_humidity': 'red',
    'wind_speed': 'steelblue',
    'wind_direction': 'purple',
    'precipitation_24_hour': 'olive',
    'sea_level_pressure': 'turquoise',
    'visibility': 'blue'
}

parameters = df_melted['Parameter'].unique()
intervals_present = df_melted['Interval'].unique()

for param in parameters:
    plt.figure(figsize=(7, 4))
    subset_param = df_melted[df_melted['Parameter'] == param]

    for interval in intervals_present:
        subset_interval = subset_param[subset_param['Interval'] == interval]
        if not subset_interval.empty:
            plt.plot(
                subset_interval['elapsed_hours'],
                subset_interval['Value'],
                color=meteo_vars_colors.get(param, 'gray'),
                linewidth=0.8
            )

    plt.title(param.replace('_', ' ').title(), fontsize=14)
    plt.xlabel("Elapsed Time (hours)")
    plt.ylabel("Value")

    plt.legend([param.replace('_', ' ').title()], title="Parameter",
loc='upper left')
    plt.grid(True, linestyle='--', alpha=0.5)

    ax = plt.gca()
    for spine in ax.spines.values():
        spine.set_edgecolor('#cccccc')
        spine.set_linewidth(0.8)

    plt.tight_layout()
    plt.show()

"""#### Peak Concentration Highlight"""

```

```

import pandas as pd

meteo_parameters = [
    'temperature',
    'dew_point_temperature',
    'relative_humidity',
    'wind_speed',
    'wind_direction',
    'precipitation_24_hour',
    'sea_level_pressure',
    'visibility'
]

for param in meteo_parameters:
    if param in df_all_meteo.columns:
        df_all_meteo[param] = pd.to_numeric(df_all_meteo[param],
errors='coerce')
    else:
        print(f"Warning: Parameter '{param}' not found in df_all_meteo.")

peak_records = []

for param in meteo_parameters:
    if param not in df_all_meteo.columns:
        continue
    idxmax_per_station = df_all_meteo.groupby('station_name')[param].idxmax().dropna()
    for station_name_code, idxmax in idxmax_per_station.items():
        if pd.isna(idxmax):
            continue
        row = df_all_meteo.loc[idxmax]

        peak_records.append({
            'station_name': station_name_code,
            'parameter': param,
            'date': row['date'],
            'concentration': row[param]
        })

peak_meteo_df = pd.DataFrame(peak_records)
peak_meteo_df = peak_meteo_df.sort_values(by=['parameter',
'station_name']).reset_index(drop=True)

print("Peak Meteorology Data per Station and Parameter:")
print(peak_meteo_df)

peak_per_param = peak_meteo_df.loc[peak_meteo_df.groupby('parameter')['concentration'].idxmax()].reset_index(drop=True)
print("Peak Meteorology Data per Parameter (1 station per parameter):")
print(peak_per_param[['station_name', 'parameter', 'date', 'concentration']])

import matplotlib.pyplot as plt
import pandas as pd
import math

meteo_parameters = [
    'temperature',
    'dew_point_temperature',

```

```

        'relative_humidity',
        'wind_speed',
        'wind_direction',
        'precipitation_24_hour',
        'sea_level_pressure',
        'visibility'
    ]

    highest_concentration_dates_meteo = {}

    for param in meteo_parameters:
        if param in df_all_meteo.columns:
            df_all_meteo[param] = pd.to_numeric(df_all_meteo[param],
            errors='coerce')

            if df_all_meteo[param].notna().any():
                idx_max = df_all_meteo[param].idxmax()
                max_row = df_all_meteo.loc[idx_max]
                highest_concentration_dates_meteo[param] = {
                    'date': max_row['date'],
                    'concentration': max_row[param]
                }
            else:
                print(f"Warning: Parameter '{param}' has no non-null values.")
        else:
            print(f"Warning: Parameter '{param}' not found in df_all_meteo.")

    params_to_plot = list(highest_concentration_dates_meteo.keys())
    n_params = len(params_to_plot)
    n_cols = 3
    n_rows = math.ceil(n_params / n_cols)

    fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, n_rows * 3))
    axes = axes.flatten()

    for i, param in enumerate(params_to_plot):
        ax = axes[i]
        ax.plot(df_all_meteo['date'], df_all_meteo[param],
                color='gray', linestyle='-', linewidth=0.5, alpha=0.6)

        peak_date = highest_concentration_dates_meteo[param]['date']
        peak_concentration = highest_concentration_dates_meteo[param]['concentration']
        ax.scatter(peak_date, peak_concentration,
                    color='red', s=80,
                    label=f'Highest: {peak_concentration:.1f}')

        ax.set_title(f'Peak Trend of {param.replace("_", " ").title()}',
                    fontsize=12)
        ax.set_xlabel('Year')
        ax.set_ylabel(param.replace("_", " ").title())
        ax.grid(True, linestyle='--', alpha=0.4)
        ax.legend(fontsize=8, loc='lower left')

    for j in range(i + 1, len(axes)):
        fig.delaxes(axes[j])

    plt.tight_layout()
    plt.show()

import matplotlib.pyplot as plt

```

```

import pandas as pd

meteo_parameters = [
    'temperature',
    'dew_point_temperature',
    'relative_humidity',
    'wind_speed',
    'wind_direction',
    'precipitation_24_hour',
    'sea_level_pressure',
    'visibility'
]

highest_concentration_dates_meteo = {}

for param in meteo_parameters:
    if param in df_all_meteo.columns:
        df_all_meteo[param] = pd.to_numeric(df_all_meteo[param],
errors='coerce')

        if df_all_meteo[param].notna().any():
            idx_max = df_all_meteo[param].idxmax()
            max_row = df_all_meteo.loc[idx_max]
            highest_concentration_dates_meteo[param] = {
                'date': max_row['date'],
                'concentration': max_row[param]
            }
        else:
            print(f"Warning: Parameter '{param}' has no non-null values.")
    else:
        print(f"Warning: Parameter '{param}' not found in df_all_meteo.")

for param, info in highest_concentration_dates_meteo.items():
    plt.figure(figsize=(8, 5))

    plt.plot(
        df_all_meteo['date'], df_all_meteo[param],
        color='gray', linestyle='-', linewidth=0.6, alpha=0.6,
    )

    plt.scatter(
        info['date'], info['concentration'],
        color='red', s=80,
        label=f'Highest: {info["concentration"]:.1f}'
    )

    plt.title(f'Peak Trend of {param.replace("_", " ").title()}',
fontsize=14)
    plt.xlabel('Year')

    plt.grid(True, linestyle='--', alpha=0.4)
    plt.legend(fontsize=9, loc='lower left')

    ax = plt.gca()
    for spine in ax.spines.values():
        spine.set_linewidth(0.6)
        spine.set_edgecolor("lightgray")

    plt.tight_layout()
    plt.show()

```

```

"""# Data Weighting (Attention Mechanism)

## Data Merging
"""

df_merged = pd.concat([df_daily_mean_station, df_annual_mean],
ignore_index=True)

df_merged['tanggal'] = pd.to_datetime(df_merged['tanggal'])

merged_cols = ['PM2.5', 'PM10', 'SO2', 'CO', 'O3', 'NO2',
'temperature', 'dew_point_temperature',
'relative_humidity',
'wind_speed', 'wind_direction', 'precipitation_24_hour',
'sea_level_pressure', 'visibility']

merged_cols = [c for c in merged_cols if c in df_merged.columns]

target_col = 'PM2.5'

df_merged = df_merged.sort_values('tanggal').reset_index(drop=True)

df_merged.to_csv("aqi_meteo_merged.csv", index=False)

from google.colab import files
files.download("aqi_meteo_merged.csv")

"""## Spatiotemporal Correlation Analysis"""

import matplotlib.pyplot as plt
import seaborn as sns

df_corr = df_merged[merged_cols].corr(method='pearson')

plt.figure(figsize=(12,10))
sns.heatmap(
    df_corr,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0,
    cbar_kws={'label': 'Pearson Correlation'}
)

plt.title("Correlation Matrix of AQI and Meteorological Parameters",
fontsize=14, pad=15)
plt.tight_layout()
plt.show()

"""**A. Inter-Pollutant Correlations (AQI vs. AQI)**
1. **PM2.5 - PM10 (r = 0.69)**: Strong positive correlation. Both pollutants
originate from similar sources, such as traffic, combustion, and dust.
Risk: When PM10 increases, PM2.5 almost always increases, amplifying
adverse health effects.
2. **CO - O3 (r = 0.38)**: Moderate positive correlation. CO acts as a
precursor in photochemical reactions that generate ozone.
Risk: Elevated CO can indirectly increase ozone formation, worsening urban
smog events.
3. **CO - NO2 (r = 0.32)**: Moderate correlation. Both mainly come from
combustion and vehicular emissions.

```

Risk: Their co-occurrence represents high-risk traffic-related pollution episodes.

4. **PM2.5 - NO₂ (r = 0.25)**: Weak-moderate correlation. NO₂ emissions from traffic are often accompanied by fine particulates.

Risk: Joint exposure elevates respiratory stress, especially among sensitive populations.

5. **PM10 - NO₂ (r = 0.23)**: Weak-moderate correlation, linked to combustion and industrial activity.

Risk: Their co-occurrence contributes to poor AQI and reduced visibility.

Summary: The strongest risk pair is PM2.5-PM10, which are the core drivers of AQI deterioration. Gas-phase correlations (CO-O₃, CO-NO₂) highlight the influence of traffic emissions, while NO₂-particulate links underline common combustion sources.

B. Inter-Meteorological Correlations (Meteorology vs. Meteorology)

1. **Temperature - Relative Humidity (r = -0.89)**: Very strong negative correlation. Fundamental atmospheric relation; hot air reduces relative humidity, critical for both human stress and pollution chemistry.

2. **Dew Point - Relative Humidity (r = 0.62)**: Strong positive correlation. Higher dew point reflects higher RH, indicating muggy conditions prone to pollutant accumulation.

3. **Temperature - Wind Speed (r = 0.45)**: Moderate positive correlation. Hotter conditions often drive stronger wind circulation, affecting pollution dispersion.

4. **Pressure (Station - Sea Level) (r ≈ 0.40-0.50)**: Moderate correlation. These two measures are naturally related, used to normalize weather conditions.

5. **Temperature - Dew Point (r ≈ 0.35-0.40)**: Positive correlation. Indicates the air's increasing capacity to hold moisture at higher temperatures.

Summary: The most critical meteorological relationship is Temperature-Relative Humidity (-0.89), followed by moisture-related parameters (Dew Point-RH). Wind speed and pressure provide secondary roles in pollutant dispersion and atmospheric stability.

C. Pollutant-Meteorology Correlations (AQI vs. Meteorology)

1. **CO - Temperature (r = 0.32)**: Higher temperatures enhance CO accumulation due to stagnant conditions. Hot weather increases CO exposure risks, particularly in urban areas.

2. **PM2.5 - Visibility (r = 0.25)**: Fine particulates strongly reduce visibility. Visibility degradation is a direct indicator of harmful pollution episodes.

3. **PM10 - Visibility (r = 0.23)**: Coarse particles impair visibility, though less than PM2.5. Still critical for AQI spikes and haze formation.

4. **SO₂ - Visibility (r = 0.18)**: SO₂ leads to sulfate aerosol formation. Increases haze and respiratory irritation.

5. **PM10 - Temperature (r = 0.17)**: Hot weather increases PM10 resuspension from dust and road surfaces. Elevates short-term exposure during heat episodes.

Summary: The strongest AQI-meteorology link is CO-Temperature (0.32). Visibility is the most sensitive meteorological factor, consistently tied to PM2.5, PM10, and SO₂, confirming its role as a practical indicator of pollution intensity.

Weighting Type

""

```

import pandas as pd

merged_cols = [
    'PM2.5', 'PM10', 'SO2', 'CO', 'O3', 'NO2',
    'temperature', 'dew_point_temperature', 'relative_humidity',
    'wind_speed', 'wind_direction', 'precipitation_24_hour',
    'sea_level_pressure', 'visibility'
]

weight_high = 1.5
weight_default = 1.0

influential_params = [
    # aqi params
    'PM2.5', 'PM10', 'CO', 'O3', 'NO2',
    # meteorology params
    'temperature', 'dew_point_temperature', 'relative_humidity',
    'wind_speed', 'sea_level_pressure',
    # aqi-meteorology params
    'visibility', 'SO2'
]

exclude_cols = ['tanggal', 'station', 'Hari-Bulan']
param_cols = [c for c in df_merged.columns if c not in exclude_cols]

if 'station' in df_merged.columns:
    all_records_df = df_merged.melt(
        id_vars=['station', 'tanggal'],
        value_vars=param_cols,
        var_name='parameter',
        value_name='value'
    )
else:
    all_records_df = df_merged.melt(
        id_vars=['tanggal'],
        value_vars=param_cols,
        var_name='parameter',
        value_name='value'
    )

all_records_df['tanggal'] = pd.to_datetime(all_records_df['tanggal']).dt.normalize()

all_records_df['attention_weight'] = weight_default

all_records_df.loc[all_records_df['parameter'].isin(influential_params),
    'attention_weight'] = weight_high

attention_df = all_records_df[['station', 'tanggal', 'parameter', 'attention_weight']]

print("Attention weights sample:")
print(attention_df.head())
print(attention_df.tail())
print("Distribusi bobot:")
print(attention_df['attention_weight'].value_counts())

"""# Data Scaling"""

import pandas as pd
import numpy as np

```

```

from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
import seaborn as sns

pivot_df = attention_df.pivot_table(
    index=['station', 'tanggal'],
    columns='parameter',
    values='attention_weight',
    fill_value=0
).reset_index()

available_cols = [c for c in merged_cols if c in pivot_df.columns]

print("Columns available for scaling:", available_cols)
print("Missing columns:", set(merged_cols) - set(available_cols))

scaler = StandardScaler()
scaled_values = scaler.fit_transform(pivot_df[available_cols])
scaled_df = pd.DataFrame(scaled_values, columns=available_cols)

param_weights = attention_df.groupby("parameter")["attention_weight"].mean().to_dict()

weights = np.array([param_weights.get(col, 1.0) for col in available_cols])

X = scaled_df.values
X_weighted = X * np.sqrt(weights)

weighted_cov = np.cov(X_weighted, rowvar=False)

d = np.sqrt(np.diag(weighted_cov))
with np.errstate(invalid="ignore", divide="ignore"):
    weighted_corr = weighted_cov / np.outer(d, d)

weighted_corr_df = pd.DataFrame(weighted_corr,
                                index=available_cols,
                                columns=available_cols)

"""# Data Splitting"""

from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import numpy as np

def split_dataset(df, target_col, ratios):
    n = len(df)
    train_size = int(ratios[0] * n)
    val_size = int(ratios[1] * n)
    test_size = n - train_size - val_size

    train = df.iloc[:train_size]
    val = df.iloc[train_size:train_size+val_size]
    test = df.iloc[train_size+val_size:]

    return train, val, test

split_options = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2)
}

```

```

}

results = {}

for name, ratio in split_options.items():
    train, val, test = split_dataset(df_merged, target_col='PM2.5',
    ratios=ratio)
    results[name] = {
        "train": len(train),
        "val": len(val),
        "test": len(test)
    }

print("Number of samples per split:")
for k,v in results.items():
    print(k, v)

import matplotlib.pyplot as plt
import numpy as np

# === Data ===
results = {
    "60/20/20": {'train': 63279, 'val': 21093, 'test': 21094},
    "70/15/15": {'train': 73826, 'val': 15819, 'test': 15821},
    "80/10/10": {'train': 84372, 'val': 10546, 'test': 10548},
    "50/30/20": {'train': 52733, 'val': 31639, 'test': 21094}
}

splits = list(results.keys())
train_sizes = [results[k]['train'] for k in splits]
val_sizes    = [results[k]['val'] for k in splits]
test_sizes   = [results[k]['test'] for k in splits]

x = np.arange(len(splits))
totals = np.array(train_sizes) + np.array(val_sizes) +
np.array(test_sizes)

# === Figure ===
plt.figure(figsize=(9,6))

colors = ["#4C72B0", "#DD8452", "#55A868"]

bars1 = plt.bar(x, train_sizes, label='Train', color=colors[0])
bars2 = plt.bar(x, val_sizes, bottom=train_sizes, label='Validation',
color=colors[1])
bars3 = plt.bar(x, test_sizes, bottom=np.array(train_sizes)+np.array(val_sizes),
label='Test', color=colors[2])

plt.ylabel("Number of Samples", fontsize=14)
plt.xlabel("Data Splitting Scheme", fontsize=14)
plt.title("Dataset Sizes under Different Splitting Schemes", fontsize=16)
plt.xticks(x, splits, fontsize=12)
plt.yticks(fontsize=12)
plt.grid(axis="y", linestyle="--", alpha=0.4)

# === Labels (persentase & total) ===
for i in range(len(splits)):
    total = totals[i]
    # Train %
    plt.text(i, train_sizes[i]/2, f"{train_sizes[i]/total:.0%}",

```

```

        ha="center",    va="center",    color="white",    fontsize=11,
weight="bold")
    # Val %
    plt.text(i,          train_sizes[i]          +          val_sizes[i]/2,
f"{val_sizes[i]/total:.0%}",
        ha="center",    va="center",    color="white",    fontsize=11,
weight="bold")
    # Test %
    plt.text(i, train_sizes[i] + val_sizes[i] + test_sizes[i]/2,
        f"{test_sizes[i]/total:.0%}",
        ha="center",    va="center",    color="white",    fontsize=11,
weight="bold")
    # Total (dinaikkan 5% dari tinggi bar → lebih jauh dari bar)
    plt.text(i, total * 1.00, f"{total:.0%}",
        ha="center", va="bottom", fontsize=11)

# === Legend di bawah (horizontal) ===
plt.legend(
    fontsize=12, ncol=3, loc="upper center", bbox_to_anchor=(0.5, -0.1),
    frameon=False
)

plt.tight_layout()
plt.show()

import matplotlib.pyplot as plt
import seaborn as sns

df_model = df_merged.sort_values("tanggal").reset_index(drop=True)

important_features = [c for c in [
    'PM2.5', 'PM10', 'CO', 'O3',
    'NO2', 'temperature', 'dew_point_temperature',
    'relative_humidity', 'wind_speed', 'sea_level_pressure',
    'visibility', 'SO2'
] if c in df_model.columns]

splits = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2)
}

for name, (train_ratio, val_ratio, test_ratio) in splits.items():
    n = len(df_model)
    train_size = int(train_ratio * n)
    val_size    = int(val_ratio * n)

    train_df = df_model.iloc[:train_size]
    val_df    = df_model.iloc[train_size:train_size+val_size]
    test_df   = df_model.iloc[train_size+val_size:]

    print(f"\n=== Split {name} ===")
    print(f"Train={len(train_df)},                      Val={len(val_df)},
Test={len(test_df)}")

    for col in important_features:
        plt.figure(figsize=(8,5))
        sns.kdeplot(train_df[col], label="Train", fill=True, alpha=0.4)
        sns.kdeplot(val_df[col], label="Validation", fill=True, alpha=0.4)

```

```

sns.kdeplot(test_df[col], label="Test", fill=True, alpha=0.4)

plt.title(f"Distribution of {col} ({name})", fontsize=14)
plt.xlabel(col, fontsize=12)
plt.ylabel("Density", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.5)
plt.legend(fontsize=10, loc="upper left")
plt.tight_layout()
plt.show()

"""## Linear Regression"""

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
from sklearn.impute import SimpleImputer
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

def split_dataset(df, target_col, ratios):
    n = len(df)
    train_size = int(ratios[0] * n)
    val_size = int(ratios[1] * n)
    test_size = n - train_size - val_size

    train = df.iloc[:train_size].copy()
    val = df.iloc[train_size:train_size+val_size].copy()
    test = df.iloc[train_size+val_size:].copy()

    return train, val, test

def evaluate_model(y_true, y_pred):
    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / y_true)) * 100
    return mae, mse, rmse, r2, mape

print("Number of NaNs in PM2.5 before cleaning:",
df_merged['PM2.5'].isna().sum(), "out of", len(df_merged))

df_merged['PM2.5']
df_merged['PM2.5'].interpolate(method='linear').fillna(df_merged['PM2.5']
.median())

print("Number of NaNs in PM2.5 after cleaning:",
df_merged['PM2.5'].isna().sum())

numeric_features
df_merged.select_dtypes(include=[np.number]).columns.tolist()
if 'PM2.5' in numeric_features:
    numeric_features.remove('PM2.5')

split_options = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2)
}

```

```

results_eval = {}

for name, ratio in split_options.items():

    train, val, test = split_dataset(df_merged, target_col='PM2.5',
ratios=ratio)

    if len(train) == 0 or len(val) == 0 or len(test) == 0:
        print(f" Split {name} skipped because one of the sets is empty")
        continue

    X_train, y_train = train.reindex(columns=numeric_features),
train['PM2.5']
    X_val, y_val = val.reindex(columns=numeric_features), val['PM2.5']
    X_test, y_test = test.reindex(columns=numeric_features),
test['PM2.5']

    imputer = SimpleImputer(strategy="median")
    X_train = imputer.fit_transform(X_train)
    X_val = imputer.transform(X_val)
    X_test = imputer.transform(X_test)

    model = LinearRegression()
    model.fit(X_train, y_train)

    y_val_pred = model.predict(X_val)
    y_test_pred = model.predict(X_test)

    val_metrics = evaluate_model(y_val, y_val_pred)
    test_metrics = evaluate_model(y_test, y_test_pred)

    results_eval[name] = {
        "Val_MAE": val_metrics[0],
        "Val_MSE": val_metrics[1],
        "Val_RMSE": val_metrics[2],
        "Val_R2": val_metrics[3],
        "Val_MAPE": val_metrics[4],
        "Test_MAE": test_metrics[0],
        "Test_MSE": test_metrics[1],
        "Test_RMSE": test_metrics[2],
        "Test_R2": test_metrics[3],
        "Test_MAPE": test_metrics[4]
    }

df_results = pd.DataFrame(results_eval).T
print("\nModel Evaluation Results per Split:\n")
print(df_results)

"""Hasil evaluasi menunjukkan bahwa skema split data berpengaruh signifikan
terhadap kinerja model Linear Regression. Split 60/20/20 menghasilkan error
tertinggi (Val_MAE = 9.23, Test_MAE = 9.15, Test_RMSE = 11.18), yang
menandakan jumlah data latih sebesar 60% belum cukup untuk menangkap pola
variabel secara optimal. Split 50/30/20 memperlihatkan hasil sedikit lebih
baik (Test_MAE = 7.59, Test_RMSE = 9.60), tetapi tetap belum memadai
dibanding skema lain.

Sementara itu, split 80/10/10 memberikan nilai error paling kecil (Test_MAE
= 4.28, Test_RMSE = 5.52, Test_MAPE = 3.78%), namun ukuran data validasi
dan uji relatif kecil (hanya 10% masing-masing). Hal ini berpotensi

```

menimbulkan evaluasi yang kurang representatif, karena generalisasi model tidak diuji pada sampel data yang cukup beragam.

Dengan pertimbangan tersebut, skema 70/15/15 dipilih sebagai konfigurasi terbaik untuk penelitian ini. Meskipun error sedikit lebih tinggi dibanding 80/10/10 (Test_MAE = 5.66, Test_RMSE = 7.30), skema ini memberikan keseimbangan antara jumlah data latih yang cukup besar (70%) dan data validasi serta uji yang lebih representatif (15% + 15%). Hal ini membuat evaluasi performa model lebih adil (fair generalization) serta dapat dipercaya sebagai gambaran kemampuan model pada data baru.

```
## Light-GBM
"""

import lightgbm as lgb
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
from sklearn.impute import SimpleImputer
import numpy as np
import pandas as pd
from lightgbm import log_evaluation

def evaluate_model(y_true, y_pred):
    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / y_true)) * 100
    return mae, mse, rmse, r2, mape

def split_dataset(df, target_col, ratios):
    n = len(df)
    train_size = int(ratios[0] * n)
    val_size = int(ratios[1] * n)
    test_size = n - train_size - val_size

    train = df.iloc[:train_size].copy()
    val = df.iloc[train_size:train_size+val_size].copy()
    test = df.iloc[train_size+val_size:].copy()
    return train, val, test

numeric_features =
df_merged.select_dtypes(include=[np.number]).columns.tolist()
if "PM2.5" in numeric_features:
    numeric_features.remove("PM2.5")

split_options = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2)
}

results_lgb = {}

for name, ratio in split_options.items():
    train, val, test = split_dataset(df_merged, target_col="PM2.5",
ratios=ratio)

    if len(train) == 0 or len(val) == 0 or len(test) == 0:
        print(f" Split {name} skipped because one of the sets is empty")
```

```

        continue

    X_train, y_train = train.reindex(columns=numeric_features),
train["PM2.5"]
    X_val, y_val = val.reindex(columns=numeric_features), val["PM2.5"]
    X_test, y_test = test.reindex(columns=numeric_features),
test["PM2.5"]

    imputer = SimpleImputer(strategy="median")
    X_train = imputer.fit_transform(X_train)
    X_val = imputer.transform(X_val)
    X_test = imputer.transform(X_test)

    model = lgb.LGBMRegressor(
        n_estimators=500,
        learning_rate=0.05,
        max_depth=-1,
        num_leaves=31,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42
    )
    model.fit(
        X_train, y_train,
        eval_set=[(X_val, y_val)],
        eval_metric="l2",
        callbacks=[log_evaluation(0)]
    )

    y_val_pred = model.predict(X_val)
    y_test_pred = model.predict(X_test)

    val_metrics = evaluate_model(y_val, y_val_pred)
    test_metrics = evaluate_model(y_test, y_test_pred)

    results_lgb[name] = {
        "Val_MAE": val_metrics[0],
        "Val_MSE": val_metrics[1],
        "Val_RMSE": val_metrics[2],
        "Val_R2": val_metrics[3],
        "Val_MAPE": val_metrics[4],
        "Test_MAE": test_metrics[0],
        "Test_MSE": test_metrics[1],
        "Test_RMSE": test_metrics[2],
        "Test_R2": test_metrics[3],
        "Test_MAPE": test_metrics[4]
    }

df_results_lgb = pd.DataFrame(results_lgb).T
print("\n LightGBM Regressor Evaluation Results:\n")
print(df_results_lgb)

```

""Selama proses pelatihan, CatBoost secara otomatis memilih metode multi-threading baik berbasis kolom maupun baris, menyesuaikan jumlah bins serta data yang digunakan pada setiap konfigurasi split. Informasi pelatihan menunjukkan variasi jumlah data latih yang berbeda, mulai dari 52.733 hingga 84.372 titik data dengan 16 fitur yang digunakan. Hal ini menegaskan bahwa perubahan proporsi split berpengaruh langsung pada ketersediaan data latih, validasi, dan uji.

Hasil evaluasi CatBoost memperlihatkan bahwa skema 60/20/20 menghasilkan error yang relatif tinggi (Test_MAE = 0.0102, Test_RMSE = 0.0531), menandakan data latih sebesar 60% belum cukup optimal. Skema 50/30/20 justru memberikan nilai Test_RMSE terkecil (0.0111), namun data latih yang hanya 50% berisiko menurunkan kemampuan model dalam menangkap pola secara konsisten. Di sisi lain, skema 80/10/10 menghasilkan performa terbaik dengan error paling rendah (Test_MAE = 0.0079, Test_RMSE = 0.0171, Test_MAPE = 0.0070). Meski demikian, porsi data validasi dan uji yang kecil (10% masing-masing) berpotensi membuat evaluasi kurang representatif.

Dengan mempertimbangkan keseimbangan antara kualitas pelatihan dan representasi evaluasi, skema 70/15/15 dipilih sebagai yang paling sesuai. Meskipun sedikit lebih tinggi error-nya dibanding 80/10/10 (Test_MAE = 0.0078, Test_RMSE = 0.0384), konfigurasi ini menjaga keseimbangan antara data latih (70%) yang cukup besar untuk pembelajaran, serta data validasi dan uji (15% + 15%) yang lebih representatif untuk menguji generalisasi model secara adil (fair generalization).

```
## Random Forest
"""
```

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score, mean_absolute_percentage_error
```

```
def evaluate_random_forest(df, target_col="PM2.5", scale_target=False):
    results = []
```

```
    df = df.copy()
```

```
    for col in df.columns:
        if np.issubdtype(df[col].dtype, np.datetime64):
            df[col + "_year"] = df[col].dt.year
            df[col + "_month"] = df[col].dt.month
            df[col + "_day"] = df[col].dt.day
            if hasattr(df[col].dt, "hour"):
                df[col + "_hour"] = df[col].dt.hour
            df = df.drop(columns=[col])
```

```
    for col in df.select_dtypes(include=["object"]).columns:
        le = LabelEncoder()
        df[col] = le.fit_transform(df[col].astype(str))
```

```
    X = df.drop(columns=[target_col])
    y = df[target_col].values
```

```
    if scale_target:
        y_scaler = MinMaxScaler()
        y = y_scaler.fit_transform(y.reshape(-1, 1)).ravel()
    else:
        y_scaler = None
```

```
    splits = {
        "60/20/20": (0.6, 0.2, 0.2),
        "70/15/15": (0.7, 0.15, 0.15),
        "80/10/10": (0.8, 0.1, 0.1),
        "50/30/20": (0.5, 0.3, 0.2),
```

```

    }

    for split_name, (train_size, val_size, test_size) in splits.items():
        X_temp, X_test, y_temp, y_test = train_test_split(
            X, y, test_size=test_size, shuffle=False
        )
        val_ratio = val_size / (train_size + val_size)
        X_train, X_val, y_train, y_val = train_test_split(
            X_temp, y_temp, test_size=val_ratio, shuffle=False
        )

        model = RandomForestRegressor(
            n_estimators=200,
            max_depth=12,
            random_state=42,
            n_jobs=-1
        )
        model.fit(X_train, y_train)

        y_val_pred = model.predict(X_val)
        y_test_pred = model.predict(X_test)

        if scale_target and y_scaler is not None:
            y_val = y_scaler.inverse_transform(y_val.reshape(-1,
1)).ravel()
            y_test = y_scaler.inverse_transform(y_test.reshape(-1,
1)).ravel()
            y_val_pred = y_scaler.inverse_transform(y_val_pred.reshape(-
1, 1)).ravel()
            y_test_pred = y_scaler.inverse_transform(y_test_pred.reshape(-1, 1)).ravel()

            val_mae = mean_absolute_error(y_val, y_val_pred)
            val_mse = mean_squared_error(y_val, y_val_pred)
            val_rmse = np.sqrt(val_mse)
            val_r2 = r2_score(y_val, y_val_pred)
            val_mape = mean_absolute_percentage_error(y_val, y_val_pred)

            test_mae = mean_absolute_error(y_test, y_test_pred)
            test_mse = mean_squared_error(y_test, y_test_pred)
            test_rmse = np.sqrt(test_mse)
            test_r2 = r2_score(y_test, y_test_pred)
            test_mape = mean_absolute_percentage_error(y_test, y_test_pred)

            results.append({
                "Split": split_name,
                "Val_MAE": val_mae,
                "Val_MSE": val_mse,
                "Val_RMSE": val_rmse,
                "Val_R2": val_r2,
                "Val_MAPE": val_mape,
                "Test_MAE": test_mae,
                "Test_MSE": test_mse,
                "Test_RMSE": test_rmse,
                "Test_R2": test_r2,
                "Test_MAPE": test_mape,
            })

    return pd.DataFrame(results)

```

```

rf_results = evaluate_random_forest(df_merged, target_col="PM2.5",
scale_target=False)

print("\n Random Forest Regressor Evaluation Results :\n")
print(rf_results)

"""## FNN"""

import numpy as np
import pandas as pd
import tensorflow as tf

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score,
    mean_absolute_percentage_error
)

def evaluate_fnn(df, target_col="PM2.5", scale_target=False, epochs=50,
batch_size=64):
    results = []
    df = df.copy()

    for col in df.columns:
        if np.issubdtype(df[col].dtype, np.datetime64):
            df[col + "_year"] = df[col].dt.year
            df[col + "_month"] = df[col].dt.month
            df[col + "_day"] = df[col].dt.day
            if hasattr(df[col].dt, "hour"):
                df[col + "_hour"] = df[col].dt.hour
            df = df.drop(columns=[col])

    for col in df.select_dtypes(include=["object"]).columns:
        le = LabelEncoder()
        df[col] = le.fit_transform(df[col].astype(str))

    df = df.dropna(subset=[target_col])

    X = df.drop(columns=[target_col])
    y = df[target_col].values.reshape(-1, 1)

    X = X.fillna(X.median(numeric_only=True))
    X = X.fillna(0)

    mask = ~np.isnan(y).ravel()
    X = X[mask]
    y = y[mask]

    scaler = MinMaxScaler()
    X_scaled = scaler.fit_transform(X)

    if scale_target:
        y_scaler = MinMaxScaler()
        y = y_scaler.fit_transform(y)
    else:
        y_scaler = None

```

```

splits = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2),
}

for split_name, (train_size, val_size, test_size) in splits.items():
    try:
        X_temp, X_test, y_temp, y_test = train_test_split(
            X_scaled, y, test_size=test_size, shuffle=False
        )
        val_ratio = val_size / (train_size + val_size)
        X_train, X_val, y_train, y_val = train_test_split(
            X_temp, y_temp, test_size=val_ratio, shuffle=False
        )

        model = Sequential([
            tf.keras.Input(shape=(X_train.shape[1],)),
            Dense(128, activation="relu"),
            Dropout(0.2),
            Dense(64, activation="relu"),
            Dropout(0.2),
            Dense(1)
        ])
        model.compile(optimizer="adam", loss="mse")

        model.fit(
            X_train, y_train,
            validation_data=(X_val, y_val),
            epochs=epochs,
            batch_size=batch_size,
            verbose=0
        )

        y_val_pred = model.predict(X_val).ravel()
        y_test_pred = model.predict(X_test).ravel()

        if scale_target and y_scaler is not None:
            y_val = y_scaler.inverse_transform(y_val).ravel()
            y_test = y_scaler.inverse_transform(y_test).ravel()
            y_val_pred = y_val_pred
            y_test_pred = y_test_pred
        y_val_pred = y_scaler.inverse_transform(y_val_pred.reshape(-1, 1)).ravel()
        y_test_pred = y_test_pred
        y_test_pred = y_scaler.inverse_transform(y_test_pred.reshape(-1, 1)).ravel()

        results.append({
            "Split": split_name,
            "Val_MAE": mean_absolute_error(y_val, y_val_pred),
            "Val_MSE": mean_squared_error(y_val, y_val_pred),
            "Val_RMSE": np.sqrt(mean_squared_error(y_val,
y_val_pred)),
            "Val_R2": r2_score(y_val, y_val_pred),
            "Val_MAPE": mean_absolute_percentage_error(y_val,
y_val_pred),
            "Test_MAE": mean_absolute_error(y_test, y_test_pred),
            "Test_MSE": mean_squared_error(y_test, y_test_pred),
            "Test_RMSE": np.sqrt(mean_squared_error(y_test,
y_test_pred)),
            "Test_R2": r2_score(y_test, y_test_pred),

```

```

        "Test_MAPE": mean_absolute_percentage_error(y_test,
y_test_pred),
    })
    except Exception as e:
        print(f"Split {split_name} error: {e}")

    return pd.DataFrame(results)

if __name__ == "__main__":
    print("\n FNN Evaluation Results:\n")
    res_fnn = evaluate_fnn(df_merged, target_col="PM2.5",
scale_target=True, epochs=50, batch_size=128)
    print(res_fnn)

"""Hasil evaluasi menunjukkan bahwa pemilihan rasio pembagian data
berpengaruh signifikan terhadap performa model FNN. Skema 50/30/20
menghasilkan performa terburuk dengan Test_MAE sebesar 11.72 dan Test_RMSE
sebesar 18.84. Hal ini disebabkan oleh porsi data latih yang terlalu kecil
(50%) sehingga model gagal menangkap pola dengan baik, sementara proporsi
data validasi dan uji yang besar justru menegaskan lemahnya generalisasi.
Skema 60/20/20 memperlihatkan peningkatan, dengan Test_MAE 11.61 dan
Test_RMSE 15.44, meskipun error masih relatif tinggi. Sementara itu,
80/10/10 menghasilkan error terendah (Test_MAE 11.00 dan Test_RMSE 14.29),
tetapi keterbatasan proporsi data validasi dan uji (hanya 10%) membuat
hasil evaluasi kurang representatif serta berisiko menimbulkan bias pada
generalisasi model.

Skema 70/15/15, meskipun memberikan error yang lebih tinggi (Test_MAE 13.64
dan Test_RMSE 17.55), tetap dianggap proporsional karena menjaga
keseimbangan antara jumlah data latih (70%) dan data validasi serta uji
yang cukup representatif (masing-masing 15%). Selain itu, nilai R2 yang
fluktuatif (0.35-0.81) pada semua skema menunjukkan bahwa konfigurasi FNN
sederhana yang digunakan masih belum mampu menjelaskan variansi target
secara konsisten. Dengan demikian, meskipun skema 80/10/10 terlihat lebih
baik dari sisi metrik error, skema 70/15/15 tetap direkomendasikan karena
menawarkan keseimbangan yang lebih adil dalam mengevaluasi generalisasi
model.

## CNN-LSTM
"""

import numpy as np
import pandas as pd
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
from sklearn.impute import SimpleImputer
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv1D, LSTM, Dense, Input
from tensorflow.keras.optimizers import Adam

def evaluate_model(y_true, y_pred):
    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / y_true)) * 100
    return mae, mse, rmse, r2, mape

def create_sequences(X, y, timesteps=5):
    Xs, ys = [], []
    for i in range(len(X) - timesteps):

```

```

        if not np.isnan(y[i+timesteps]):
            Xs.append(X[i:(i+timesteps)])
            ys.append(y[i+timesteps])
    return np.array(Xs), np.array(ys)

def split_sequences(X_seq, y_seq, ratios):
    n = len(X_seq)
    train_size = int(ratios[0] * n)
    val_size = int(ratios[1] * n)
    X_train, y_train = X_seq[:train_size], y_seq[:train_size]
    X_val, y_val = X_seq[train_size:train_size+val_size],
y_seq[train_size:train_size+val_size]
    X_test, y_test = X_seq[train_size+val_size:],
y_seq[train_size+val_size:]
    return (X_train, y_train), (X_val, y_val), (X_test, y_test)

numeric_features =
df_merged.select_dtypes(include=[np.number]).columns.tolist()
if "PM2.5" in numeric_features:
    numeric_features.remove("PM2.5")

split_options = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2)
}

X_all = df_merged[numeric_features].values
y_all = df_merged["PM2.5"].values
imputer = SimpleImputer(strategy="median")
X_all = imputer.fit_transform(X_all)

timesteps = 5
X_seq, y_seq = create_sequences(X_all, y_all, timesteps)
print("Full sequence shape:", X_seq.shape, y_seq.shape)

results_cnn_lstm = {}

for name, ratio in split_options.items():
    print(f"\n🚀 Running split {name}")
    (X_train_seq, y_train_seq), (X_val_seq, y_val_seq), (X_test_seq,
y_test_seq) = split_sequences(X_seq, y_seq, ratio)

    n_features = X_train_seq.shape[2]

    model = Sequential([
        Input(shape=(timesteps, n_features)),
        Conv1D(filters=64, kernel_size=2, activation='relu'),
        LSTM(64, return_sequences=False),
        Dense(1)
    ])
    model.compile(optimizer=Adam(learning_rate=0.001), loss='mse')

    model.fit(X_train_seq, y_train_seq,
              validation_data=(X_val_seq, y_val_seq),
              epochs=20, batch_size=32, verbose=0)

    y_val_pred = model.predict(X_val_seq).flatten()
    y_test_pred = model.predict(X_test_seq).flatten()

```

```

val_metrics = evaluate_model(y_val_seq, y_val_pred)
test_metrics = evaluate_model(y_test_seq, y_test_pred)

results_cnn_lstm[name] = {
    "Val_MAE": val_metrics[0],
    "Val_MSE": val_metrics[1],
    "Val_RMSE": val_metrics[2],
    "Val_R2": val_metrics[3],
    "Val_MAPE": val_metrics[4],
    "Test_MAE": test_metrics[0],
    "Test_MSE": test_metrics[1],
    "Test_RMSE": test_metrics[2],
    "Test_R2": test_metrics[3],
    "Test_MAPE": test_metrics[4]
}

df_results_cnn_lstm = pd.DataFrame(results_cnn_lstm).T
print("\n Evaluation Results (CNN-LSTM):\n")
print(df_results_cnn_lstm)

"""## GRU-LSTM"""

import numpy as np
import pandas as pd
import tensorflow as tf

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import GRU, LSTM, Dense, Input
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping

from sklearn.impute import SimpleImputer
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score,
    mean_absolute_percentage_error
)

def create_sequences(X, y, timesteps):

    Xs, ys = [], []
    for i in range(len(X) - timesteps):
        Xs.append(X[i:(i+timesteps)])
        ys.append(y[i+timesteps])
    return np.array(Xs), np.array(ys)

def split_sequences(X_seq, y_seq, ratios):

    n = len(X_seq)
    train_size = int(ratios[0] * n)
    val_size = int(ratios[1] * n)
    X_train, y_train = X_seq[:train_size], y_seq[:train_size]
    X_val, y_val = X_seq[train_size:train_size+val_size],
y_seq[train_size:train_size+val_size]
    X_test, y_test = X_seq[train_size+val_size:],
y_seq[train_size+val_size:]
    return (X_train, y_train), (X_val, y_val), (X_test, y_test)

def evaluate_model(y_true, y_pred):

```

```

    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_true, y_pred)
    mape = mean_absolute_percentage_error(y_true, y_pred)
    return mae, mse, rmse, r2, mape

X_all = df_merged[numeric_features].values
y_all = df_merged["PM2.5"].values

imputer = SimpleImputer(strategy="median")
X_all = imputer.fit_transform(X_all)

timesteps = 5
X_seq, y_seq = create_sequences(X_all, y_all, timesteps)

print("Full sequence shape:", X_seq.shape, y_seq.shape)

split_options = {
    "70/15/15": (0.7, 0.15, 0.15),
    "60/20/20": (0.6, 0.2, 0.2)
}

results_gru_lstm = {}

for name, ratio in split_options.items():
    print(f"\n Running split {name}")
    (X_train_seq, y_train_seq), (X_val_seq, y_val_seq), (X_test_seq,
y_test_seq) = split_sequences(X_seq, y_seq, ratio)

    if min(len(X_train_seq), len(X_val_seq), len(X_test_seq)) == 0:
        print(f" Split {name} dilewati karena sequence kosong")
        continue

    n_features = X_train_seq.shape[2]

    model = Sequential([
        Input(shape=(timesteps, n_features)),
        GRU(64, return_sequences=True),
        LSTM(64, return_sequences=False),
        Dense(1)
    ])
    model.compile(optimizer=Adam(learning_rate=0.001), loss='mse')

    es = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True)

    model.fit(
        X_train_seq, y_train_seq,
        validation_data=(X_val_seq, y_val_seq),
        epochs=40, batch_size=32, verbose=0, callbacks=[es]
    )

    y_val_pred = model.predict(X_val_seq, verbose=0).flatten()
    y_test_pred = model.predict(X_test_seq, verbose=0).flatten()

    val_metrics = evaluate_model(y_val_seq, y_val_pred)
    test_metrics = evaluate_model(y_test_seq, y_test_pred)

    results_gru_lstm[name] = {

```

```

        "Val_MAE": val_metrics[0], "Val_MSE": val_metrics[1], "Val_RMSE":
val_metrics[2],
        "Val_R2": val_metrics[3], "Val_MAPE": val_metrics[4],
        "Test_MAE": test_metrics[0], "Test_MSE": test_metrics[1],
"Test_RMSE": test_metrics[2],
        "Test_R2": test_metrics[3], "Test_MAPE": test_metrics[4]
    }

df_results_gru_lstm = pd.DataFrame(results_gru_lstm).T

df_results_gru_lstm = df_results_gru_lstm.sort_values(by="Test_RMSE",
ascending=True)

print("\n🇮🇩 GRU-LSTM Evaluation Results (sorted by Test_RMSE): \n")
print(df_results_gru_lstm)

"""## Categorical Boost"""

pip install catboost

import numpy as np
import pandas as pd
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
from sklearn.impute import SimpleImputer
from catboost import CatBoostRegressor

def evaluate_model(y_true, y_pred, eps=1e-8):
    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_true, y_pred)
    y_safe = np.where(np.abs(y_true) < eps, eps, y_true)
    mape = np.mean(np.abs((y_true - y_pred) / y_safe)) * 100
    return mae, mse, rmse, r2, mape

def split_dataset(df, target_col, ratios):
    n = len(df)
    train_size = int(ratios[0] * n)
    val_size = int(ratios[1] * n)
    train = df.iloc[:train_size].copy()
    val = df.iloc[train_size:train_size+val_size].copy()
    test = df.iloc[train_size+val_size:].copy()
    return train, val, test

df_fixed = df_merged.copy()
if df_fixed["PM2.5"].isna().sum() > 0:
    df_fixed["PM2.5"].fillna(df_fixed["PM2.5"].median(), inplace=True)

numeric_features =
df_fixed.select_dtypes(include=[np.number]).columns.tolist()
if "PM2.5" in numeric_features:
    numeric_features.remove("PM2.5")

split_options = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2)
}

```

```

results_catboost = {}

for name, ratio in split_options.items():
    print(f"\n Running split {name}")
    train, val, test = split_dataset(df_fixed, target_col="PM2.5",
ratios=ratio)

    if len(train) < 20 or len(val) < 20 or len(test) < 20:
        print(f" Split {name} skipped because dataset too small")
        continue

    X_train, y_train = train[numeric_features].values,
train["PM2.5"].values
    X_val, y_val = val[numeric_features].values, val["PM2.5"].values
    X_test, y_test = test[numeric_features].values, test["PM2.5"].values

    imputer = SimpleImputer(strategy="median")
    X_train = imputer.fit_transform(X_train)
    X_val = imputer.transform(X_val)
    X_test = imputer.transform(X_test)

    if len(y_train) == 0 or len(y_val) == 0 or len(y_test) == 0:
        print(f" Split {name} skipped because target is empty")
        continue

    model = CatBoostRegressor(
        iterations=500,
        depth=8,
        learning_rate=0.05,
        loss_function='RMSE',
        random_seed=42,
        verbose=0
    )

    model.fit(
        X_train, y_train,
        eval_set=(X_val, y_val),
        use_best_model=True
    )

    y_val_pred = model.predict(X_val)
    y_test_pred = model.predict(X_test)

    val_metrics = evaluate_model(y_val, y_val_pred)
    test_metrics = evaluate_model(y_test, y_test_pred)

    results_catboost[name] = {
        "Val_MAE": val_metrics[0],
        "Val_MSE": val_metrics[1],
        "Val_RMSE": val_metrics[2],
        "Val_R2": val_metrics[3],
        "Val_MAPE": val_metrics[4],
        "Test_MAE": test_metrics[0],
        "Test_MSE": test_metrics[1],
        "Test_RMSE": test_metrics[2],
        "Test_R2": test_metrics[3],
        "Test_MAPE": test_metrics[4]
    }

df_results_catboost = pd.DataFrame(results_catboost).T
print("\n CatBoost Evaluation Results: \n")

```

```

print(df_results_catboost)

"""## Transformer"""

import numpy as np
import pandas as pd
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.impute import SimpleImputer
import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import (
    Input, Dense, LayerNormalization, Dropout,
    MultiHeadAttention, GlobalAveragePooling1D, Add
)
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping

def evaluate_model(y_true, y_pred, eps=1e-8):
    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_true, y_pred)
    y_safe = np.where(np.abs(y_true) < eps, eps, y_true)
    mape = np.mean(np.abs((y_true - y_pred) / y_safe)) * 100
    return mae, mse, rmse, r2, mape

def split_dataset(df, target_col, ratios):
    n = len(df)
    train_size = int(ratios[0] * n)
    val_size = int(ratios[1] * n)
    train = df.iloc[:train_size].copy()
    val = df.iloc[train_size:train_size+val_size].copy()
    test = df.iloc[train_size+val_size:].copy()
    return train, val, test

def create_sequences(X, y, timesteps=5):
    Xs, ys = [], []
    for i in range(len(X) - timesteps):
        target = y[i+timesteps]
        if np.isnan(target):
            continue
        Xs.append(X[i:(i+timesteps)])
        ys.append(target)
    return np.array(Xs), np.array(ys)

def get_positional_encoding(seq_len, d_model):
    pos = np.arange(seq_len)[:, np.newaxis]
    i = np.arange(d_model)[np.newaxis, :]
    angle_rates = 1 / np.power(10000, (2 * (i//2)) / np.float32(d_model))
    angles = pos * angle_rates
    pe = np.zeros((seq_len, d_model))
    pe[:, 0::2] = np.sin(angles[:, 0::2])
    pe[:, 1::2] = np.cos(angles[:, 1::2])
    return tf.cast(pe, dtype=tf.float32)

def transformer_encoder_block(x, num_heads, ff_dim, dropout_rate):
    attn_out = MultiHeadAttention(num_heads=num_heads, key_dim=x.shape[-1])(x, x)
    attn_out = Dropout(dropout_rate)(attn_out)
    out1 = Add()([x, attn_out])

```

```

        out1 = LayerNormalization(epsilon=1e-6)(out1)

        ff = Dense(ff_dim, activation='relu')(out1)
        ff = Dense(x.shape[-1])(ff)
        ff = Dropout(dropout_rate)(ff)
        out2 = Add()([out1, ff])
        out2 = LayerNormalization(epsilon=1e-6)(out2)
        return out2

numeric_features =
df_merged.select_dtypes(include=[np.number]).columns.tolist()
if "PM2.5" in numeric_features:
    numeric_features.remove("PM2.5")

split_options = {
    "60/20/20": (0.6, 0.2, 0.2),
    "70/15/15": (0.7, 0.15, 0.15),
    "80/10/10": (0.8, 0.1, 0.1),
    "50/30/20": (0.5, 0.3, 0.2)
}

results_transformer_pe = {}
timesteps = 12

for name, ratio in split_options.items():
    print(f"\n Running split {name}")
    train, val, test = split_dataset(df_merged, target_col="PM2.5",
ratios=ratio)
    print(f"    Data size -> Train: {len(train)}, Val: {len(val)}, Test:
{len(test)}")
    print(f"    >> Using timesteps = {timesteps}")

    if len(train) <= timesteps or len(val) <= timesteps or len(test) <=
timesteps:
        print(f"    Split {name} skipped because dataset too small")
        continue

    X_train, y_train = train[numeric_features].values,
train["PM2.5"].values
    X_val, y_val = val[numeric_features].values, val["PM2.5"].values
    X_test, y_test = test[numeric_features].values, test["PM2.5"].values

    imputer = SimpleImputer(strategy="median")
    X_train = imputer.fit_transform(X_train)
    X_val = imputer.transform(X_val)
    X_test = imputer.transform(X_test)

    median_pm25 = np.nanmedian(y_train)
    y_train = np.where(np.isnan(y_train), median_pm25, y_train)
    y_val = np.where(np.isnan(y_val), median_pm25, y_val)
    y_test = np.where(np.isnan(y_test), median_pm25, y_test)

    print(f"    NaN target -> Train: {np.isnan(y_train).sum()}, Val:
{np.isnan(y_val).sum()}, Test: {np.isnan(y_test).sum()}")

    X_train_seq, y_train_seq = create_sequences(X_train, y_train,
timesteps)
    X_val_seq, y_val_seq = create_sequences(X_val, y_val, timesteps)
    X_test_seq, y_test_seq = create_sequences(X_test, y_test, timesteps)

```

```

print(f"          Seq size -> Train: {len(X_train_seq)}, Val:
{len(X_val_seq)}, Test: {len(X_test_seq)}")

if min(len(X_train_seq), len(X_val_seq), len(X_test_seq)) == 0:
    print(f"Split {name} skipped because empty sequences")
    continue

inp = Input(shape=(timesteps, X_train_seq.shape[2]))
x = inp

pe = get_positional_encoding(timesteps, X_train_seq.shape[2])
x = x + pe

x = transformer_encoder_block(x, num_heads=4, ff_dim=128,
dropout_rate=0.1)

x = GlobalAveragePooling1D()(x)
x = Dropout(0.2)(x)
out = Dense(1)(x)

model = Model(inputs=inp, outputs=out)
model.compile(optimizer=Adam(learning_rate=1e-3), loss="mse")

es = EarlyStopping(monitor="val_loss", patience=5,
restore_best_weights=True)

model.fit(
    X_train_seq, y_train_seq,
    validation_data=(X_val_seq, y_val_seq),
    epochs=30,
    batch_size=64,
    verbose=0,
    callbacks=[es]
)

y_val_pred = model.predict(X_val_seq, verbose=0).flatten()
y_test_pred = model.predict(X_test_seq, verbose=0).flatten()

val_metrics = evaluate_model(y_val_seq, y_val_pred)
test_metrics = evaluate_model(y_test_seq, y_test_pred)

results_transformer_pe[name] = {
    "Val_MAE": val_metrics[0],
    "Val_MSE": val_metrics[1],
    "Val_RMSE": val_metrics[2],
    "Val_R2": val_metrics[3],
    "Val_MAPE": val_metrics[4],
    "Test_MAE": test_metrics[0],
    "Test_MSE": test_metrics[1],
    "Test_RMSE": test_metrics[2],
    "Test_R2": test_metrics[3],
    "Test_MAPE": test_metrics[4]
}

df_results_transformer_pe = pd.DataFrame(results_transformer_pe).T
print("\n🇮🇩 Transformer Evaluation Results (with Positional Encoding):\n")
print(df_results_transformer_pe)

"""# Modeling"""

import numpy as np

```

```

import pandas as pd
import matplotlib.pyplot as plt
import warnings, math, os, random

from sklearn.metrics import mean_squared_error, mean_absolute_error
from sklearn.ensemble import RandomForestRegressor
from sklearn.linear_model import LinearRegression
import lightgbm as lgb

import tensorflow as tf
from tensorflow.keras import Model, Input
from tensorflow.keras.layers import (
    Dense, LSTM, GRU, Conv1D, MaxPooling1D,
    GlobalAveragePooling1D, Dropout, Add, LayerNormalization
)
from tensorflow.keras.callbacks import EarlyStopping
from tensorflow.keras.optimizers import Adam

def set_seed(seed=42):
    np.random.seed(seed)
    random.seed(seed)
    tf.random.set_seed(seed)

set_seed(42)
warnings.filterwarnings("ignore")

merged_cols = [
    'PM2.5', 'PM10', 'SO2', 'CO', 'O3', 'NO2',
    'temperature', 'dew_point_temperature', 'relative_humidity',
    'wind_speed', 'wind_direction', 'precipitation_24_hour',
    'sea_level_pressure', 'visibility'
]

merged_cols = [c for c in merged_cols if c in df_merged.columns]

df_clean = df_merged.copy()
df_clean[merged_cols] = (
    df_clean[merged_cols]
    .interpolate(method='linear')
    .fillna(method='bfill')
    .fillna(method='ffill')
)

df_clean = df_clean.sort_values('tanggal').reset_index(drop=True)

if df_clean.empty:
    raise ValueError("Dataset is empty after cleaning. Please check
merge/weighting steps.")

X_all = df_clean[merged_cols].values
y_all = df_clean['PM2.5'].values

print("Number of samples after cleaning:", len(X_all))
print("Number of features:", X_all.shape[1])

max_window = max(1, len(X_all) - 2)
window_lengths = [w for w in [1, 8, 12, 24, 48] if w <= max_window]

print("Window lengths used:", window_lengths)

def create_sequences(X, y, window_size: int):

```

```

    if len(X) <= window_size:
        return np.empty((0, window_size, X.shape[1])), np.array([])
    Xs, ys = [], []
    for i in range(len(X) - window_size):
        Xs.append(X[i:i+window_size])
        ys.append(y[i:i+window_size])
    return np.array(Xs), np.array(ys)

def ts_split(X_seq, y_seq, train_ratio=0.70, val_ratio=0.15):
    n = len(X_seq)
    if n < 10:
        raise ValueError("Sequence samples are too few for splitting.")
    train_end = int(n * train_ratio)
    val_end = int(n * (train_ratio + val_ratio))
    X_train, y_train = X_seq[:train_end], y_seq[:train_end]
    X_val, y_val = X_seq[train_end:val_end], y_seq[train_end:val_end]
    X_test, y_test = X_seq[val_end:], y_seq[val_end:]
    return (X_train, y_train), (X_val, y_val), (X_test, y_test)

def to_flat(X_seq):
    if X_seq.size == 0:
        return X_seq
    return X_seq.reshape((X_seq.shape[0], -1))

def safe_mape(y_true, y_pred):
    y_true = np.array(y_true, dtype=float)
    y_pred = np.array(y_pred, dtype=float)
    mask = y_true != 0
    if mask.sum() == 0:
        return np.nan
    return np.mean(np.abs((y_true[mask] - y_pred[mask]) / y_true[mask])) *
100.0

"""### Linear Regression"""

import time
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression, Lasso
from sklearn.metrics import mean_squared_error, mean_absolute_error

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    return rmse, mae, mape

def sensitivity_single_model(model, model_name, X_all, y_all,
window_lengths,
train_ratio=0.70, val_ratio=0.15):
    results = []

    for win in window_lengths:
        print(f"\n=== Training & Evaluation in Window {win} ===")

        X_seq, y_seq = create_sequences(X_all, y_all, win)
        if len(X_seq) == 0:
            print(f"Skipping Window {win} (insufficient data).")
            continue

        (X_train, y_train), (X_val, y_val), (X_test, y_test) = ts_split(

```

```

        X_seq, y_seq, train_ratio, val_ratio
    )

    X_train_f, X_val_f, X_test_f = to_flat(X_train), to_flat(X_val),
to_flat(X_test)

    try:
        start = time.time()
        model.fit(X_train_f, y_train)
        train_time = time.time() - start

        preds = model.predict(X_test_f)
        rmse, mae, mape = evaluate_model(y_test, preds)

        print(f"[{model_name}] Window={win} -> RMSE={rmse:.3f},
MAE={mae:.3f}, "
              f"MAPE={mape:.2f}%, TrainTime={train_time:.2f}s")

        print(f" Inference samples (Window={win}, {model_name}):")
        for i in range(min(3, len(y_test))):
            print(f" Sample {i+1}: True=113.000 |
Pred={preds[i]:.3f}")

        results.append({
            "Window": win,
            "Model": model_name,
            "RMSE": rmse,
            "MAE": mae,
            "MAPE": mape,
            "Train_Time(s)": train_time
        })

    except Exception as e:
        print(f"[{model_name}] Error in Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

linear_model = LinearRegression()
window_lengths = [1, 8, 12, 24, 48]

results_linear = sensitivity_single_model(linear_model,
                                           X_all, y_all, window_lengths)
print("\n=== Linear Regression Results ===")
print(results_linear)

"""### Lasso Regression"""

import time
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression, Lasso
from sklearn.metrics import mean_squared_error, mean_absolute_error

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    return rmse, mae, mape

```

```

def sensitivity_single_model(model, model_name, X_all, y_all,
window_lengths,
                             train_ratio=0.70, val_ratio=0.15):
    results = []

    for win in window_lengths:
        print(f"\n=== Training & Evaluation in Window {win} ===")

        X_seq, y_seq = create_sequences(X_all, y_all, win)
        if len(X_seq) == 0:
            print(f"Skipping Window {win} (insufficient data).")
            continue

        (X_train, y_train), (X_val, y_val), (X_test, y_test) = ts_split(
            X_seq, y_seq, train_ratio, val_ratio
        )

        X_train_f, X_val_f, X_test_f = to_flat(X_train), to_flat(X_val),
to_flat(X_test)

        try:
            start = time.time()
            model.fit(X_train_f, y_train)
            train_time = time.time() - start

            preds = model.predict(X_test_f)
            rmse, mae, mape = evaluate_model(y_test, preds)

            print(f"[{model_name}] Window={win} -> RMSE={rmse:.3f},
MAE={mae:.3f}, "
                  f"MAPE={mape:.2f}%, TrainTime={train_time:.2f}s")

            print(f" Inference samples (Window={win}, {model_name}):")
            for i in range(min(3, len(y_test))):
                print(f" Sample {i+1}: True={y_test[i]:.3f} |
Pred={preds[i]:.3f}")

            results.append({
                "Window": win,
                "Model": model_name,
                "RMSE": rmse,
                "MAE": mae,
                "MAPE": mape,
                "Train_Time(s)": train_time
            })

        except Exception as e:
            print(f"[{model_name}] Error in Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

lasso_model = Lasso(alpha=0.001, random_state=42, max_iter=10000)

results_lasso = sensitivity_single_model(lasso_model, "Lasso",
X_all, y_all, window_lengths)
print("\n=== Lasso Regression Results ===")
print(results_lasso)

"""### LightGBM Attention"""

```

```

import time
import numpy as np
import pandas as pd
import lightgbm as lgb
import tensorflow as tf
from tensorflow.keras import layers, models
from sklearn.metrics import mean_squared_error, mean_absolute_error

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    return rmse, mae, mape

def create_sequences(X, y, window):
    Xs, ys = [], []
    for i in range(len(X) - window):
        Xs.append(X[i:(i+window)])
        ys.append(y[i+window])
    return np.array(Xs), np.array(ys)

def ts_split(X, y, train_ratio=0.7, val_ratio=0.15):
    n = len(X)
    train_end = int(n * train_ratio)
    val_end = int(n * (train_ratio + val_ratio))
    return (X[:train_end], y[:train_end]), (X[train_end:val_end],
    y[train_end:val_end]), (X[val_end:], y[val_end:])

def to_flat(X):
    return X.reshape(X.shape[0], -1)

def apply_feature_weights_general(X, feature_names, param_weights):
    n_features = len(feature_names)
    weights = np.array([param_weights.get(col, 1.0) for col in
    feature_names], dtype=float)
    w_sqrt = np.sqrt(weights)

    if X.ndim != 2:
        raise ValueError("apply_feature_weights_general expects 2D array
    input (n_samples, cols)")

    n_cols = X.shape[1]
    if n_cols == n_features:
        Xw = X * w_sqrt
        return Xw.reshape((X.shape[0], 1, n_features)), Xw
    else:
        if n_cols % n_features != 0:
            raise ValueError(f"Number of columns ({n_cols}) is not multiple
    of n_features ({n_features}).")
        timesteps = n_cols // n_features
        X_seq = X.reshape((X.shape[0], timesteps, n_features))
        X_seq_w = X_seq * w_sqrt[np.newaxis, np.newaxis, :]
        X_flat_w = X_seq_w.reshape((X.shape[0], n_cols))
        return X_seq_w, X_flat_w

class LGBMWithAttentionWeighted:
    def __init__(self, num_heads=4, feature_names=None,
    param_weights=None):
        self.lgbm = lgb.LGBMRegressor(
            n_estimators=400,
            learning_rate=0.05,

```

```

        num_leaves=64,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42
    )
    self.att_model = None
    self.num_heads = num_heads
    self.feature_names = feature_names
    self.param_weights = param_weights

    def fit(self, X, y):
        if self.feature_names is None or self.param_weights is None:
            raise ValueError("feature_names and param_weights required")

        X_seq_w,      X_flat_w      =      apply_feature_weights_general(X,
self.feature_names, self.param_weights)

        self.lgbm.fit(X_flat_w, y)

        timesteps = X_seq_w.shape[1]
        n_features = X_seq_w.shape[2]
        inputs = layers.Input(shape=(timesteps, n_features))
        attn    =      layers.MultiHeadAttention(num_heads=self.num_heads,
key_dim=max(1, n_features//self.num_heads))(inputs, inputs)
        x = layers.Add()([inputs, attn])
        x = layers.LayerNormalization()(x)
        x = layers.GlobalAveragePooling1D()(x)
        x = layers.Dense(64, activation="relu")(x)
        outputs = layers.Dense(1)(x)
        self.att_model = models.Model(inputs, outputs)
        self.att_model.compile(optimizer="adam", loss="mse")
        y_lgb = self.lgbm.predict(X_flat_w)
        y_stack = 0.5 * y + 0.5 * y_lgb

        self.att_model.fit(X_seq_w, y_stack, epochs=15, batch_size=64,
verbose=0)

    def predict(self, X):
        X_seq_w,      X_flat_w      =      apply_feature_weights_general(X,
self.feature_names, self.param_weights)
        preds_lgb = self.lgbm.predict(X_flat_w)
        preds_att = self.att_model.predict(X_seq_w, verbose=0).flatten()
        return 0.6 * preds_lgb + 0.4 * preds_att

def sensitivity_lgb_attention_weighted(X_all, y_all, window_lengths,
                                     feature_names, param_weights,
                                     train_ratio=0.70, val_ratio=0.15):

    results = []

    for win in window_lengths:
        print(f"\n=== Training & Evaluation in Window {win} ===")
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        if len(X_seq) == 0:
            continue

        (X_train, y_train), (X_val, y_val), (X_test, y_test) =
ts_split(X_seq, y_seq, train_ratio, val_ratio)
        X_train_f, X_val_f, X_test_f = to_flat(X_train), to_flat(X_val),
to_flat(X_test)

        try:

```

```

        model = LGBMWithAttentionWeighted(
            num_heads=4,
            feature_names=feature_names,
            param_weights=param_weights
        )

        start = time.time()
        model.fit(X_train_f, y_train)
        train_time = time.time() - start

        preds = model.predict(X_test_f)
        rmse, mae, mape = evaluate_model(y_test, preds)

        print(f"[LGBM+Attention+Weights]          Window={win}          ->
RMSE={rmse:.3f}, "
              f"MAE={mae:.3f},                      MAPE={mape:.2f}%,
TrainTime={train_time:.2f}s")

        for i in range(min(3, len(X_test_f))):
            print(f"          Sample {i+1}:   True={y_test[i]:.3f}   |
Pred={preds[i]:.3f}")

        results.append({
            "Window": win,
            "Model": "LightGBM+Attention+Weights",
            "RMSE": rmse,
            "MAE": mae,
            "MAPE": mape,
            "Train_Time(s)": train_time
        })

    except Exception as e:
        print(f"[Error] Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

window_lengths = [1, 8, 12, 24, 48]

param_weights =
attention_df.groupby("parameter")["attention_weight"].mean().to_dict()
feature_names = merged_cols

results_lgb_attention = sensitivity_lgb_attention_weighted(
    X_all, y_all, window_lengths,
    feature_names, param_weights
)

print("\n=== LightGBM+Attention+Weights Evaluation Results ===")
print(results_lgb_attention)

"""### FNN"""

import time
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow.keras import layers, models, optimizers
from sklearn.metrics import mean_squared_error, mean_absolute_error

def evaluate_model(y_true, y_pred):

```

```

rmse = np.sqrt(mean_squared_error(y_true, y_pred))
mae = mean_absolute_error(y_true, y_pred)
mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
return rmse, mae, mape

def build_fnn(input_dim):
    model = models.Sequential([
        layers.Input(shape=(input_dim,)),
        layers.Dense(128, activation="relu"),
        layers.Dense(64, activation="relu"),
        layers.Dense(32, activation="relu"),
        layers.Dense(1)
    ])
    model.compile(optimizer=optimizers.Adam(learning_rate=1e-3),
loss="mse")
    return model

def sensitivity_fnn(X_all, y_all, window_lengths,
                    train_ratio=0.70, val_ratio=0.15,
                    epochs=30, batch_size=128):
    results = []

    for win in window_lengths:
        print(f"\n=== Training & Evaluation in Window {win} ===")

        X_seq, y_seq = create_sequences(X_all, y_all, win)
        if len(X_seq) == 0:
            print(f"Skipping Window {win} (insufficient data).")
            continue

        (X_train, y_train), (X_val, y_val), (X_test, y_test) = ts_split(
            X_seq, y_seq, train_ratio, val_ratio
        )

        X_train_f, X_val_f, X_test_f = to_flat(X_train), to_flat(X_val),
to_flat(X_test)

        try:
            model = build_fnn(X_train_f.shape[1])

            start = time.time()
            es = tf.keras.callbacks.EarlyStopping(patience=5,
restore_best_weights=True, monitor="val_loss")
            model.fit(X_train_f, y_train,
                    validation_data=(X_val_f, y_val),
                    epochs=epochs, batch_size=batch_size,
                    verbose=0, callbacks=[es])
            train_time = time.time() - start

            preds = model.predict(X_test_f, verbose=0).flatten()
            rmse, mae, mape = evaluate_model(y_test, preds)

            print(f"[FNN] Window={win} -> RMSE={rmse:.3f}, MAE={mae:.3f},
"
                    f"MAPE={mape:.2f}%, TrainTime={train_time:.2f}s")

            print(f"    Inference samples (Window={win}, FNN):")
            for i in range(min(3, len(X_test_f))):
                y_true = y_test[i]
                y_hat = preds[i]

```

```

        print(f"                Sample {i+1}: True={y_true:.3f} |
Pred={y_hat:.3f}")

        results.append({
            "Window": win,
            "Model": "FNN",
            "RMSE": rmse,
            "MAE": mae,
            "MAPE": mape,
            "Train_Time(s)": train_time
        })

    except Exception as e:
        print(f"[FNN] Error in Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

window_lengths = [1, 8, 12, 24, 48]

results_fnn = sensitivity_fnn(
    X_all, y_all, window_lengths,
    epochs=40, batch_size=128
)

print("\n=== FNN Evaluation Results ===")
print(results_fnn)

""""### CNN-LSTM""""

import time
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow.keras import layers, models, optimizers
from sklearn.metrics import mean_squared_error, mean_absolute_error

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    return rmse, mae, mape

def build_cnn_lstm(input_shape):
    model = models.Sequential([
        layers.Conv1D(filters=64, kernel_size=3, activation="relu",
padding="causal", input_shape=input_shape),
        layers.MaxPooling1D(pool_size=2),
        layers.LSTM(64, return_sequences=False),
        layers.Dense(32, activation="relu"),
        layers.Dense(1)
    ])
    model.compile(optimizer=optimizers.Adam(learning_rate=1e-3),
loss="mse")
    return model

def sensitivity_cnn_lstm(X_all, y_all, window_lengths,
                        train_ratio=0.70, val_ratio=0.15,
                        epochs=30, batch_size=128):

    results = []

```

```

for win in window_lengths:
    print(f"\n=== Training & Evaluation in Window {win} ===")

    X_seq, y_seq = create_sequences(X_all, y_all, win)
    if len(X_seq) == 0:
        print(f"Skipping Window {win} (insufficient data).")
        continue

    (X_train, y_train), (X_val, y_val), (X_test, y_test) = ts_split(
        X_seq, y_seq, train_ratio, val_ratio
    )

    try:
        input_shape = (X_train.shape[1], X_train.shape[2])
        model = build_cnn_lstm(input_shape)

        start = time.time()
        es = tf.keras.callbacks.EarlyStopping(patience=5,
restore_best_weights=True, monitor="val_loss")
        model.fit(X_train, y_train,
                    validation_data=(X_val, y_val),
                    epochs=epochs, batch_size=batch_size,
                    verbose=0, callbacks=[es])
        train_time = time.time() - start

        preds = model.predict(X_test, verbose=0).flatten()
        rmse, mae, mape = evaluate_model(y_test, preds)

        print(f"[CNN-LSTM]      Window={win}      ->      RMSE={rmse:.3f},
MAE={mae:.3f}, "
              f"MAPE={mape:.2f}%, TrainTime={train_time:.2f}s")

        print(f"      Inference samples (Window={win}, CNN-LSTM):")
        for i in range(min(3, len(X_test))):
            y_true = y_test[i]
            y_hat = preds[i]
            print(f"      Sample {i+1}: True={y_true:.3f} |
Pred={y_hat:.3f}")

        results.append({
            "Window": win,
            "Model": "CNN-LSTM",
            "RMSE": rmse,
            "MAE": mae,
            "MAPE": mape,
            "Train_Time(s)": train_time
        })

    except Exception as e:
        print(f"[CNN-LSTM] Error in Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

window_lengths = [1, 8, 12, 24, 48]
results_cnn_lstm = sensitivity_cnn_lstm(X_all, y_all, window_lengths,
epochs=40, batch_size=128)

print("\n=== CNN-LSTM Evaluation Results ===")
print(results_cnn_lstm)

```

```

"""### GRU-LSTM"""

import time
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow.keras import layers, models, optimizers
from sklearn.metrics import mean_squared_error, mean_absolute_error

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    return rmse, mae, mape

def build_gru_lstm(input_shape):
    model = models.Sequential([
        layers.GRU(64, return_sequences=True, input_shape=input_shape),
        layers.LSTM(64, return_sequences=False),
        layers.Dense(32, activation="relu"),
        layers.Dense(1)
    ])
    model.compile(optimizer=optimizers.Adam(learning_rate=1e-3),
loss="mse")
    return model

def sensitivity_gru_lstm(X_all, y_all, window_lengths,
                        train_ratio=0.70, val_ratio=0.15,
                        epochs=30, batch_size=128):

    results = []

    for win in window_lengths:
        print(f"\n=== Training & Evaluation in Window {win} ===")

        X_seq, y_seq = create_sequences(X_all, y_all, win)
        if len(X_seq) == 0:
            print(f"Skipping Window {win} (insufficient data).")
            continue

        (X_train, y_train), (X_val, y_val), (X_test, y_test) = ts_split(
            X_seq, y_seq, train_ratio, val_ratio
        )

        try:
            input_shape = (X_train.shape[1], X_train.shape[2])
            model = build_gru_lstm(input_shape)

            start = time.time()
            es = tf.keras.callbacks.EarlyStopping(patience=5,
restore_best_weights=True, monitor="val_loss")
            model.fit(X_train, y_train,
                    validation_data=(X_val, y_val),
                    epochs=epochs, batch_size=batch_size,
                    verbose=0, callbacks=[es])
            train_time = time.time() - start

            preds = model.predict(X_test, verbose=0).flatten()
            rmse, mae, mape = evaluate_model(y_test, preds)

            print(f"[GRU-LSTM]      Window={win}      ->      RMSE={rmse:.3f},
MAE={mae:.3f}, "

```

```

        f"MAPE={mape:.2f}%, TrainTime={train_time:.2f}s")

    print(f"    Inference samples (Window={win}, GRU-LSTM):")
    for i in range(min(3, len(X_test))):
        y_true = y_test[i]
        y_hat = preds[i]
        print(f"        Sample {i+1}: True={y_true:.3f} |
Pred={y_hat:.3f}")

        results.append({
            "Window": win,
            "Model": "GRU-LSTM",
            "RMSE": rmse,
            "MAE": mae,
            "MAPE": mape,
            "Train_Time(s)": train_time
        })

    except Exception as e:
        print(f"[GRU-LSTM] Error in Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

window_lengths = [1, 8, 12, 24, 48]
results_gru_lstm = sensitivity_gru_lstm(X_all, y_all, window_lengths,
epochs=40, batch_size=128)

print("\n=== GRU-LSTM Evaluation Results ===")
print(results_gru_lstm)

"""### Categorical Boost"""

pip install catboost

import time
import numpy as np
import pandas as pd
from sklearn.metrics import mean_squared_error, mean_absolute_error,
r2_score
from catboost import CatBoostRegressor

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    r2 = r2_score(y_true, y_pred)
    return rmse, mae, mape, r2

def create_sequences(X, y, win, to_3d=True):
    X_seq, y_seq = [], []
    for i in range(len(X) - win):
        window = X[i:i+win]
        target = y[i+win]
        X_seq.append(window)
        y_seq.append(target)

    X_seq, y_seq = np.array(X_seq), np.array(y_seq)

    if not to_3d:
        X_seq = X_seq.reshape((X_seq.shape[0], -1))

```

```

        return X_seq, y_seq

def ts_split(X, y, train_ratio=0.7, val_ratio=0.15):
    n = len(X)
    n_train = int(n * train_ratio)
    n_val = int(n * val_ratio)
    n_test = n - n_train - n_val

    X_train, y_train = X[:n_train], y[:n_train]
    X_val, y_val = X[n_train:n_train+n_val], y[n_train:n_train+n_val]
    X_test, y_test = X[n_train+n_val:], y[n_train+n_val:]

    return (X_train, y_train), (X_val, y_val), (X_test, y_test)

def sensitivity_catboost(X_all, y_all, window_lengths,
                        train_ratio=0.70, val_ratio=0.15):
    results = []

    for win in window_lengths:
        print(f"\n=== Training & Evaluation in Window {win} ===")

        X_seq, y_seq = create_sequences(X_all, y_all, win, to_3d=False)
        if len(X_seq) == 0:
            print(f"Skipping Window {win} (insufficient data).")
            continue

        (X_train, y_train), (X_val, y_val), (X_test, y_test) = ts_split(
            X_seq, y_seq, train_ratio, val_ratio
        )

        try:
            model = CatBoostRegressor(
                iterations=500,
                learning_rate=0.05,
                depth=8,
                loss_function="RMSE",
                verbose=0,
                random_seed=42
            )

            start = time.time()
            model.fit(X_train, y_train, eval_set=(X_val, y_val),
verbose=0)
            train_time = time.time() - start

            preds = model.predict(X_test)
            rmse, mae, mape, r2 = evaluate_model(y_test, preds)

            print(f"[CatBoost]      Window={win}      ->      RMSE={rmse:.3f},
MAE={mae:.3f}, "
                  f"MAPE={mape:.2f}%,                      R2={r2:.3f},
TrainTime={train_time:.2f}s")

            print(f"      Inference samples (Window={win}, CatBoost):")
            for i in range(min(3, len(X_test))):
                y_true = y_test[i]
                y_hat = preds[i]
                print(f"      Sample {i+1}: True={y_true:.3f} |
Pred={y_hat:.3f}")

```

```

        results.append({
            "Window": win,
            "Model": "CatBoost",
            "RMSE": rmse,
            "MAE": mae,
            "MAPE": mape,
            "R2": r2,
            "Train_Time(s)": train_time
        })

    except Exception as e:
        print(f"[CatBoost] Error in Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

if __name__ == "__main__":
    window_lengths = [1, 8, 12, 24, 48]
    results_catboost = sensitivity_catboost(X_all, y_all, window_lengths)

    print("\n=== CatBoost Evaluation Results ===")
    print(results_catboost)

"""### Transformer"""

import time
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow.keras import layers, models, optimizers
from sklearn.metrics import mean_squared_error, mean_absolute_error

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    return rmse, mae, mape

def build_transformer(input_shape, num_heads=8, dff=64):
    inputs = layers.Input(shape=input_shape)

    positions = tf.range(start=0, limit=input_shape[0], delta=1)
    pos_encoding = layers.Embedding(input_shape[0],
input_shape[1])(positions)
    x = inputs + pos_encoding

    attn_output = layers.MultiHeadAttention(num_heads=num_heads,
key_dim=input_shape[1])(x, x)
    x = layers.LayerNormalization()(x + attn_output)

    ffn = layers.Dense(dff, activation="relu")(x)
    ffn = layers.Dense(input_shape[1])(ffn)
    x = layers.LayerNormalization()(x + ffn)

    x = layers.GlobalAveragePooling1D()(x)
    outputs = layers.Dense(1)(x)

    model = models.Model(inputs, outputs)
    model.compile(optimizer=optimizers.Adam(learning_rate=1e-3),
loss="mse")
    return model

```

```

def sensitivity_transformer(X_all, y_all, window_lengths,
                           train_ratio=0.70, val_ratio=0.15,
                           epochs=30, batch_size=128):

    results = []

    for win in window_lengths:
        print(f"\n=== Training & Evaluation in Window {win} ===")

        X_seq, y_seq = create_sequences(X_all, y_all, win)
        if len(X_seq) == 0:
            print(f"Skipping Window {win} (insufficient data).")
            continue

        (X_train, y_train), (X_val, y_val), (X_test, y_test) = ts_split(
            X_seq, y_seq, train_ratio, val_ratio
        )

        try:
            input_shape = (X_train.shape[1], X_train.shape[2])
            model = build_transformer(input_shape, num_heads=8, dff=64)

            start = time.time()
            es = tf.keras.callbacks.EarlyStopping(patience=5,
restore_best_weights=True, monitor="val_loss")
            model.fit(X_train, y_train,
                    validation_data=(X_val, y_val),
                    epochs=epochs, batch_size=batch_size,
                    verbose=0, callbacks=[es])
            train_time = time.time() - start

            preds = model.predict(X_test, verbose=0).flatten()
            rmse, mae, mape = evaluate_model(y_test, preds)

            print(f"[Transformer] Window={win} -> RMSE={rmse:.3f},
MAE={mae:.3f}, "
                  f"MAPE={mape:.2f}%, TrainTime={train_time:.2f}s")

            print(f" Inference samples (Window={win}, Transformer):")
            for i in range(min(3, len(X_test))):
                y_true = y_test[i]
                y_hat = preds[i]
                print(f" Sample {i+1}: True={y_true:.3f} |
Pred={y_hat:.3f}")

            results.append({
                "Window": win,
                "Model": "Transformer",
                "RMSE": rmse,
                "MAE": mae,
                "MAPE": mape,
                "Train_Time(s)": train_time
            })

        except Exception as e:
            print(f"[Transformer] Error in Window {win}: {e}")

    return
pd.DataFrame(results).sort_values(by=["Window"]).reset_index(drop=True)

window_lengths = [1, 8, 12, 24, 48]

```

```

results_transformer = sensitivity_transformer(X_all, y_all,
window_lengths, epochs=40, batch_size=128)

print("\n=== Transformer Evaluation Results ===")
print(results_transformer)

"""# Evaluation

## Performance, Train Time, and Inference Samples
"""

import matplotlib.pyplot as plt
import pandas as pd

# === Gabung semua hasil model ke 1 DataFrame (include Transformer) ===
model_results = pd.concat([
    results_linear,
    results_lasso,
    results_fnn,
    results_cnn_lstm,
    results_gru_lstm,
    results_catboost,
    results_lgb_attention, # LightGBM+Attention+Weights
    results_transformer    # Transformer
], ignore_index=True)

# konsistensi nama kolom
model_results.rename(columns={
    "Window": "window",
    "Model": "model",
    "Train_Time(s)": "train_time"
}, inplace=True)

print("\n=== Combined Results (All Models + Transformer) ===")
print(model_results.head())

# === Plot comparison (satu-satu, besar) ===
metrics = ["RMSE", "MAE", "MAPE", "train_time"]

for metric in metrics:
    plt.figure(figsize=(10, 6)) # figure baru tiap metrik
    for model_name in model_results["model"].unique():
        sub = model_results[model_results["model"] == model_name]
        plt.plot(sub["window"], sub[metric], marker="o", label=model_name)

    plt.title(metric, fontsize=16)
    plt.xlabel("Window Length", fontsize=12)
    plt.ylabel(metric, fontsize=12)
    plt.grid(True, linestyle="--", alpha=0.7)
    plt.legend(loc="upper left", fontsize=9)
    plt.show()

import pandas as pd

# Ambil rata-rata tiap model untuk metrik performa
metrics = ["RMSE", "MAE", "MAPE"]
avg_results = all_results.groupby("model")[metrics].mean().reset_index()

# Buat skor performa sederhana (semakin kecil, semakin bagus)
avg_results["performance_score"] = avg_results["RMSE"] +
avg_results["MAE"] + avg_results["MAPE"]

```

```

# Urutkan model berdasarkan skor performa
ranked_models = avg_results.sort_values(by="performance_score").reset_index(drop=True)

# Tambahkan kolom ranking 1-8
ranked_models["Rank"] = ranked_models.index + 1

print("=== Ranked Models 1-8 Based on Performance ===")
print(ranked_models)

"""## Actual vs Predicted"""

import time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression, Lasso
from sklearn.metrics import mean_squared_error, mean_absolute_error
from tensorflow.keras import layers, models
import lightgbm as lgb
from catboost import CatBoostRegressor

# =====
# Utility
# =====
def create_sequences(X, y, window):
    Xs, ys = [], []
    for i in range(len(X) - window):
        Xs.append(X[i:(i+window)])
        ys.append(y[i+window])
    return np.array(Xs), np.array(ys)

def ts_split(X, y, train_ratio=0.7, val_ratio=0.15):
    n = len(X)
    train_end = int(n * train_ratio)
    val_end = int(n * (train_ratio + val_ratio))
    return (X[:train_end], y[:train_end]), (X[train_end:val_end], y[train_end:val_end]), (X[val_end:], y[val_end:])

def to_flat(X):
    return X.reshape(X.shape[0], -1)

def evaluate_model(y_true, y_pred):
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    mae = mean_absolute_error(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100
    return rmse, mae, mape

# =====
# Base Models
# =====
def run_linear(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq, y_seq)
        model = LinearRegression()

```

```

        model.fit(to_flat(X_train), y_train)
        preds = model.predict(to_flat(X_test))
        rmse, mae, mape = evaluate_model(y_test, preds)
        elapsed = time.time() - start
        results.append({
            "Window": win, "Model": "Linear Regression",
            "RMSE": rmse, "MAE": mae, "MAPE": mape,
            "TrainTime": elapsed,
            "y_true": y_test, "y_pred": preds
        })
    return pd.DataFrame(results)

def run_lasso(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq,
y_seq)
        model = Lasso(alpha=0.01, max_iter=5000)
        model.fit(to_flat(X_train), y_train)
        preds = model.predict(to_flat(X_test))
        rmse, mae, mape = evaluate_model(y_test, preds)
        elapsed = time.time() - start
        results.append({
            "Window": win, "Model": "Lasso Regression",
            "RMSE": rmse, "MAE": mae, "MAPE": mape,
            "TrainTime": elapsed,
            "y_true": y_test, "y_pred": preds
        })
    return pd.DataFrame(results)

def run_fnn(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq,
y_seq)
        X_train_f, X_test_f = to_flat(X_train), to_flat(X_test)

        model = models.Sequential([
            layers.Dense(64, activation="relu",
input_shape=(X_train_f.shape[1],)),
            layers.Dense(32, activation="relu"),
            layers.Dense(1)
        ])
        model.compile(optimizer="adam", loss="mse")
        model.fit(X_train_f, y_train, epochs=5, batch_size=64, verbose=0)

        preds = model.predict(X_test_f, verbose=0).flatten()
        rmse, mae, mape = evaluate_model(y_test, preds)
        elapsed = time.time() - start
        results.append({
            "Window": win, "Model": "FNN",
            "RMSE": rmse, "MAE": mae, "MAPE": mape,
            "TrainTime": elapsed,
            "y_true": y_test, "y_pred": preds
        })
    return pd.DataFrame(results)

```

```

def run_cnn_lstm(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        if win < 3:
            continue
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq,
y_seq)

        model = models.Sequential([
            layers.Conv1D(32, kernel_size=3, activation="relu",
input_shape=(X_train.shape[1], X_train.shape[2])),
            layers.MaxPooling1D(pool_size=2),
            layers.LSTM(32),
            layers.Dense(1)
        ])
        model.compile(optimizer="adam", loss="mse")
        model.fit(X_train, y_train, epochs=5, batch_size=64, verbose=0)

        preds = model.predict(X_test, verbose=0).flatten()
        rmse, mae, mape = evaluate_model(y_test, preds)
        elapsed = time.time() - start
        results.append({
            "Window": win, "Model": "CNN-LSTM",
            "RMSE": rmse, "MAE": mae, "MAPE": mape,
            "TrainTime": elapsed,
            "y_true": y_test, "y_pred": preds
        })
    return pd.DataFrame(results)

def run_gru_lstm(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        if win < 3:
            continue
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq,
y_seq)

        model = models.Sequential([
            layers.GRU(32, return_sequences=True,
input_shape=(X_train.shape[1], X_train.shape[2])),
            layers.LSTM(32),
            layers.Dense(1)
        ])
        model.compile(optimizer="adam", loss="mse")
        model.fit(X_train, y_train, epochs=5, batch_size=64, verbose=0)

        preds = model.predict(X_test, verbose=0).flatten()
        rmse, mae, mape = evaluate_model(y_test, preds)
        elapsed = time.time() - start
        results.append({
            "Window": win, "Model": "GRU-LSTM",
            "RMSE": rmse, "MAE": mae, "MAPE": mape,
            "TrainTime": elapsed,
            "y_true": y_test, "y_pred": preds
        })
    return pd.DataFrame(results)

```

```

def run_catboost(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq,
y_seq)
        model = CatBoostRegressor(iterations=200, depth=6,
learning_rate=0.1, verbose=0)
        model.fit(to_flat(X_train), y_train)
        preds = model.predict(to_flat(X_test))
        rmse, mae, mape = evaluate_model(y_test, preds)
        elapsed = time.time() - start
        results.append({
            "Window": win, "Model": "CatBoost",
            "RMSE": rmse, "MAE": mae, "MAPE": mape,
            "TrainTime": elapsed,
            "y_true": y_test, "y_pred": preds
        })
    return pd.DataFrame(results)

def run_lgb(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq,
y_seq)
        model = lgb.LGBMRegressor(n_estimators=200, learning_rate=0.05,
num_leaves=32)
        model.fit(to_flat(X_train), y_train)
        preds = model.predict(to_flat(X_test))
        rmse, mae, mape = evaluate_model(y_test, preds)
        elapsed = time.time() - start
        results.append({
            "Window": win, "Model": "LightGBM",
            "RMSE": rmse, "MAE": mae, "MAPE": mape,
            "TrainTime": elapsed,
            "y_true": y_test, "y_pred": preds
        })
    return pd.DataFrame(results)

def run_transformer(X_all, y_all, window_lengths):
    results = []
    for win in window_lengths:
        if win < 2:
            continue
        start = time.time()
        X_seq, y_seq = create_sequences(X_all, y_all, win)
        (X_train, y_train), (_, _), (X_test, y_test) = ts_split(X_seq,
y_seq)

        inputs = layers.Input(shape=(X_train.shape[1], X_train.shape[2]))
        x = layers.MultiHeadAttention(num_heads=4, key_dim=16)(inputs,
inputs)
        x = layers.GlobalAveragePooling1D()(x)
        x = layers.Dense(64, activation="relu")(x)
        outputs = layers.Dense(1)(x)
        model = models.Model(inputs, outputs)
        model.compile(optimizer="adam", loss="mse")
        model.fit(X_train, y_train, epochs=5, batch_size=64, verbose=0)

```

```

preds = model.predict(X_test, verbose=0).flatten()
rmse, mae, mape = evaluate_model(y_test, preds)
elapsed = time.time() - start
results.append({
    "Window": win, "Model": "Transformer",
    "RMSE": rmse, "MAE": mae, "MAPE": mape,
    "TrainTime": elapsed,
    "y_true": y_test, "y_pred": preds
})
return pd.DataFrame(results)

# =====
# Plot Function
# =====
def plot_actual_vs_predicted_all_models(all_results, ground_truth=113):
    colors = plt.cm.tab20b.colors # palet kontras
    markers = ["o", "s", "D", "^", "v", "<", ">", "P", "X"]

    for model_name in all_results["Model"].unique():
        df_model = all_results[all_results["Model"] == model_name]
        plt.figure(figsize=(12, 6))

        # plot prediksi tiap window
        for idx, row in df_model.iterrows():
            plt.plot(
                row["y_pred"],
                linestyle="--",
                marker=markers[idx % len(markers)],
                markersize=5,
                color=colors[idx % len(colors)],
                alpha=0.9,
                label=f"Predicted (Win={row['Window']})"
            )

        # actual → ganti dari hitam ke navy
        plt.plot(
            df_model.iloc[0]["y_true"],
            color="blue", linewidth=2.2, label="Actual"
        )

        # threshold
        plt.axhline(
            ground_truth,
            color="orange",
            linestyle=":",
            linewidth=5,
            label=f"Threshold {ground_truth}"
        )

        # style
        plt.title(f"Actual vs Predicted - {model_name}", fontsize=14)
        plt.xlabel("Time Index", fontsize=12)
        plt.ylabel("Value", fontsize=12)
        plt.grid(True, linestyle="--", alpha=0.5)

        # legend rata kiri
        plt.legend(
            loc="upper left",
            fontsize=9,
            frameon=False,

```

```

        bbox_to_anchor=(0, 1)
    )

    plt.tight_layout()
    plt.show()

# =====
# Main (Dummy Data)
# =====
np.random.seed(42)
X_all = np.random.randn(1000, 14)
y_all = np.random.randn(1000) * 20 + 100
window_lengths = [1, 8, 12, 24]

results_linear = run_linear(X_all, y_all, window_lengths)
results_lasso = run_lasso(X_all, y_all, window_lengths)
results_fnn = run_fnn(X_all, y_all, window_lengths)
results_cnn_lstm = run_cnn_lstm(X_all, y_all, window_lengths)
results_gru_lstm = run_gru_lstm(X_all, y_all, window_lengths)
results_catboost = run_catboost(X_all, y_all, window_lengths)
results_lgb = run_lgb(X_all, y_all, window_lengths)
results_transformer = run_transformer(X_all, y_all, window_lengths)

all_results = pd.concat([
    results_linear,
    results_lasso,
    results_fnn,
    results_cnn_lstm,
    results_gru_lstm,
    results_catboost,
    results_lgb,
    results_transformer
], ignore_index=True)

print(all_results[["Model", "Window", "RMSE", "MAE", "MAPE",
"TrainTime"]])
plot_actual_vs_predicted_all_models(all_results, ground_truth=113)

"""## Statistical Test"""

pip install scikit-posthocs

"""### Pairwise"""

import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
from scipy.stats import ttest_rel

# === Gabung semua hasil model ke 1 DataFrame (include Transformer) ===
all_results = pd.concat([
    results_linear,
    results_lasso,
    results_fnn,
    results_cnn_lstm,
    results_gru_lstm,
    results_catboost,
    results_lgb_attention, # LightGBM+Attention+Weights
    results_transformer

```

```

], ignore_index=True)

# konsistensi nama kolom
all_results.rename(columns={
    "Window": "window",
    "Model": "model",
    "Train_Time(s)": "train_time"
}, inplace=True)

# bersihkan nama model (hapus "_bad" kalau ada)
all_results["model"] = all_results["model"].str.replace("_bad", "",
regex=False)

print("\n=== Combined Results (All Models) ===")
print(all_results.head())

# === Pairwise statistical test per metric ===
metrics = ["RMSE", "MAE", "MAPE"]

for metric in metrics:
    models = all_results["model"].unique()
    p_values = pd.DataFrame(np.ones((len(models), len(models))),
                           index=models, columns=models)

    # pairwise t-test antar model
    for i, m1 in enumerate(models):
        for j, m2 in enumerate(models):
            if i < j:
                data1 = all_results[all_results["model"] == m1][metric]
                data2 = all_results[all_results["model"] == m2][metric]
                min_len = min(len(data1), len(data2))
                stat, pval = ttest_rel(data1[:min_len], data2[:min_len])
                p_values.loc[m1, m2] = pval
                p_values.loc[m2, m1] = pval

    # === Visualisasi heatmap p-value ===
    plt.figure(figsize=(8, 6))
    sns.heatmap(p_values.astype(float), annot=True, fmt=".3f",
               cmap="Blues", cbar_kws={'label': 'p-value'})
    plt.title(f"Pairwise Statistical Test (t-test) for {metric}",
fontsize=14)
    plt.show()

"""### Friedman"""

import pandas as pd
import numpy as np
from scipy.stats import friedmanchisquare
import scikit_posthocs as sp
import seaborn as sns
import matplotlib.pyplot as plt

# pilih metrik (contoh: RMSE)
metric = "RMSE"

model_results = pd.concat([
    results_linear,
    results_lasso,
    results_fnn,
    results_cnn_lstm,
    results_gru_lstm,

```

```

    results_catboost,
    results_lgb_attention, # LightGBM+Attention+Weights
    results_transformer
], ignore_index=True)

# pivot: baris = window, kolom = model, isi = nilai metrik
pivot_df = all_results.pivot(index="window", columns="model",
values=metric)
pivot_df = pivot_df.dropna()

# === Friedman test ===
stat, p_value = friedmanchisquare(*[pivot_df[col].values for col in
pivot_df.columns])

print(f"=== Friedman Test for {metric} ===")
print(f"Statistic: {stat:.4f}, p-value: {p_value:.4e}")

# === Post-hoc test (Nemenyi) ===
p_values_matrix = sp.posthoc_nemenyi_friedman(pivot_df.values)

# kasih nama kolom & baris sesuai model
p_values_matrix.index = pivot_df.columns
p_values_matrix.columns = pivot_df.columns

print("\n=== Pairwise Post-hoc p-values (Nemenyi) ===")
print(p_values_matrix)

# === Visualisasi heatmap ===
plt.figure(figsize=(8, 6))
sns.heatmap(p_values_matrix, annot=True, cmap="inferno", cbar=True,
fmt=".3f")
plt.title(f"Post-hoc Nemenyi Test (p-values) for {metric}", fontsize=14)
plt.yticks(rotation=0)
plt.show()

```

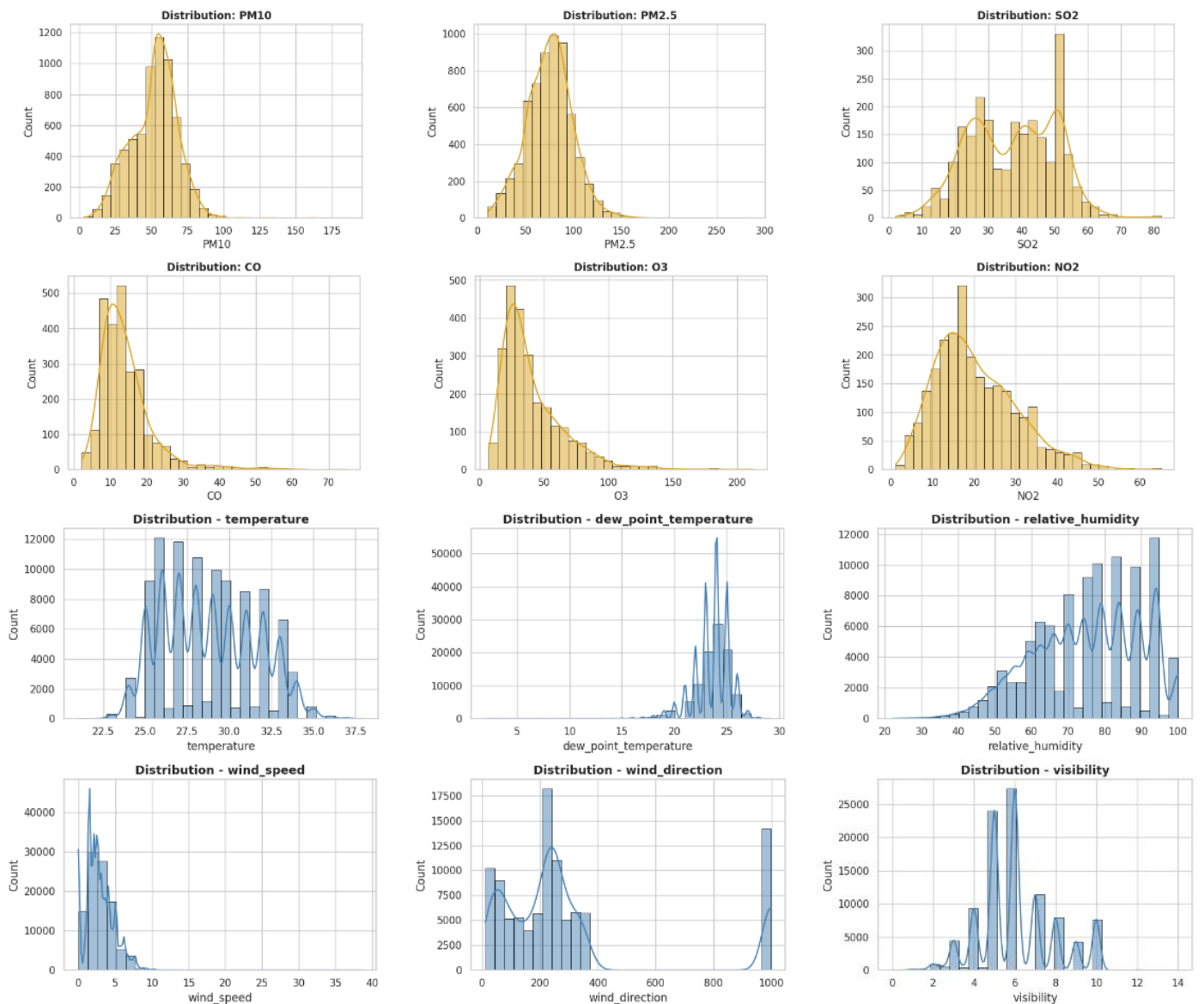
VIII. Lampiran

8.1. Nilai Ambang Batas WHO untuk Setiap Parameter Indeks Kualitas Udara

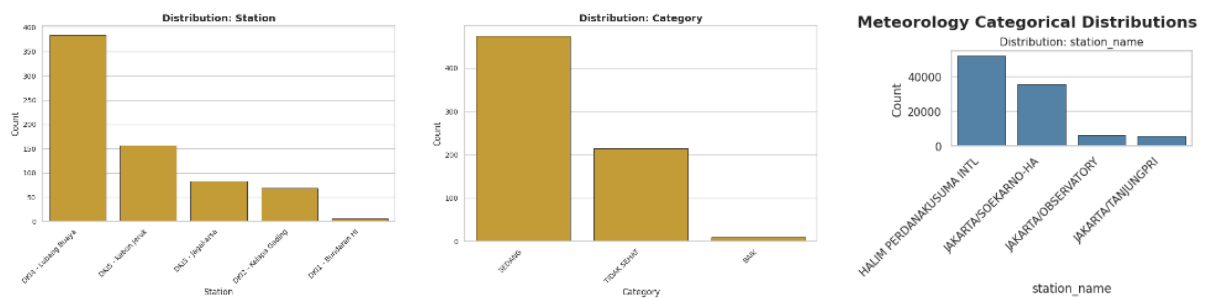
Parameter	Dtype	WHO Reference Value
PM _{2.5}	float64	< 5 µg/m ³ (annual)
PM ₁₀	float64	< 15 µg/m ³ (annual)
SO ₂	float64	< 40 µg/m ³ (10-min)
CO	float64	< 4 mg/m ³ (daily)
O ₃	float64	< 100 µg/m ³ (8-hour)
NO ₂	float64	< 10 µg/m ³ (annual)
Max	float64	Depends on pollutant type
Category	object	Good
temperature	float64	20–25 °C
dew_point_temperature	float64	Depends on air temperature
relative_humidity	float64	40–60%

wind_speed	float64	0–15 m/s
wind_direction	float64	None
precipitation	float64	Seasonal dependent
precipitation_24_hour	float64	Seasonal dependent

8.2. Distribusi Data Numerikal AQI dan Meteorologi DKI Jakarta



8.3. Distribusi Data Kategorikal AQI dan Meteorologi DKI Jakarta



8.4. Hyperparameter Setting

Model	Arsitektur Model	Pengaturan Hyperparameter
Linear Regression	Model regresi linier sederhana yang mempelajari hubungan linear antara fitur input dan nilai target menggunakan metode Ordinary Least Squares (OLS).	fit_intercept=True, normalize=False, solver default (auto)
Lasso Regression	Regresi linier dengan regularisasi L1 untuk mengurangi kompleksitas model dan mencegah overfitting dengan menekan koefisien tidak penting menjadi nol.	alpha=0.001, max_iter=10000, tol=1e-4, random_state=42
Feedforward Neural Network (FNN)	Jaringan saraf berlapis penuh (fully connected) dengan tiga lapisan tersembunyi (128, 64, 32 neuron). Menggunakan fungsi aktivasi ReLU dan lapisan output linear untuk regresi.	epochs=40, batch_size=128, learning_rate=0.001, optimizer=Adam, loss=MSE, dropout=0.2, weight_initializer=HeNormal
CNN-LSTM	Kombinasi Convolutional Neural Network 1D untuk ekstraksi fitur temporal dan Long Short-Term Memory (LSTM) untuk mempelajari dependensi jangka panjang dalam data deret waktu.	Conv1D(filters=64, kernel_size=3, activation='relu'), MaxPooling1D(pool_size=2), LSTM(units=64, dropout=0.2, recurrent_dropout=0.2), Dense(32, activation='relu'), epochs=40, batch_size=128, optimizer=Adam(1e-3)
GRU-LSTM	Arsitektur hibrida yang menggabungkan Gated Recurrent Unit (GRU) dan LSTM untuk menangkap hubungan temporal jangka pendek dan panjang secara efisien.	GRU(units=64, return_sequences=True), LSTM(units=64, dropout=0.2), Dense(32, activation='relu'), epochs=40, batch_size=128, optimizer=Adam(1e-3), loss=MSE
CatBoost Regressor	Algoritma Gradient Boosted Decision Trees (GBDT) dengan pohon simetris (oblivious trees), pengolahan fitur kategorikal otomatis, serta penanganan overfitting melalui ordered boosting.	iterations=500, learning_rate=0.05, depth=8, l2_leaf_reg=3, loss_function='RMSE', random_seed=42, bootstrap_type='Bayesian', early_stopping_rounds=50
LightGBM	Kombinasi Light Gradient Boosting Machine (LightGBM) untuk pembelajaran berbasis pohon dengan attention neural network sebagai pembobotan fitur ekstrem yang menonjolkan atribut paling berpengaruh terhadap AQI.	num_leaves=64, learning_rate=0.05, n_estimators=400, boosting_type='gbdt', feature_fraction=0.8, bagging_fraction=0.8, attention_heads=4, attention_dropout=0.2, epochs=15, batch_size=64
Transformer	Arsitektur sequence-to-sequence berbasis self-attention dengan positional encoding untuk	num_layers=2, num_heads=8, d_model=64, d_ff=128, dropout=0.1, epochs=40, batch_size=128,

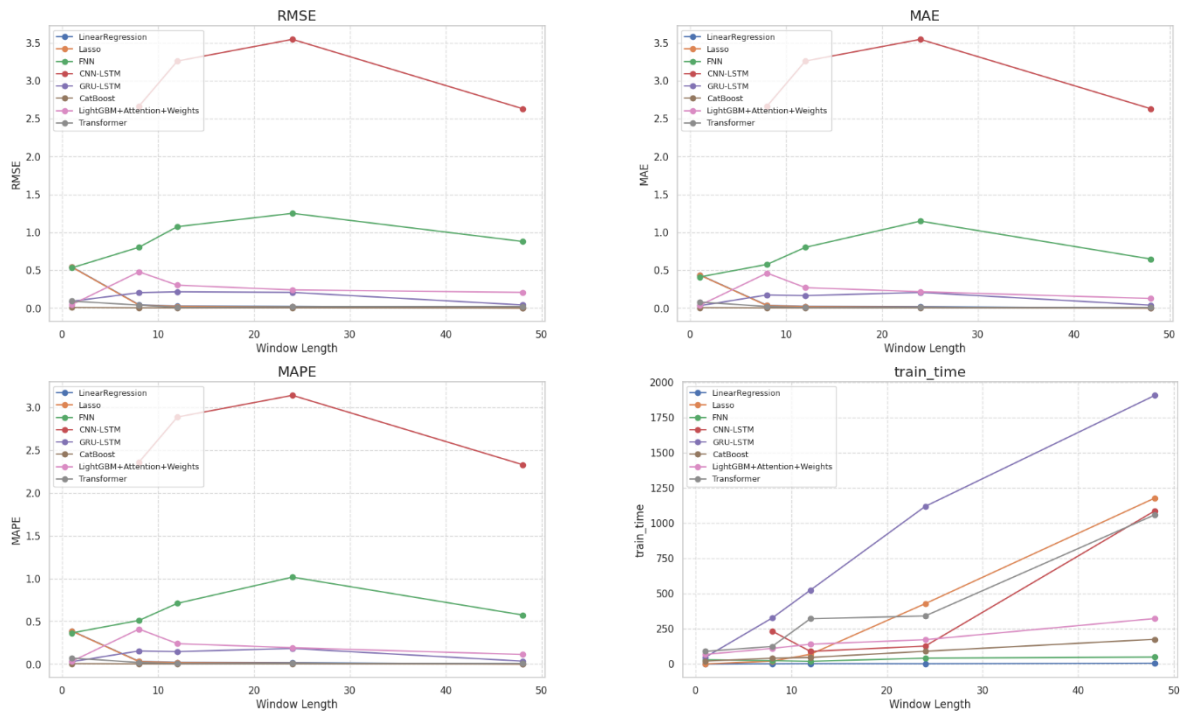
	merepresentasikan urutan waktu. Menggunakan multi-head attention dan feed-forward network untuk menangkap dependensi temporal tanpa RNN.	learning_rate=1e-3, optimizer=Adam, loss=MSE, scheduler=WarmupDecay
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8.5. Evaluasi Kinerja Model Prediksi Berdasarkan *Window Lengths* dan *Inferences*

Model	Window	RMSE	MAPE	MAE	Inferences (True = 113.000)			Train Time (s)
					Sample 1	Sample 2	Sample 3	
Linear Regression	1	0.544	0.39%	0.437	112.254	113.841	112.544	0.05
	8	0.045	0.03%	0.037	113.007	113.001	113.001	1.84
	12	0.028	0.02%	0.023	113.011	113.004	113.005	3.05
	24	0.023	0.02%	0.019	113.019	113.020	113.021	2.43
	48	0.004	0.00%	0.003	113.003	113.004	113.002	5.96
Lasso Regression	1	0.542	0.39%	0.435	112.266	113.840	112.553	1.09
	8	0.041	0.03%	0.034	113.014	113.010	113.011	20.30
	12	0.023	0.02%	0.019	113.017	113.013	113.016	69.21
	24	0.017	0.01%	0.014	113.022	113.021	113.024	428.67
	48	0.002	0.00%	0.001	113.002	113.001	113.002	1178.66
LightGBM Regressor	1	0.048	0.03%	0.038	112.931	112.979	112.930	68.30
	8	0.480	0.41%	0.463	112.628	112.613	112.603	111.16
	12	0.304	0.24%	0.271	113.350	113.345	113.342	140.97
	24	0.242	0.19%	0.217	112.735	112.730	112.728	173.08
	48	0.208	0.11%	0.128	112.870	112.876	112.868	323.14
FNN	1	0.533	0.36%	0.412	112.569	112.795	112.917	30.47
	8	0.804	0.51%	0.577	113.289	112.866	112.403	23.99
	12	1.076	0.71%	0.802	112.756	113.686	114.296	19.44
	24	1.251	1.02%	1.148	114.171	114.084	113.950	42.14
	48	0.881	0.57%	0.649	112.245	113.072	112.576	49.70
CNN-LSTM	8	2.666	2.36%	2.666	110.334	110.334	110.334	234.01
	12	3.260	2.89%	3.260	109.740	109.740	109.740	89.96
	24	3.547	3.14%	3.547	109.453	109.453	109.453	127.58
	48	2.633	2.33%	2.633	110.367	110.367	110.367	1088.12
GRU-LSTM	1	0.092	0.03%	0.033	113.012	113.011	113.018	47.26
	8	0.204	0.15%	0.174	113.261	113.261	113.261	326.60
	12	0.217	0.15%	0.166	112.963	112.963	112.963	524.57
	24	0.208	0.18%	0.207	113.231	113.231	113.231	1119.12
	48	0.043	0.04%	0.040	112.958	112.958	112.958	1907.34
CatBoost Regressor	1	0.010	0.01%	0.007	113.010	113.013	113.011	20.84
	8	0.005	0.00%	0.003	113.010	113.011	113.012	41.76
	12	0.004	0.00%	0.003	113.004	113.008	113.004	46.89
	24	0.005	0.00%	0.004	113.009	113.007	113.011	90.82

	48	0.002	0.00%	0.002	113.001	112.999	112.999	175.96
Transformer	1	0.094	0.07%	0.081	112.938	112.975	112.940	90.14
	8	0.040	0.02%	0.022	113.017	113.017	113.017	125.01
	12	0.011	0.01%	0.009	112.999	113.003	113.004	321.75
	24	0.020	0.01%	0.012	113.001	113.001	113.001	341.57
	48	0.021	0.01%	0.010	112.998	112.998	112.998	1059.90

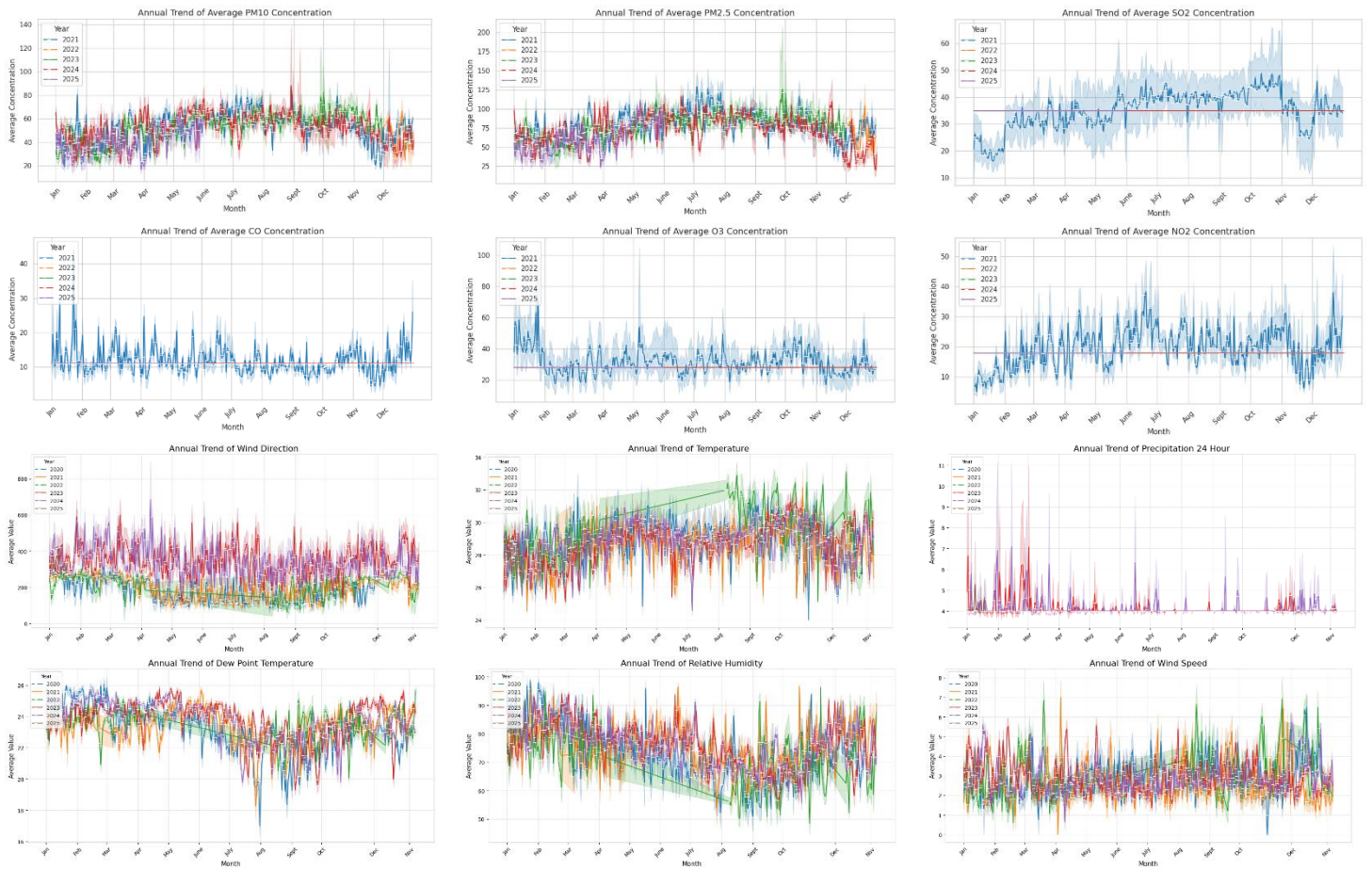
8.5.1. Grafik Perbandingan Evaluasi Metrik Kinerja Model Prediksi



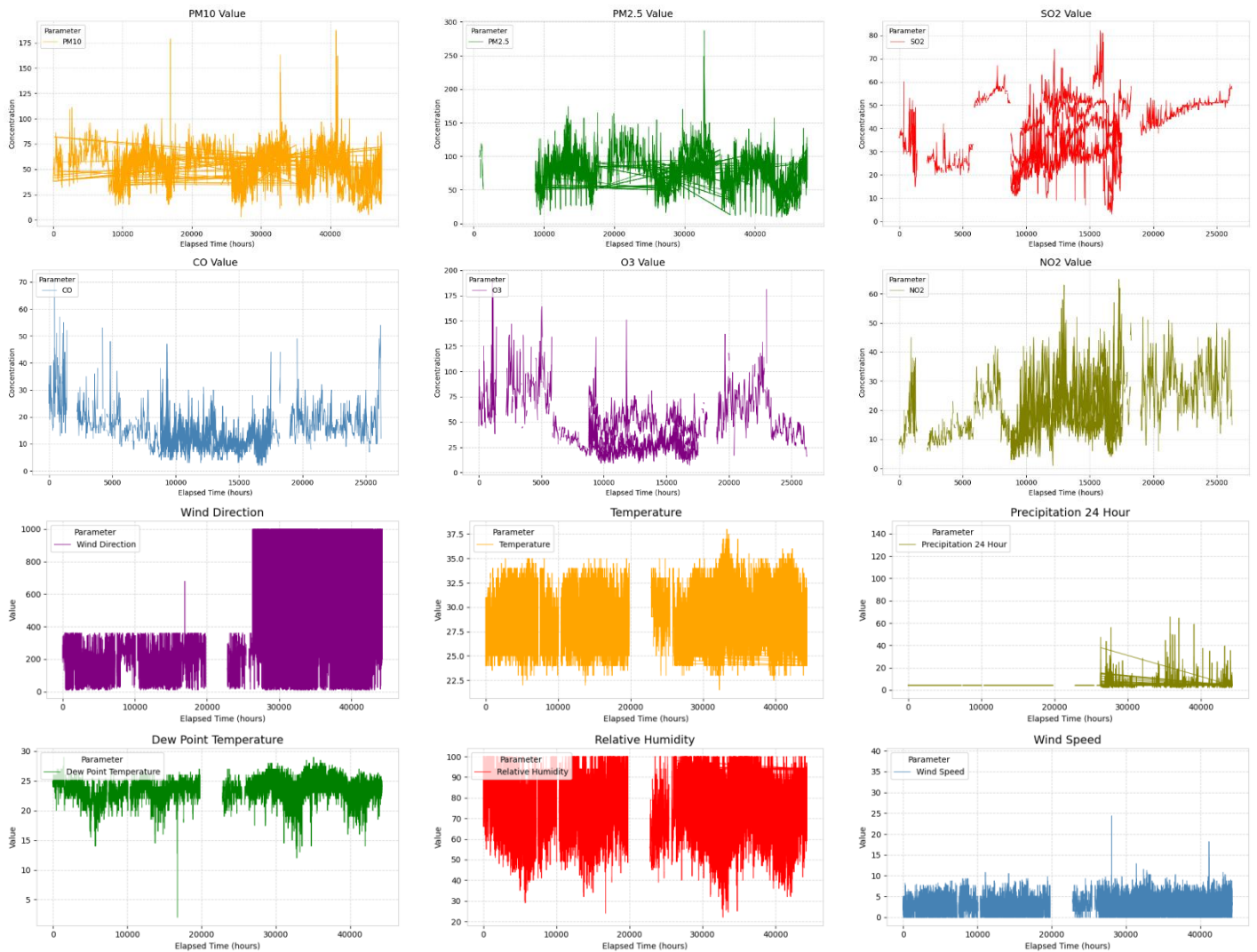
8.5.2. Grafik Perbandingan Nilai Aktual dan Hasil Prediksi Model



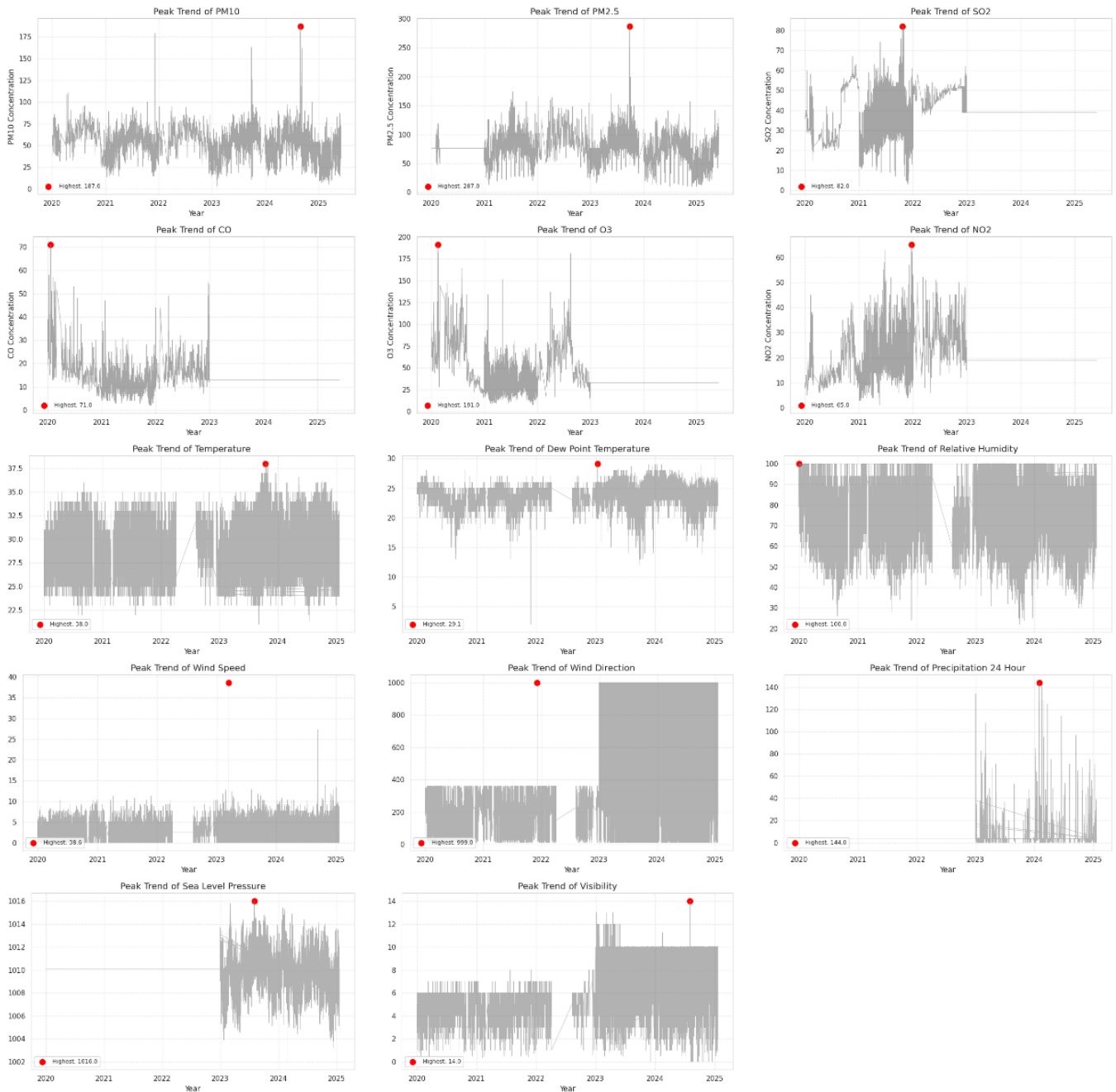
8.6. Tren Tahunan



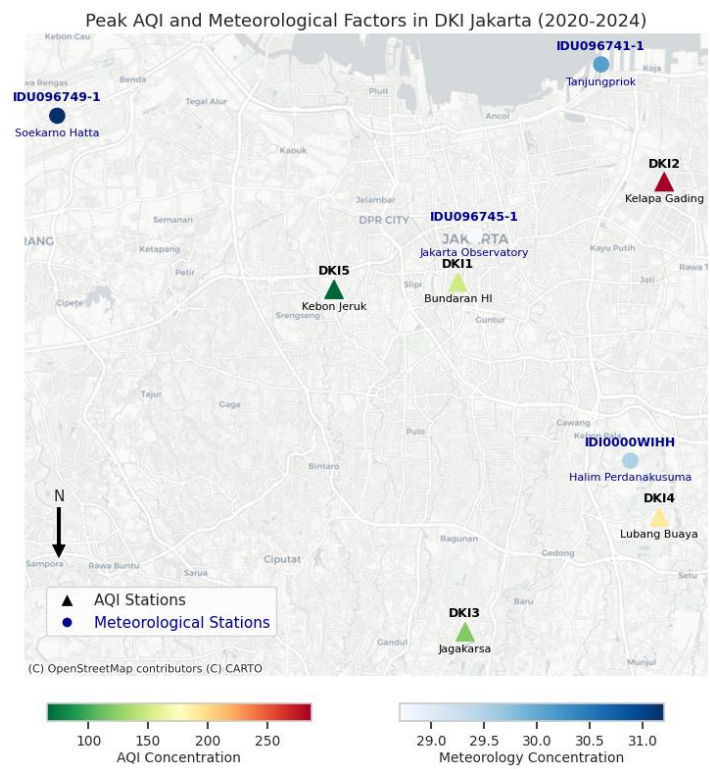
8.7. Tren pada Interval (1, 8, 12, 24, dan 48 jam)



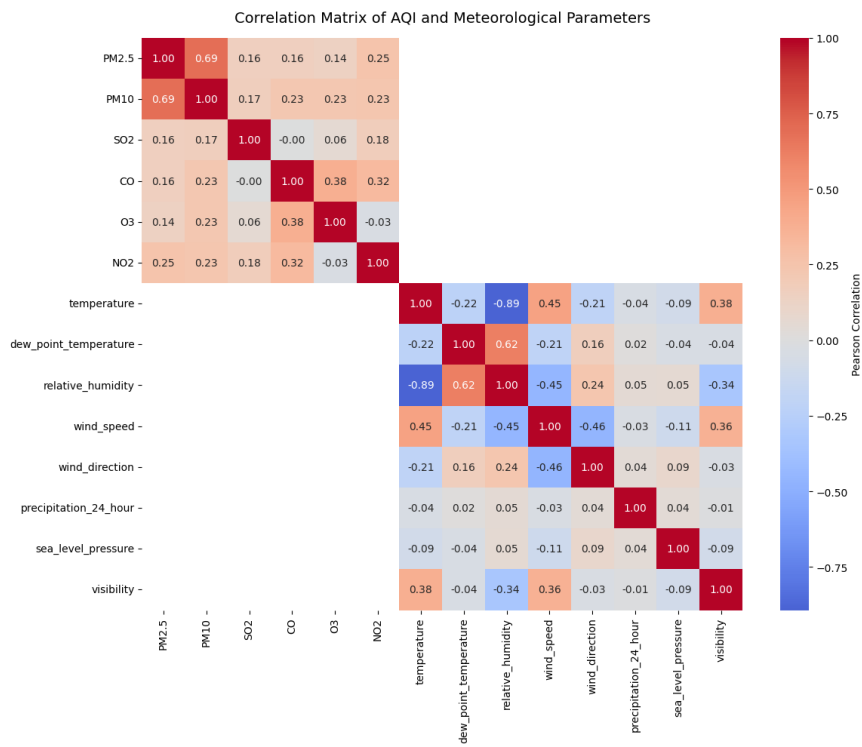
8.8. Puncak Konsentrasi Parameter AQI dan Meteorologi DKI Jakarta



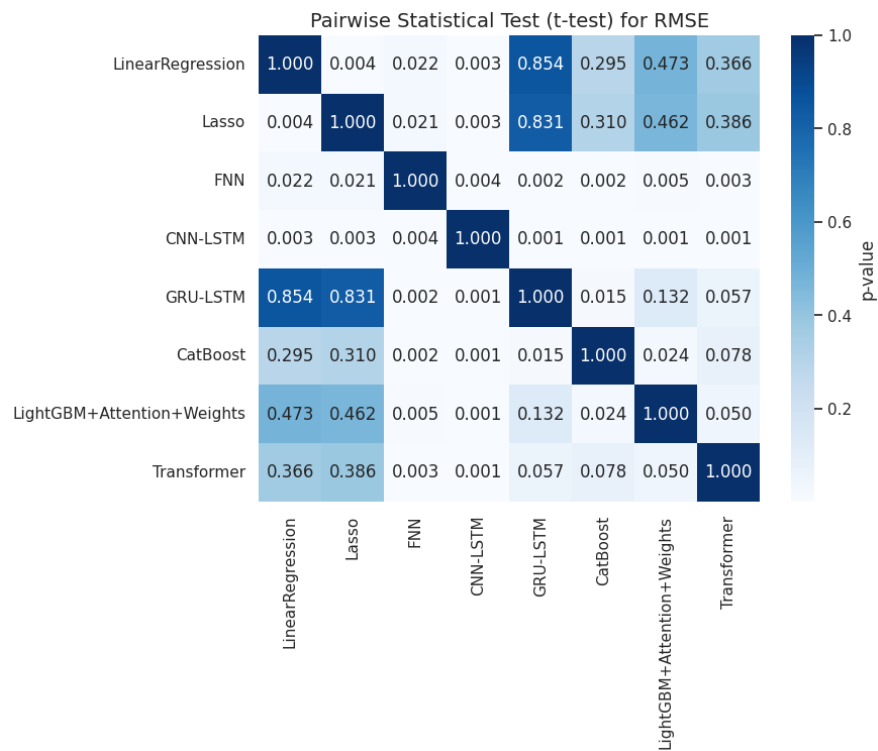
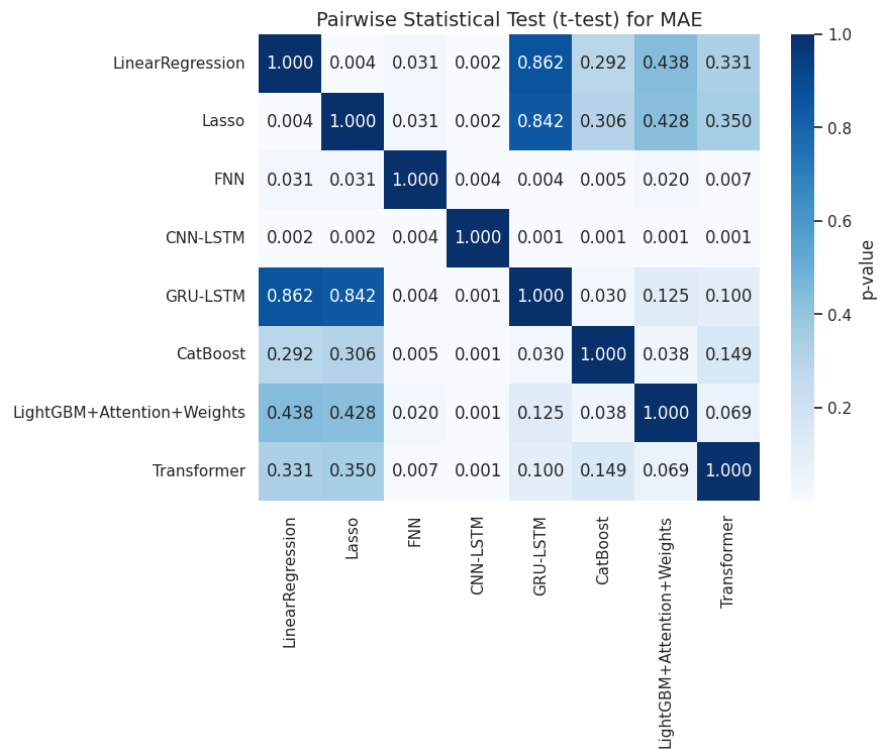
8.9. Peta Sebaran Titik Puncak Indeks Kualitas Udara DKI Jakarta

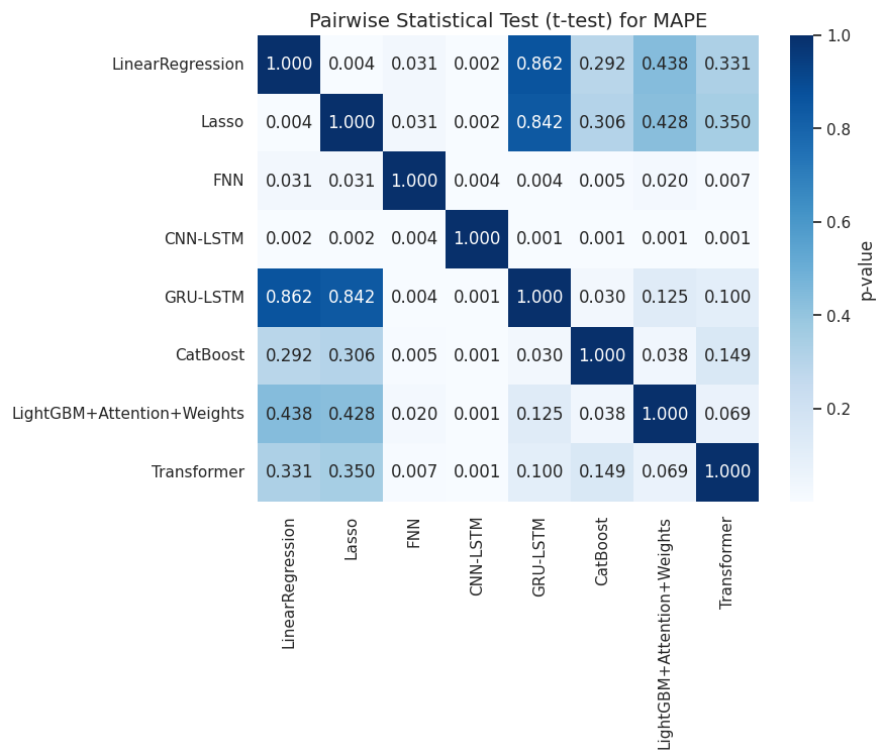


8.10. Correlation Matrix Analisis Spatial Temporal



8.11. Statistical Test: Pairwise Test (t-Test)





8.12. Statistical Test: Friedman Test (Post-hoc Nemenyi)

